

How the World Is Organised

A Visual Account

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Preface

This book is a catalogue of the visual forms the sciences use to organise what they know — roughly a hundred of them, from Playfair’s bars to the Higgs bump. Each form is reproduced from data and code, not reprinted; each carries a citable account of where it came from; and the whole set is arranged in a single teaching order that runs from the raw perceptual channels a mark can exploit, through the Gestalt principles that let marks group into structure, to the dynamic and uncertain territory beyond the static frame.

Two disciplines govern every entry. The figure is **reproduced, never reprinted** — the historical originals are frequently under copyright, so each is redrawn from data or a faithful model instance, with the original described in prose and cited. And the provenance is **citable, never asserted** — an entry is not finished until its canonical-introduction citation resolves in the bibliography. Where the first appearance and the canonical reference are different documents (they often are), both are given and kept distinct.

The book renders from one set of sources to two targets: the online edition you may be reading now, and a print edition built with LuaLaTeX. Every figure is generated by the executable code shown with it.

How to read the Atlas

Each entry follows one fixed template — provenance class, canonical introduction, domain, a gloss of one or two sentences, then the reproduction

Preface

— so the set reads as a catalogue, not an anthology. The provenance classes are defined at the head of the Atlas. Entries are numbered 1–103 and sit under unit headings that carry the perceptual argument; the AIATSIS map of Indigenous Australia stands before them all as an acknowledgement of the country on which this book was written.

1 The Argument

A chart is a claim about how the world is organised. The scatter that became the Hertzsprung–Russell diagram claims that a star’s brightness and colour are not independent; the map Snow drew of Broad Street claims that cholera deaths have a centre; Anscombe’s four little panels claim that a summary statistic is not a substitute for looking. The history of scientific visualisation is not a history of decoration — it is a history of structural claims that happened to be made with ink.

The claims succeed or fail through the eye. Human vision resolves a small set of channels before attention gets involved — position, length, size, colour, orientation, motion — and resolves them with sharply different precision; position on a common scale is read almost losslessly, area and colour saturation are read with systematic bias (Cleveland and McGill 1984). Bertin catalogued these channels as a grammar of marks (Bertin 1967), and the catalogue explains why the same data can be legible in one encoding and mute in another. The first part of the Atlas walks through the channels one at a time, each illustrated not by a toy but by the canonical figure that made the channel famous.

Perception does more than measure single marks — it groups them. Marks that connect read as one structure; marks that flow read as a path; marks that sit close, or look alike, read as a cluster; a boundary read as enclosure turns space itself into an encoding. These Gestalt principles are the load-bearing walls of the network diagram, the Sankey, the heatmap, and the map, and they organise the second part. The third part leaves the static frame: systems that move, figures that animate, and the honest depiction of what is not known — uncertainty drawn rather than suppressed.

1 *The Argument*

Provenance is treated as data. *First* and *canonical* are distinct facts: the treemap is introduced at a 1991 conference but cited from a 1992 journal article, and conflating the two is the most common error in visual-history writing. Each Atlas entry therefore declares its provenance class — a named first, a canonical reference, a shared genre idiom with no single inventor, or an attribution that remains contested — and backs the claim with a citation that must resolve for the entry to exist. Some famous forms are excluded on principle: the supply–demand cross and the political compass are conceptual diagrams, not depictions of measured data, and this book catalogues only forms that show the world rather than a theory of it.

What the catalogue adds up to is the book’s argument: the sciences do not share a subject matter, but they share a visual language, and that language is small. A few perceptual channels, a few grouping principles, a few honest ways to draw a distribution or a flow or a doubt — the whole of organised knowledge is drawn with them.

2 Atlas

Every entry below follows one fixed template — a provenance class, a citable canonical introduction, the domain the form organises, a gloss of one or two sentences, and a reproduction generated from data or a faithful model instance. The set is meant to read as a catalogue, not an anthology: the shape repeats so that the differences between forms, not the differences between write-ups, do the work. Each entry carries a *citable canonical introduction* rather than an asserted origin, and where the earliest instance and the cited introduction are different documents both are named.

Class	Means
Named first	A specific, attributable earliest instance.
Canonical reference	The publication the field cites as the introduction, which may post-date the first.
Shared genre idiom	A form with no single inventor; emerged across multiple hands.
Uncertain attribution	Contested or unverified origin; flagged, not resolved.

The order is a teaching order, not a chronology and not a subject index. It runs from the raw perceptual channels a single mark can exploit (Part A), through the Gestalt principles that let many marks group into structure (Part B), to the dynamic and uncertain territory beyond the static frame (Part C). Entries keep a running number from 1 to 103 and sit under

2 *Atlas*

unit headings that carry the argument. The AIATSIS map of Indigenous Australia stands before them all.

Every figure is redrawn from data or a reproduced model instance, with one deliberate exception: the AIATSIS map that opens the book is reproduced with attribution. Every other historical image is described in prose and cited. Reproductions use representative data where the original series is not to hand — enough to show the form the field made canonical, never passed off as the original measurement.

3 Prelude — Acknowledgement of Country

3.1 AIATSIS Map of Indigenous Australia

Class: named first

Canonical introduction: (Horton 1996)

Domain: country, language and nation groups across the Australian continent

This book was written on Aboriginal land. David Horton's map for the Australian Institute of Aboriginal and Torres Strait Islander Studies renders the general locations of more than 250 language, social and nation groups across the continent — an attempt to show, on one sheet, that the land was never empty (Horton 1996). Its boundaries are deliberately soft: they are a general guide, contested in places, and explicitly not for use in native-title claims. The map is reproduced below with attribution; the problem of drawing a curved continent onto a flat sheet returns as a formal question at entry 81.

3 Prelude — Acknowledgement of Country

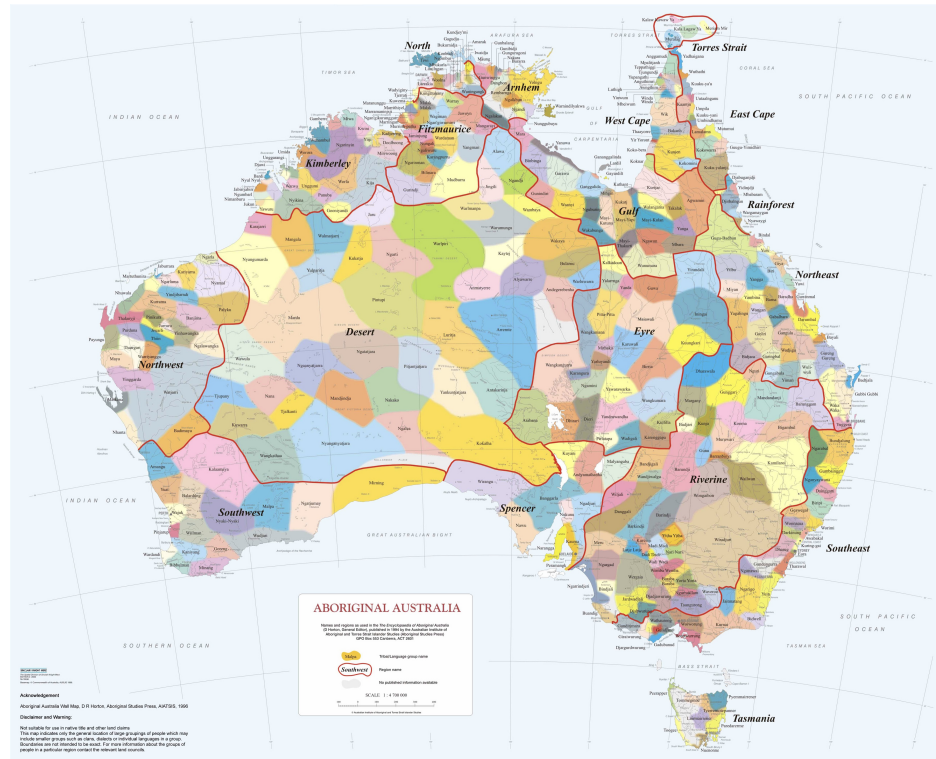


Figure 3.1: The AIATSIS map of Indigenous Australia. Aboriginal Australia Wall Map, David R Horton (creator), © Aboriginal Studies Press, AIATSIS and Auslig/Sinclair, Knight, Merz, 1996. The map indicates only the general location of larger groupings of people; boundaries are not intended to be exact, and it is not suitable for use in native title or other land claims.

4 Day One — cold open

Two figures open the book because they make its case before any theory does. Anscombe's quartet (entry 18) holds four datasets with the same mean, variance, correlation and regression line, and four completely different shapes — the argument for why a summary statistic is not a substitute for a picture. John Snow's cholera map (entry 78) is the consequence: a picture that was an argument, and that argument changed what a city did about its water. Both live in their proper units below; they are named here because everything between them is an elaboration of the same two claims — *look at the data*, and *a good picture is a claim about cause*.

5 Part A — Encoding & the pre-attentive channels

The pre-attentive channels — position, length, size, colour, orientation — are read before attention lands, and each is read at a sharply different precision. This part walks them one at a time, each illustrated by the canonical figure that made the channel famous.

5.1 A0. The channels themselves

After Bertin, Sémiologie graphique (1967).

5.1.1 1 — Bertin’s visual-variables chart

Class: named first

Canonical introduction: (Bertin 1967)

Domain: the grammar of marks

Every mark on a plane carries information through a fixed set of retinal variables — position, size, value, texture, colour, orientation, shape — and Bertin was the first to name them as a system and rank their perceptual power (Bertin 1967). The chart holds every variable constant but one, so each row shows a single channel varying in isolation.

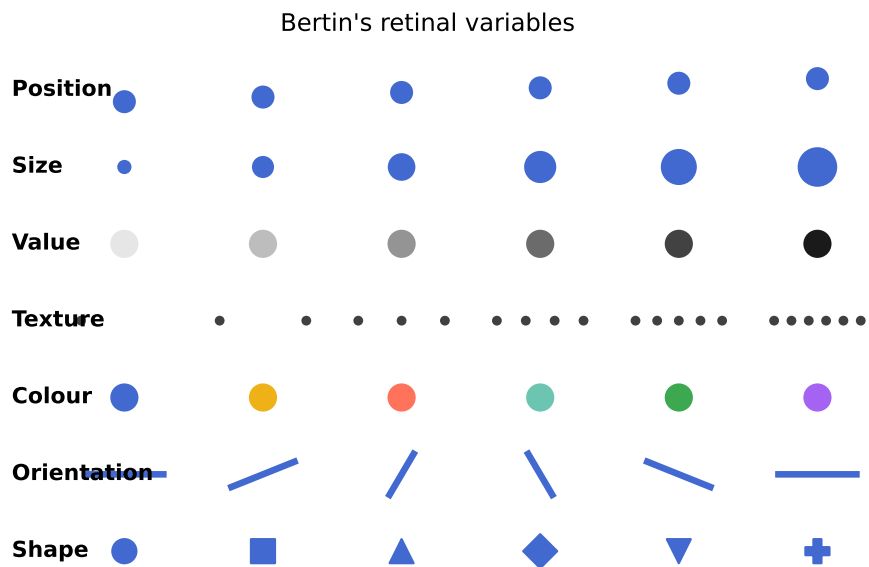


Figure 5.1: The retinal variables, each varied in isolation along a row.

5.2 A1. Magnitude — length, size, and the biased area channel

After Cleveland & McGill, “Graphical Perception” (1984).

5.2.1 2 — Mohs hardness scale

Class: named first

Canonical introduction: (Mohs 1825)

Domain: rank as position

5.2 A1. Magnitude — length, size, and the biased area channel

Mohs ordered ten reference minerals so that each scratches the one below, giving a clean ordinal ladder from talc to diamond (Mohs 1825). Placed against absolute hardness the ladder betrays itself — the rungs are evenly spaced by rank yet the Vickers numbers explode near the top, so rank is not magnitude.

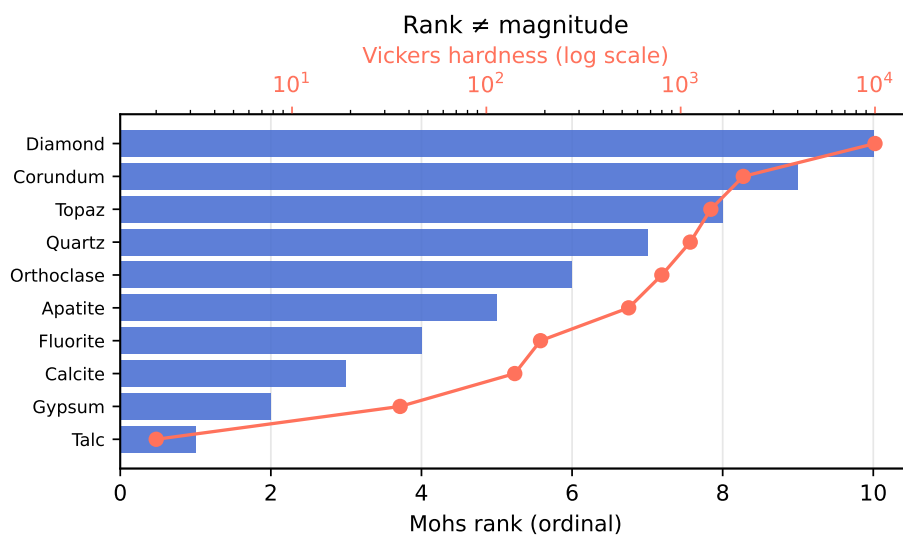


Figure 5.2: Mohs rank (evenly spaced) against absolute Vickers hardness (wildly nonlinear).

5.2.2 3 — Population pyramids

Class: uncertain attribution

Canonical introduction: (Walker 1874)

Domain: a distribution split by category

Back-to-back horizontal bars set male against female across age bands, so the shape of a population — youthful and broad, or ageing and top-heavy

— is read at a glance. The form appears in Francis Walker’s census atlas (Walker 1874), though who drew the first such pyramid is not settled.

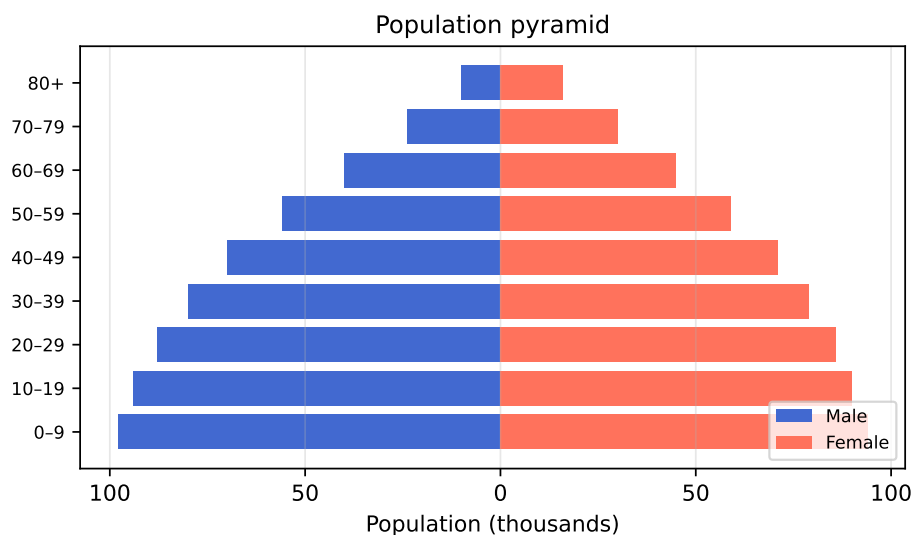


Figure 5.3: Back-to-back age–sex bars: male left, female right, counts tapering with age.

5.2.3 4 — Cosmic abundance of the elements

Class: named first

Canonical introduction: (Suess and Urey 1956)

Domain: the odd–even sawtooth

Plotting abundance against atomic number on a log scale shows two facts at once: a steep overall decline, and a persistent sawtooth in which even- Z elements sit above their odd neighbours — the Oddo–Harkins rule of nuclear stability. Suess and Urey compiled the reference abundance table that fixed the curve’s shape (Suess and Urey 1956).

5.2 A1. Magnitude — length, size, and the biased area channel

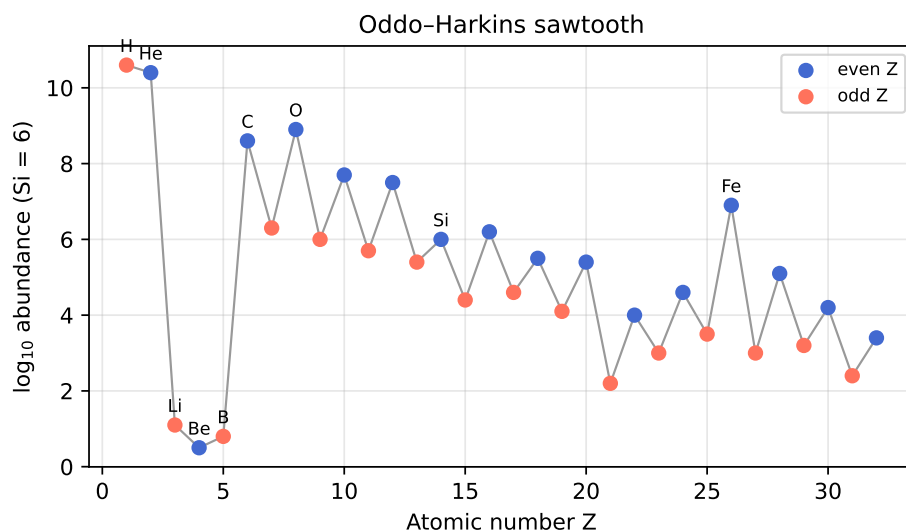


Figure 5.4: Cosmic abundance vs atomic number Z; even-Z elements ride above odd (Oddo–Harkins).

5.2.4 5 — Box plot

Class: named first

Canonical introduction: (Tukey 1977)

Domain: robust summary read by length

The box plot compresses a batch to five robust numbers — median, quartiles, and whiskers — so spread and skew are read from length rather than computed. Tukey introduced it as part of exploratory data analysis (Tukey 1977); the runs shown here are representative of Michelson’s 1879 speed-of-light experiment.

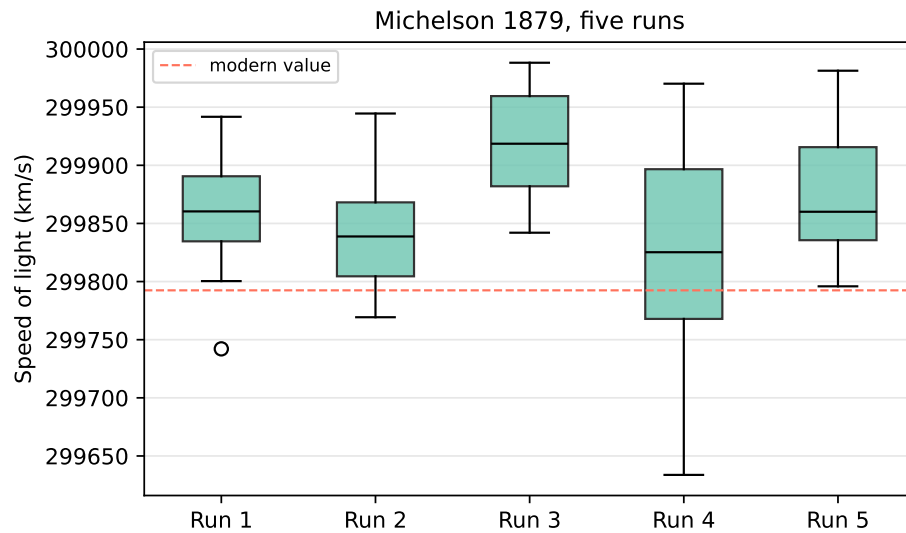


Figure 5.5: Box plots of five representative Michelson (1879) speed-of-light runs.

5.2.5 6 — Lorenz curve + Gini

Class: named first

Canonical introduction: (Lorenz 1905)

Domain: inequality as the area between curves

Cumulative income share plotted against cumulative population share bows below the 45° line of equality, and the area of that gap measures concentration. Lorenz proposed the curve to compare distributions of wealth (Lorenz 1905); twice the shaded area is the Gini coefficient.

5.3 A2. The single curve — distributions and profiles

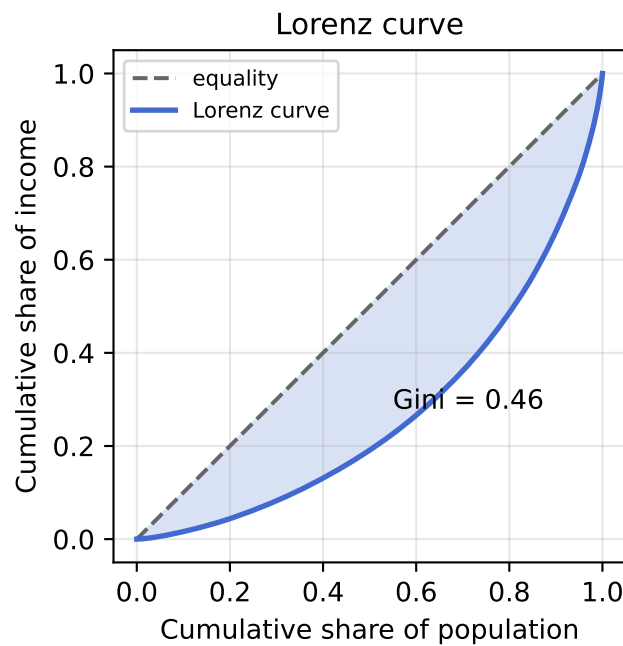


Figure 5.6: Lorenz curve against the line of equality; shaded gap gives the Gini coefficient.

5.3 A2. The single curve — distributions and profiles

After Tufte, The Visual Display of Quantitative Information (1983).

5.3.1 7 — The bell curve (Gaussian)

Class: named first + canonical reference (distinct documents)

Canonical introduction: (Gauss 1809)

Named first: (Moivre 1733) **Domain:** additive error

5 Part A — Encoding & the pre-attentive channels

The normal curve is the shape additive errors pile into, and its bell is the single most recognisable profile in statistics. De Moivre derived it as the limiting form of the binomial (Moivre 1733); Gauss gave it its canonical role as the law of observational error (Gauss 1809).

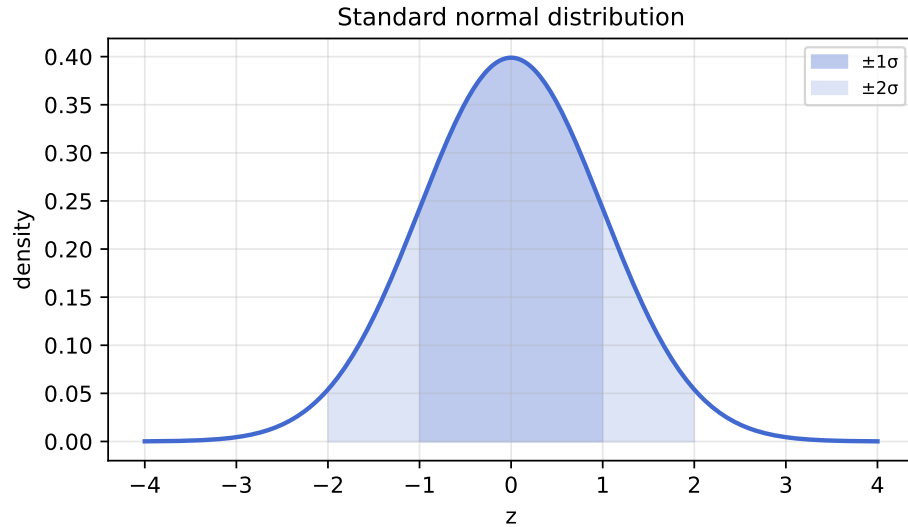


Figure 5.7: The standard normal pdf, with ± 1 and ± 2 bands shaded.

5.3.2 8 — The log-normal

Class: uncertain attribution

Canonical introduction: (Aitchison and Brown 1957)

Named first: (Galton 1879) **Domain:** multiplicative processes

When effects multiply rather than add, the logarithm of the outcome is normal and the outcome itself is right-skewed with a long upper tail. Galton described the geometric mean's distribution (Galton 1879); Aitchison and

5.3 A2. The single curve — distributions and profiles

Brown gave the log-normal its canonical monograph treatment (Aitchison and Brown 1957).

```
/var/folders/mk/gq1zr39x11jg5qqz1cqb199c0000gn/T/ipykernel_65279/3370549604.py:23: UserWarning:
plt.tight_layout()
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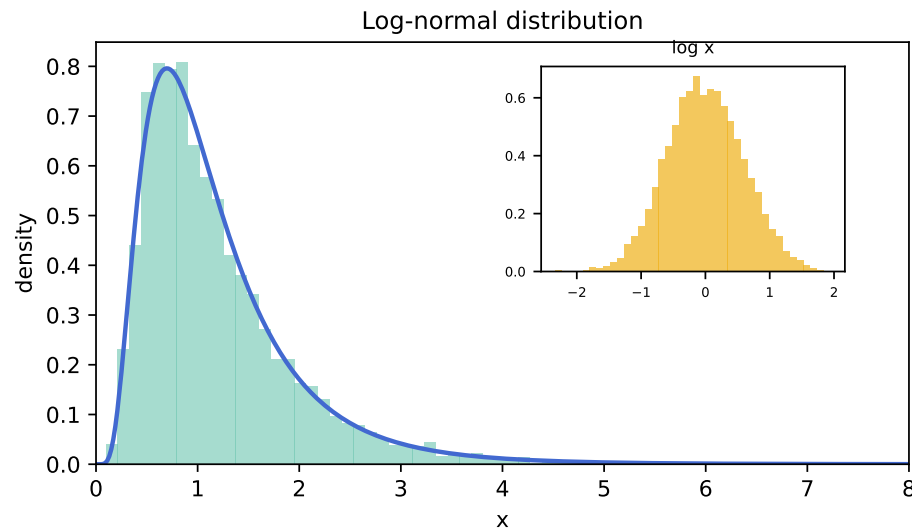


Figure 5.8: Right-skewed log-normal density; inset shows it normalised under the log transform.

5.3.3 9 — The power law / heavy tail

Class: named first

Canonical introduction: (Pareto 1896)

Domain: the signature of complexity

5 Part A — Encoding & the pre-attentive channels

A power law is a straight line on log-log axes, its slope the exponent, and its tail decays far more slowly than an exponential's. Pareto found the form in the distribution of incomes (Pareto 1896); it recurs across city sizes, word frequencies, and network degrees.

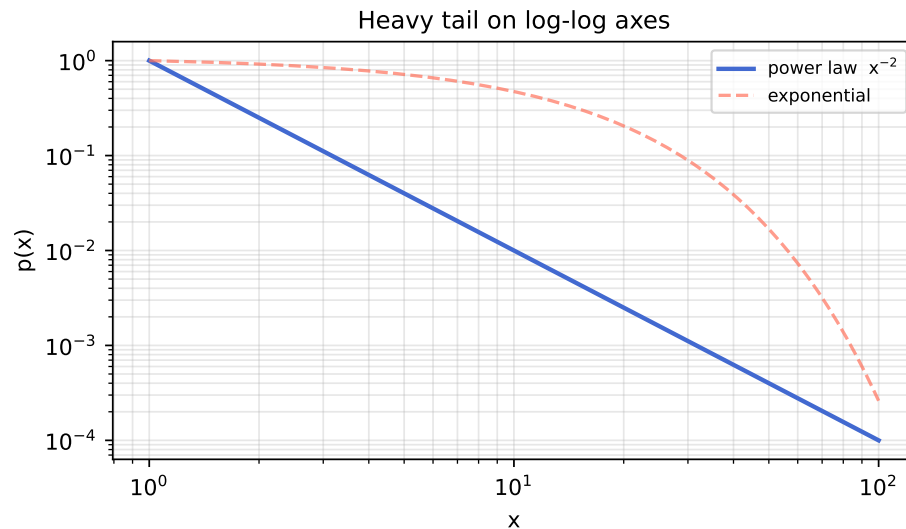


Figure 5.9: A power law is straight on log-log axes; an exponential curves away.

5.3.4 10 — Species-abundance distribution

Class: named first

Canonical introduction: (Fisher et al. 1943)

Domain: ecology's heavy tail

Binning species by abundance octave yields the hollow curve every community shares: a wall of rare species and a thin scatter of common ones.

5.3 A2. The single curve — distributions and profiles

Fisher, Corbet, and Williams fitted the log-series that made the pattern quantitative (Fisher et al. 1943).

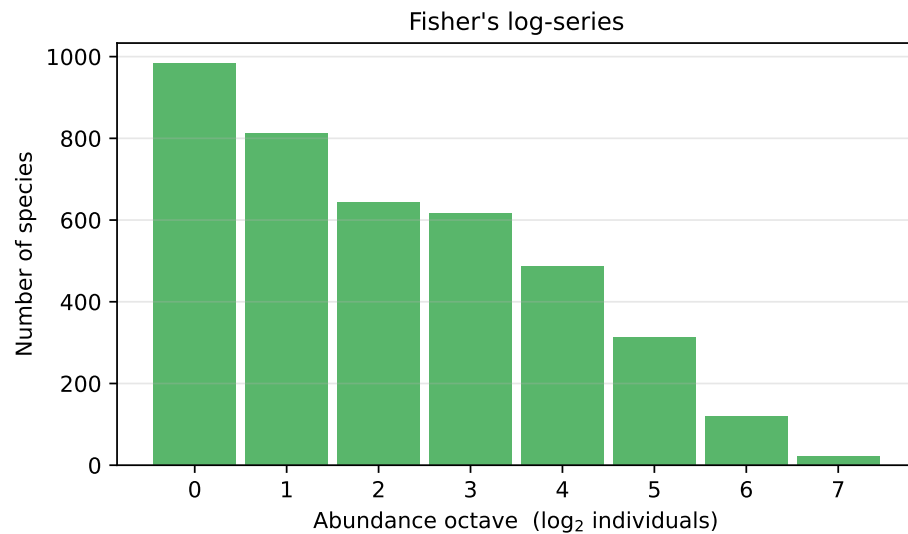


Figure 5.10: Species-abundance distribution: many rare species, few common (abundance octaves).

5.3.5 11 — The S-curve / logistic

Class: named first

Canonical introduction: (Verhulst 1838)

Domain: bounded growth

Growth that starts exponential but is checked by a ceiling traces an S, accelerating then saturating at the carrying capacity K . Verhulst named the logistic and wrote its closed-form solution (Verhulst 1838).

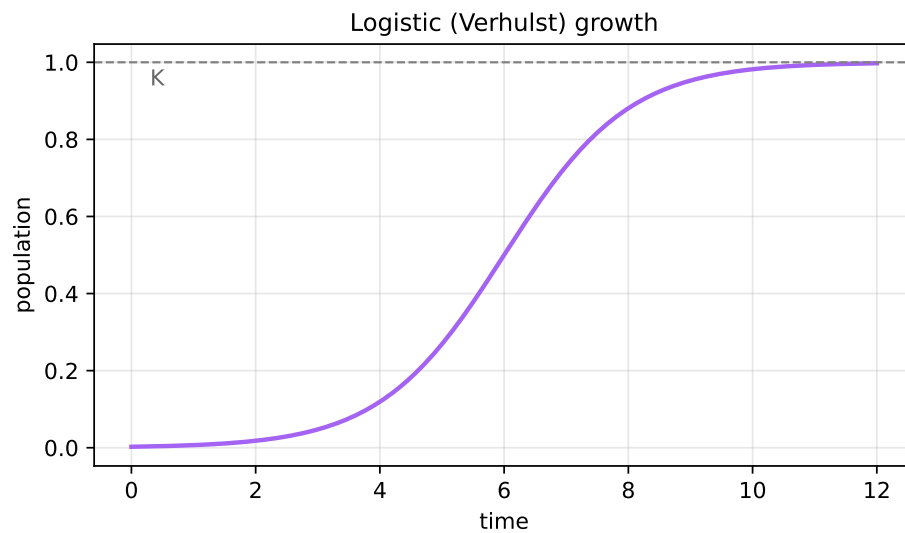


Figure 5.11: The logistic curve rising from near zero and saturating at carrying capacity K .

5.3.6 12 — Violin plot

Class: named first

Canonical introduction: (Hintze and Nelson 1998)

Domain: a density shape mirrored

The violin mirrors a kernel density estimate about its axis, so multimodality and skew that a box plot hides become visible in the outline. Hintze and Nelson introduced the form and its name (Hintze and Nelson 1998).

5.3 A2. The single curve — distributions and profiles

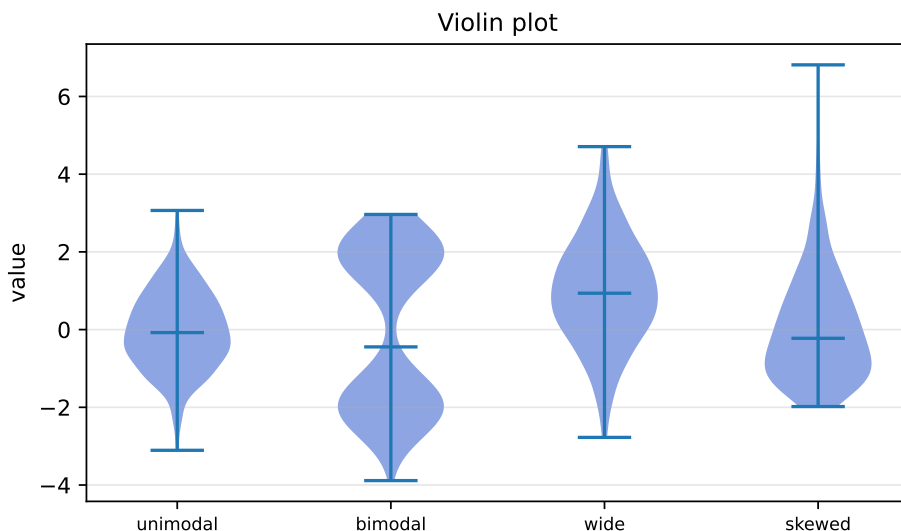


Figure 5.12: Violin plots of four groups; the mirrored density reveals bimodal shape.

5.3.7 13 — Reaction energy-coordinate diagram

Class: shared genre idiom

Canonical introduction: (Evans and Polanyi 1935)

Domain: an energy landscape

Potential energy plotted against a reaction coordinate shows reactants and products as wells separated by a transition-state hump whose height is the activation energy. Evans and Polanyi gave the profile its canonical thermodynamic reading (Evans and Polanyi 1935); the barrier sets the rate, the well depth difference sets the equilibrium.

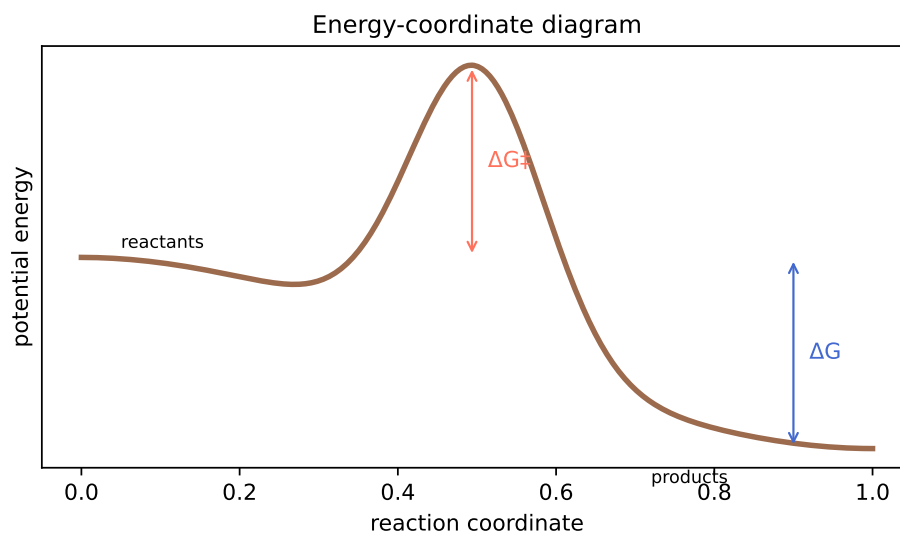


Figure 5.13: Energy vs reaction coordinate: reactant well, activation hump, lower product well.

5.4 A3. Position — relating two variables

Cleveland, The Elements of Graphing Data (1985), ranks position along a common scale as the channel the eye decodes most accurately — so relating two variables begins with points in a plane.

5.4.1 14 — Hertzsprung–Russell diagram

Class: named first + canonical reference (distinct documents)

Canonical introduction: (Russell 1914)

Named first: (Hertzsprung 1911) **Domain:** stellar evolution
read as a sequence, not a snapshot

5.4 A3. Position — relating two variables

Plotting luminosity against temperature turns the whole stellar population into a shape — a dense main sequence running diagonally, a giant clump, a scatter of white dwarfs — that later reads as an evolutionary track. Hertzsprung tabulated the luminosity–colour relation first (Hertzsprung 1911); Russell drew the diagram the field now cites (Russell 1914).

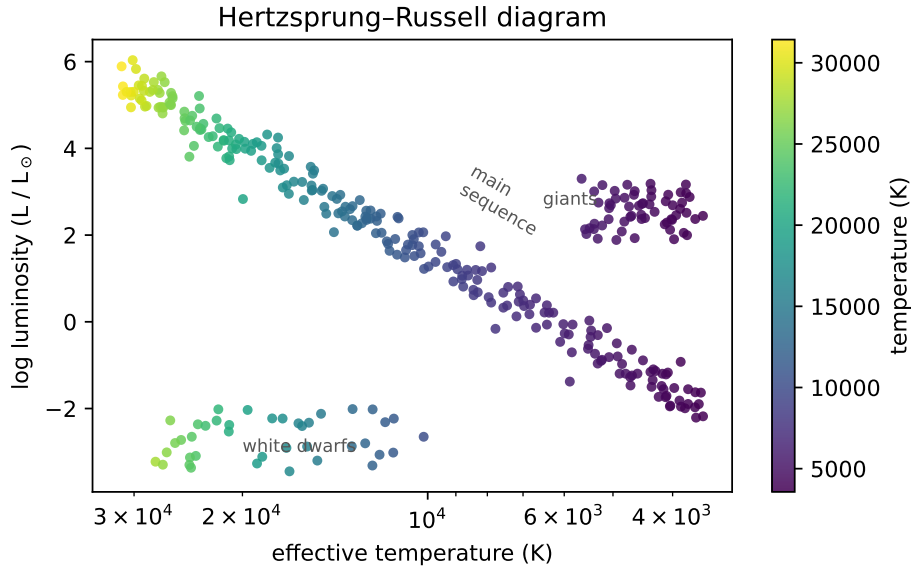


Figure 5.14: Synthetic Hertzsprung–Russell diagram: luminosity against temperature, hot stars to the left, brightness upward.

5.4.2 15 — Hubble diagram

Class: named first

Canonical introduction: (Hubble 1929)

Domain: the expansion of the universe as a straight line

A galaxy’s recession velocity rises in proportion to its distance, and the

slope of that line is the expansion rate itself. Hubble's 1929 plot fixed the relation with a scatter of a few dozen galaxies through a line near the origin (Hubble 1929).

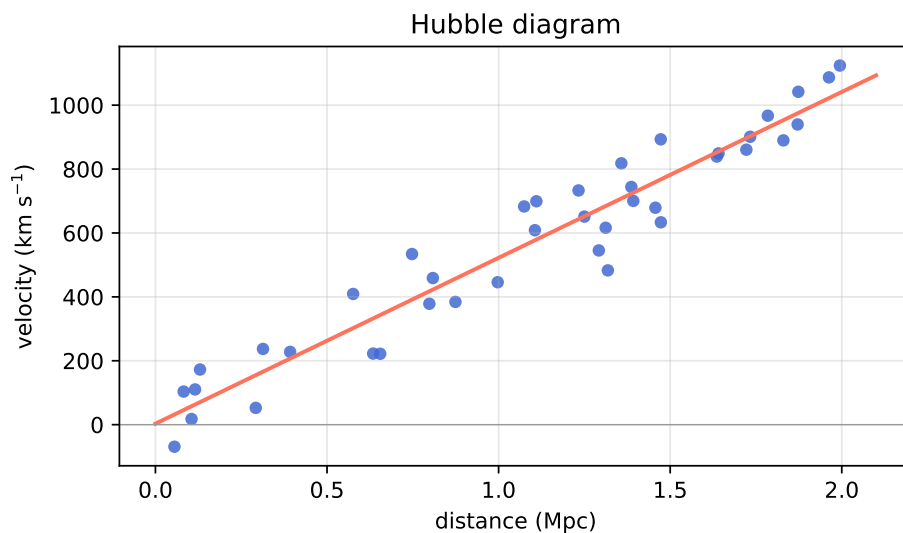


Figure 5.15: Recession velocity against distance at 1929 scale, with a linear fit through the origin.

5.4.3 16 — Kleiber's law

Class: named first

Canonical introduction: (Kleiber 1932)

Domain: metabolic rate against body mass across the tree of life

On log-log axes the metabolic rates of animals spanning six orders of magnitude in mass fall on a single line of slope three-quarters. Kleiber

5.4 A3. Position — relating two variables

measured the exponent and named the regularity that carries his name (Kleiber 1932).

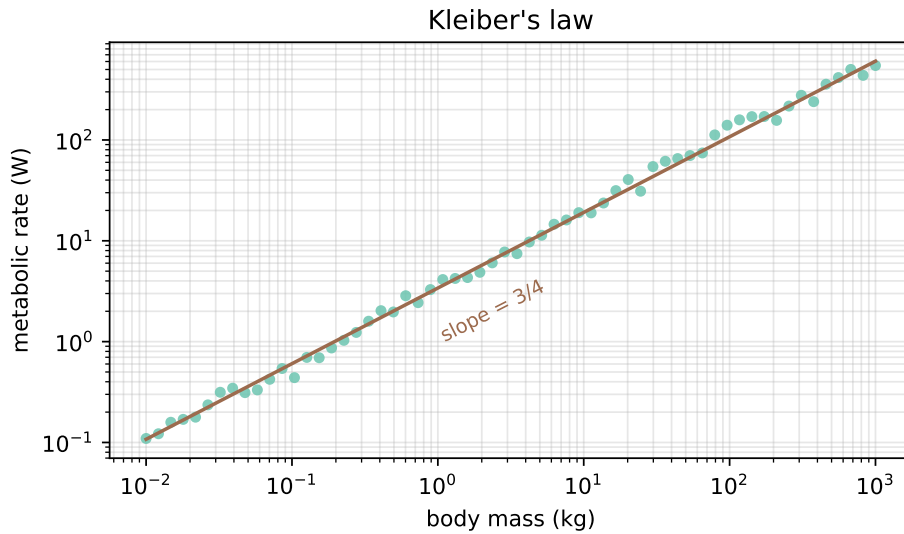


Figure 5.16: Metabolic rate against body mass, log–log, on a line of slope $3/4$.

5.4.4 17 — Zipf's law of city sizes

Class: named first + canonical reference (distinct documents)

Canonical introduction: (Zipf 1949)

Named first: (Auerbach 1913) **Domain:** rank against size in a heavy-tailed population

Rank a country's cities from largest down and plot size against rank on log–log axes: the points follow a straight line of slope near -1 , so the second city is roughly half the first. Auerbach first noticed the rank–size

regularity (Auerbach 1913); Zipf made it a general law of human systems (Zipf 1949).

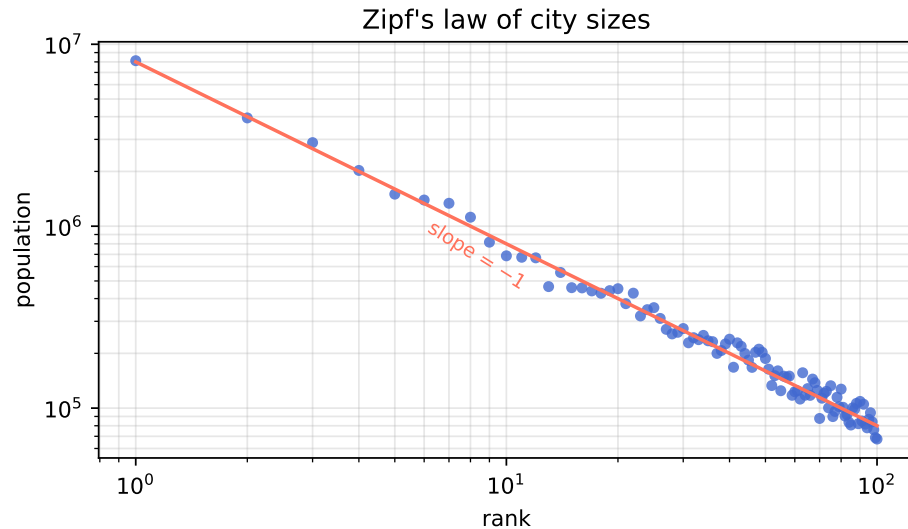


Figure 5.17: City population against rank, log-log, on a line of slope near -1 .

5.4.5 18 — Anscombe's quartet

Class: named first

Canonical introduction: (Anscombe 1973)

Domain: the argument for plotting the data at all

Four datasets share the same mean, variance, correlation and fitted line $y = 3 + 0.5x$, yet a linear trend, a curve, an outlier-bent line and a single leverage point could not look less alike. This is the book's Day-One cold open — Anscombe built the quartet precisely to show that a summary statistic is no substitute for the picture (Anscombe 1973).

5.4 A3. Position — relating two variables

Anscombe's quartet — identical statistics, different shapes

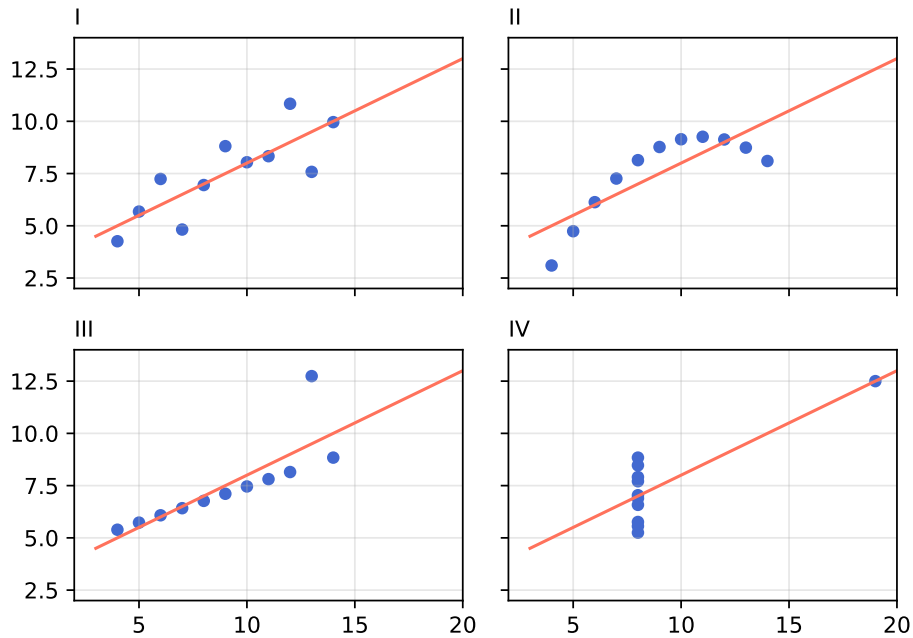


Figure 5.18: Anscombe's four datasets, each with the identical fit $y = 3 + 0.5x$.

5.4.6 19 — P–T phase diagram

Class: canonical reference

Canonical introduction: (Gibbs 1876)

Domain: the state space of matter

Pressure against temperature carves the plane into solid, liquid and gas, with sublimation, fusion and vaporisation curves meeting at a triple point

and the liquid–gas line ending abruptly at a critical point. Gibbs’s phase rule gave the diagram its coordinates and its constraints (Gibbs 1876).

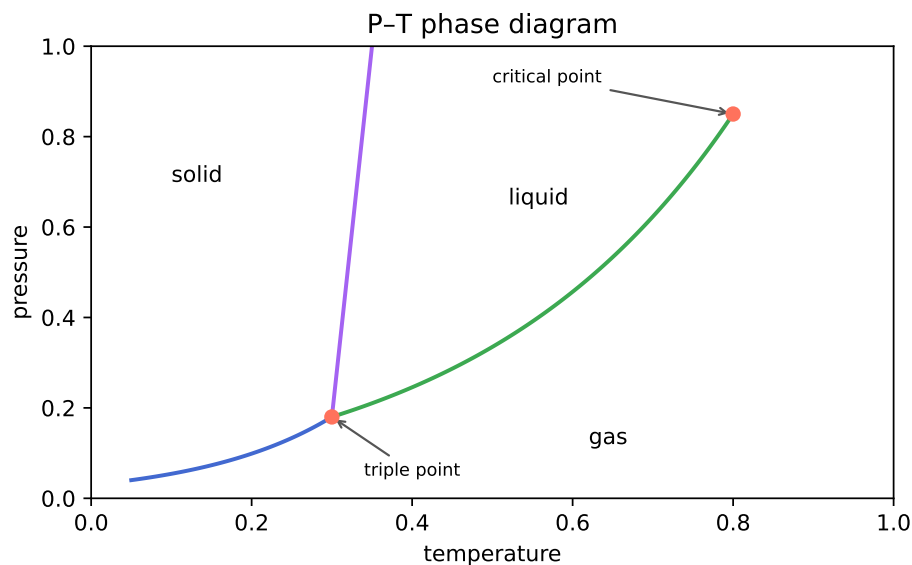


Figure 5.19: Schematic pressure–temperature phase diagram with triple and critical points.

5.4.7 20 — Ramachandran plot

Class: named first

Canonical introduction: (Ramachandran et al. 1963)

Domain: the allowed backbone conformations of a protein

The two backbone dihedral angles φ and ψ span a torus, and steric clashes forbid most of it — leaving tight clusters at the α -helix and β -sheet conformations. Ramachandran computed the allowed regions from hard-sphere contacts and gave the plot its name (Ramachandran et al. 1963).

5.5 A4. The line over time — trends

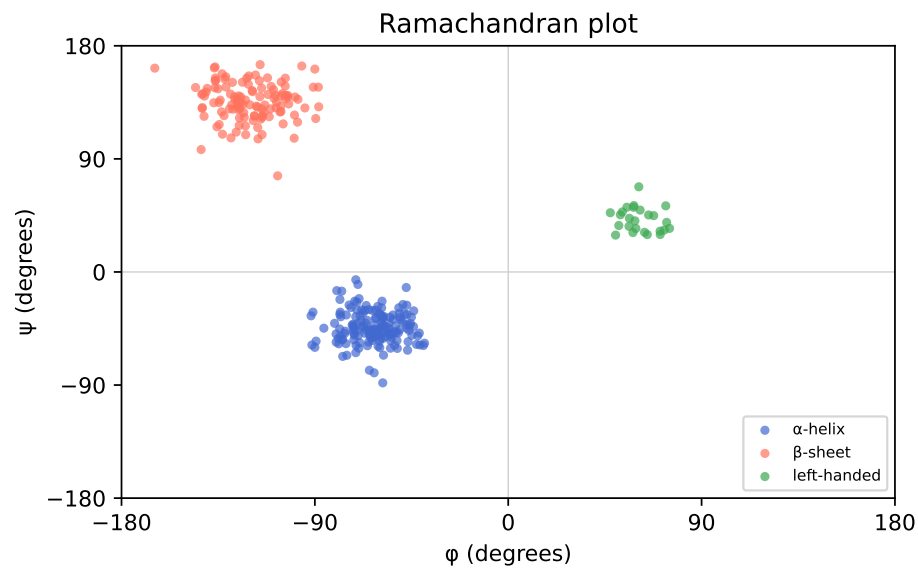


Figure 5.20: Backbone dihedral angles ψ against ϕ , clustered in the allowed regions.

5.5 A4. The line over time — trends

Playfair's time-series line, as Tufte reads it, turns a column of dated numbers into a single continuous trend the eye follows without effort.

5.5.1 21 — Keeling curve

Class: named first

Canonical introduction: (Keeling 1960)

Domain: the line that moved climate policy

5 Part A — Encoding & the pre-attentive channels

Atmospheric CO₂ climbs from about 315 ppm in 1958 toward 420 ppm, with an annual sawtooth from the breathing of Northern-Hemisphere forests riding the upward trend. Keeling’s continuous Mauna Loa record established both the rise and its seasonal signature (Keeling 1960).

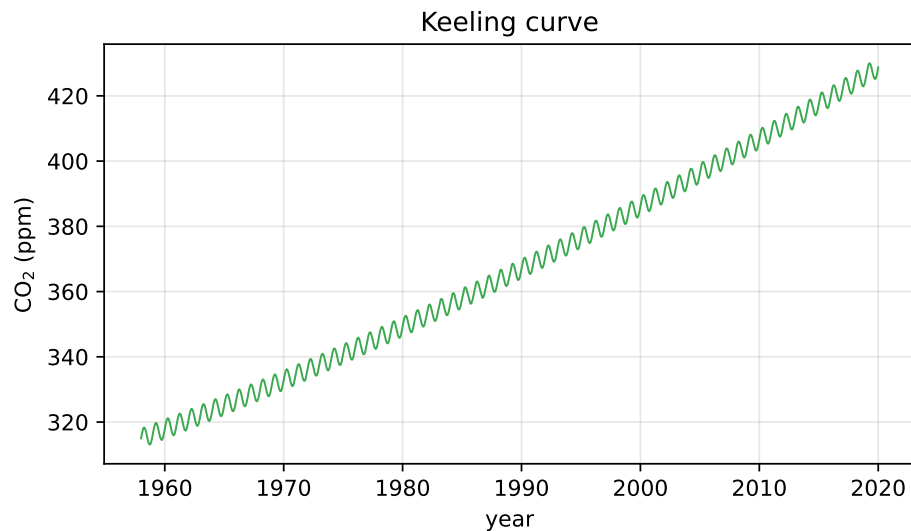


Figure 5.21: Atmospheric CO₂ rising with an annual sawtooth (representative shape).

5.5.2 22 — Hockey stick

Class: named first

Canonical introduction: (Mann et al. 1998)

Domain: a millennium of reconstructed temperature

A temperature anomaly reconstructed from proxies stays roughly flat for nine centuries, then bends sharply upward in the twentieth — the blade of the stick. Mann, Bradley and Hughes assembled the multiproxy

5.5 A4. The line over time — trends

reconstruction that fixed the shape and its uncertainty band (Mann et al. 1998).

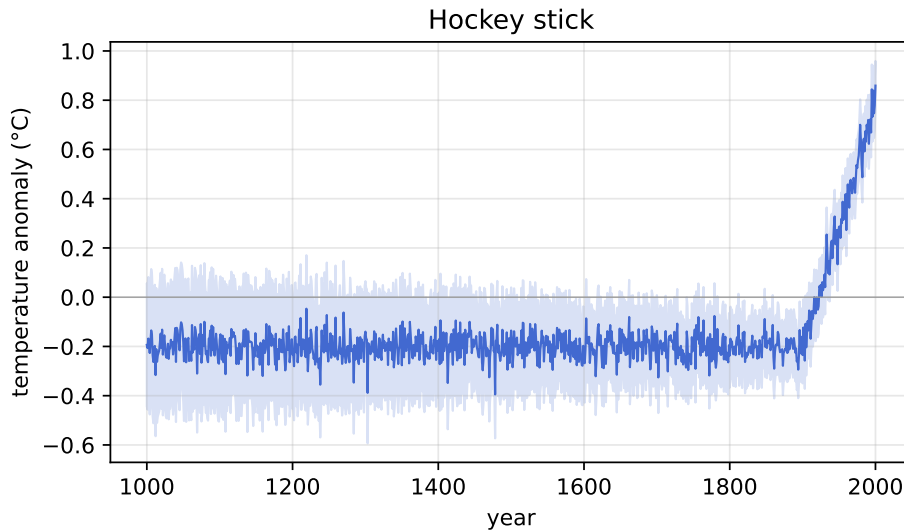


Figure 5.22: Reconstructed temperature anomaly, flat then sharply rising, with an uncertainty band.

5.5.3 23 — Sea-ice annual-cycle graph

Class: canonical reference

Canonical introduction: (Fetterer et al. 2017)

Domain: one calendar year, a line per year

Drawing sea-ice extent month by month, one line per year, stacks the seasonal max–min cycle and lets a downward drift across decades show as later lines sitting lower. The National Snow and Ice Data Center’s sea-ice index is the reference series for this now-standard spaghetti plot (Fetterer et al. 2017).

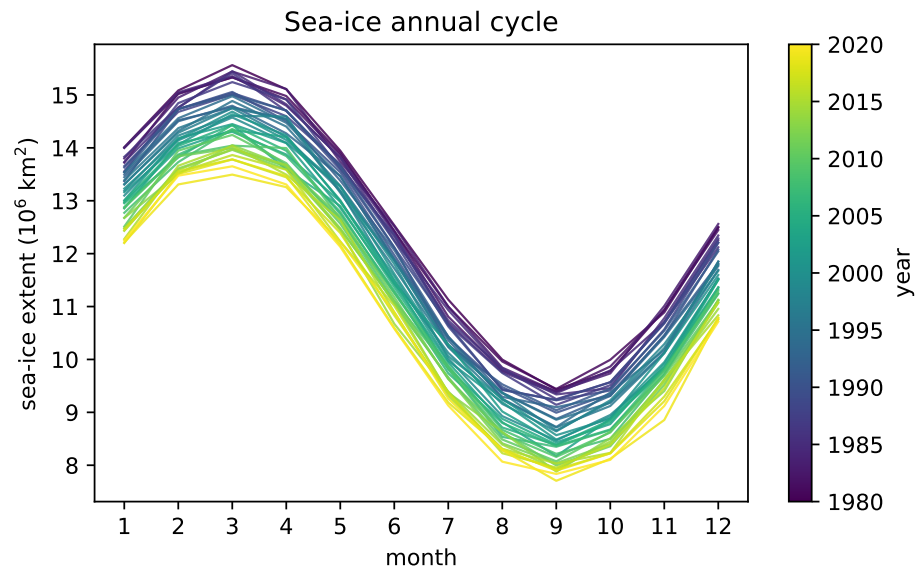


Figure 5.23: Monthly sea-ice extent, one line per year, coloured by year (representative shape).

5.5.4 24 — Kyoto cherry-blossom bloom dates

Class: named first

Canonical introduction: (Aono and Kazui 2008)

Domain: twelve centuries of phenology

Kyoto’s court diaries record the day of peak cherry bloom back to the ninth century, giving a noisy line that plunges to markedly earlier dates in the warming modern era. Aono and Kazui reconstructed the full-flowering series and the March temperatures behind it (Aono and Kazui 2008).

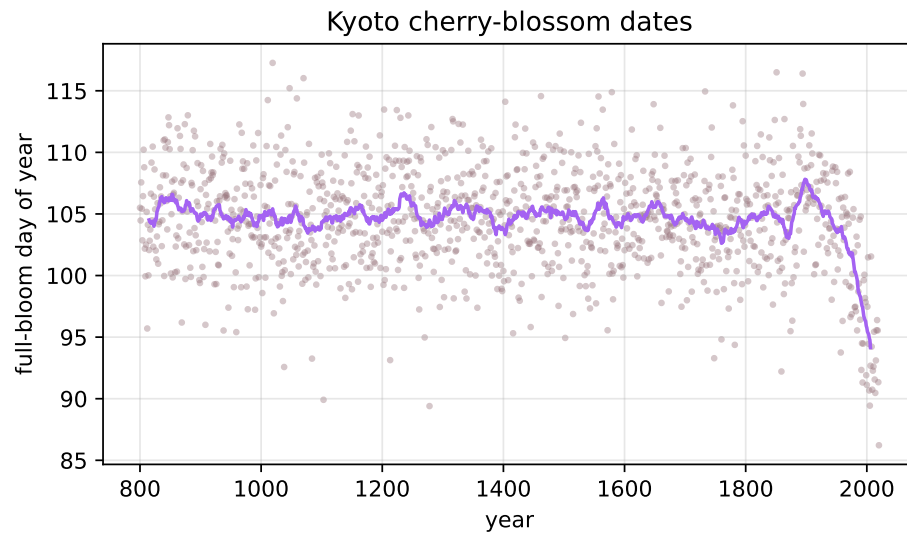


Figure 5.24: Kyoto full-bloom day of year, 800–2020, with a rolling mean (representative shape).

5.5.5 25 — Candlestick / financial series

Class: named first

Canonical introduction: (Nison 1991)

Domain: four prices per session in one dense mark

Each session becomes a candle: a body from open to close, thin wicks to the high and low, coloured by whether the price rose or fell — four numbers per bar with almost no ink. Nison brought the Japanese rice-trading technique to Western markets and named it for the West (Nison 1991).

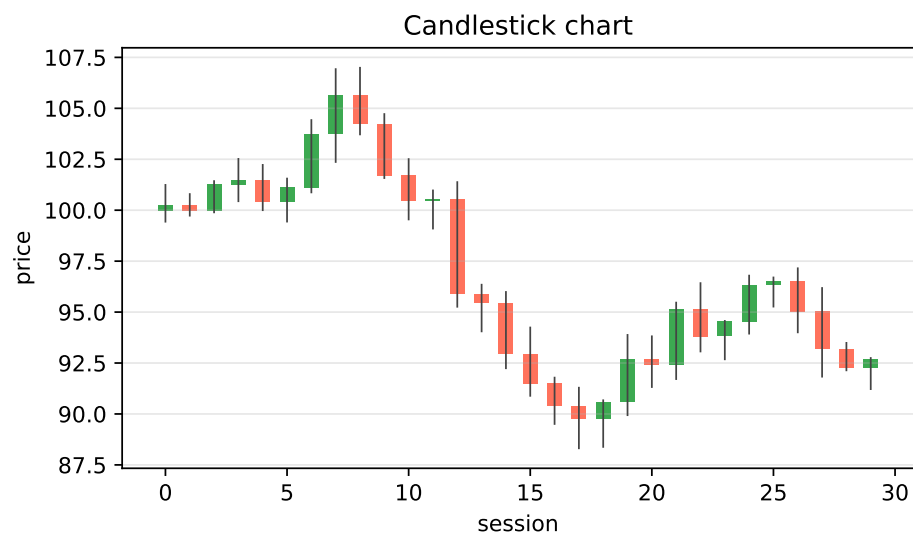


Figure 5.25: OHLC candlesticks for thirty synthetic sessions; up sessions green, down red.

5.5.6 26 — Culturomics / Google Books Ngram

Class: named first

Canonical introduction: (Michel et al. 2011)

Domain: a word's frequency across centuries

Relative frequency against year traces the life of a word — one term rising as its rival falls, their curves crossing where usage changed hands. Michel and colleagues built the culturomic corpus that made these trajectories a quantitative object (Michel et al. 2011).

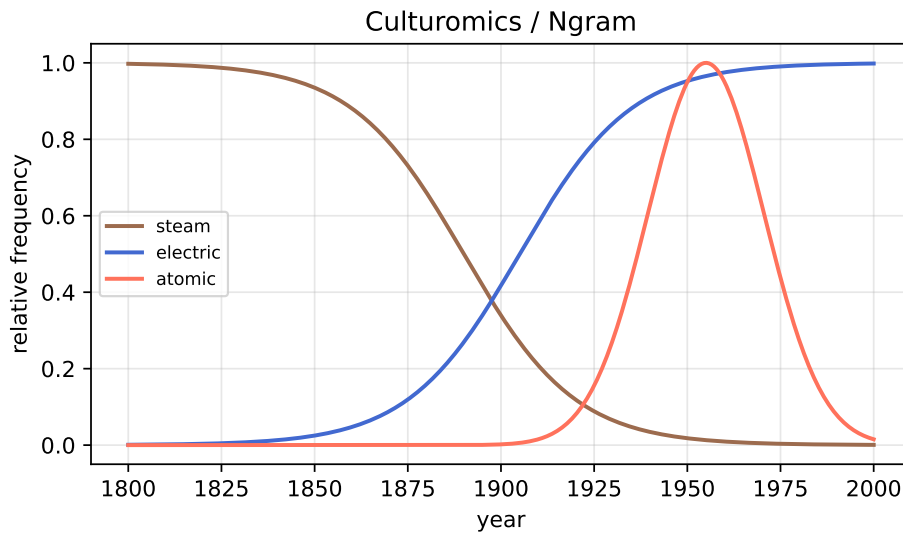


Figure 5.26: Relative frequency of three competing terms crossing over (synthetic).

5.5.7 27 — LIGO gravitational-wave chirp

Class: named first

Canonical introduction: (Abbott et al. 2016)

Domain: a signal in time

The strain from two merging black holes sweeps up in frequency and amplitude — the chirp — then rings down in a few cycles once the horizons fuse. The LIGO collaboration’s GW150914 detection was the first direct observation of the waveform (Abbott et al. 2016).

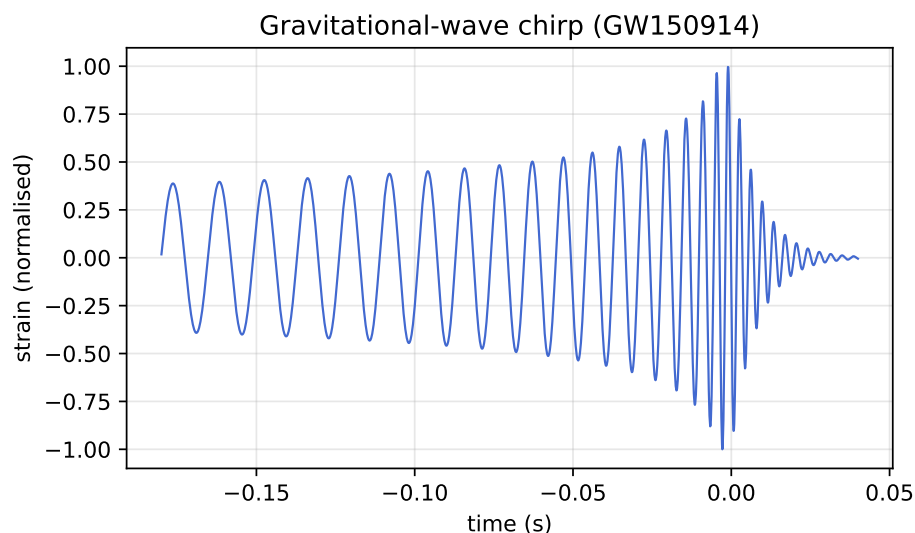


Figure 5.27: Synthetic gravitational-wave chirp: strain against time, sweeping up then ringing down.

5.5.8 28 — Exoplanet transit & radial-velocity curves

Class: named first

Canonical introduction: (Charbonneau et al. 2000)

Domain: the dimming and the wobble

A transiting planet cuts a flat-bottomed dip in the star's light while its gravity pulls the star through a radial-velocity sinusoid — two independent curves for one orbit. Charbonneau and colleagues measured the first transit of a known exoplanet, joining the photometric dip to the spectroscopic wobble (Charbonneau et al. 2000).

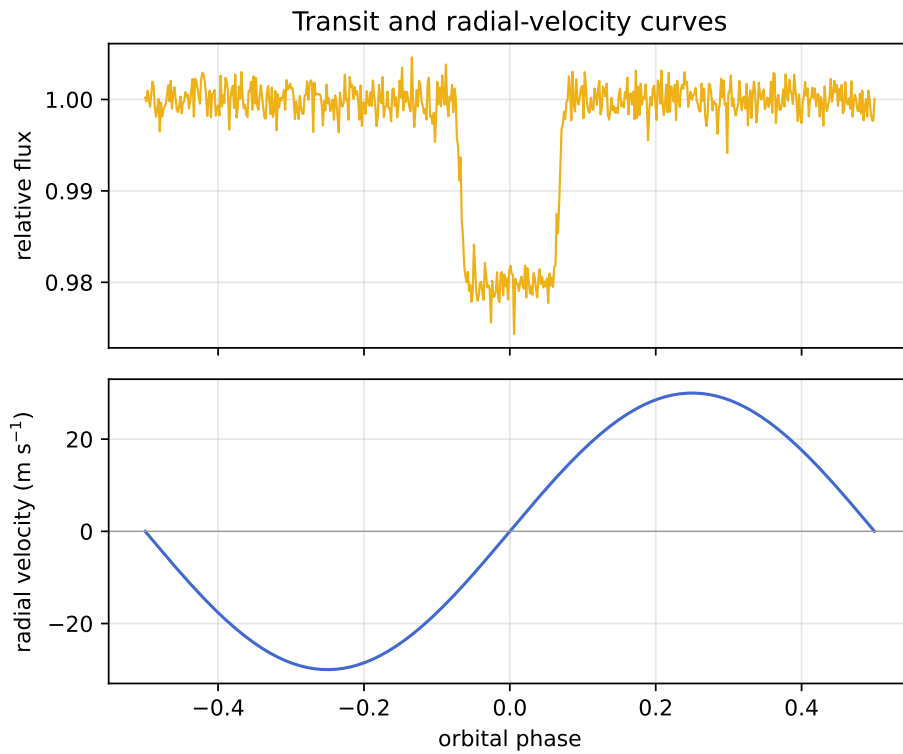


Figure 5.28: Stacked transit light curve and radial-velocity curve against orbital phase.

5.5.9 29 — Yield curve

Class: shared genre idiom

Canonical introduction: (Macaulay 1938)

Domain: a cross-sectional curve read as a signal

Yield against maturity is normally an upward slope; when it flattens or inverts — short rates above long — the shape itself is read as a recession

5 Part A — Encoding & the pre-attentive channels

signal. Macaulay's study of interest-rate movements gave the term-structure curve its analytical footing (Macaulay 1938).

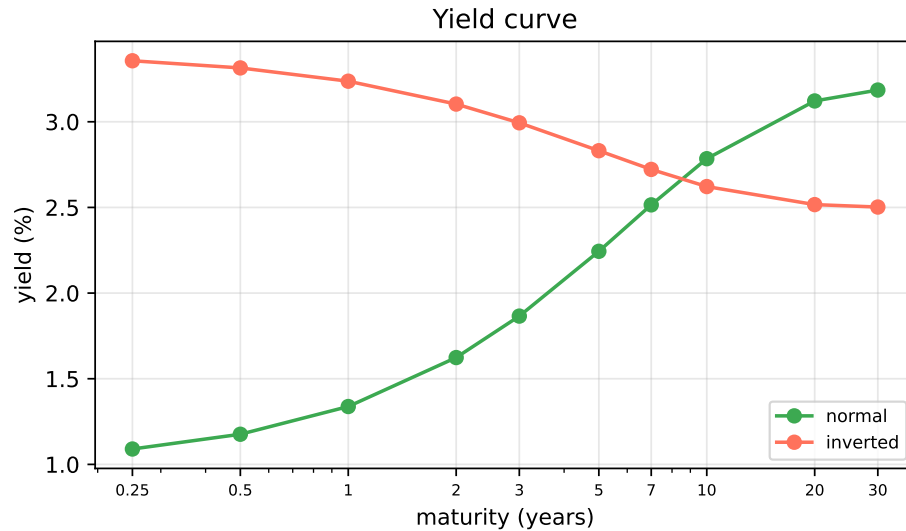


Figure 5.29: Yield against maturity for a normal and an inverted date.

5.6 A5. Colour — luminance, hue, the wheel, and colour-blindness

Ware, Information Visualization (2004), treats colour as three separable channels — luminance, hue, saturation — each with its own perceptual budget and its own failure under colour-blindness.

5.6 A5. Colour — luminance, hue, the wheel, and colour-blindness

5.6.1 30 — Colour wheel

Class: named first

Canonical introduction: (Newton 1704)

Domain: the organisation of colour into a closed ring

Bending the spectrum into a circle joins violet back to red and turns hue into an angle — the first diagram to treat colour as a cyclic quantity. Newton drew the wheel in the *Opticks* (Newton 1704); Dalton's 1798 account of his own colour-blindness stands as the conscience of every palette choice that follows.

Colour wheel

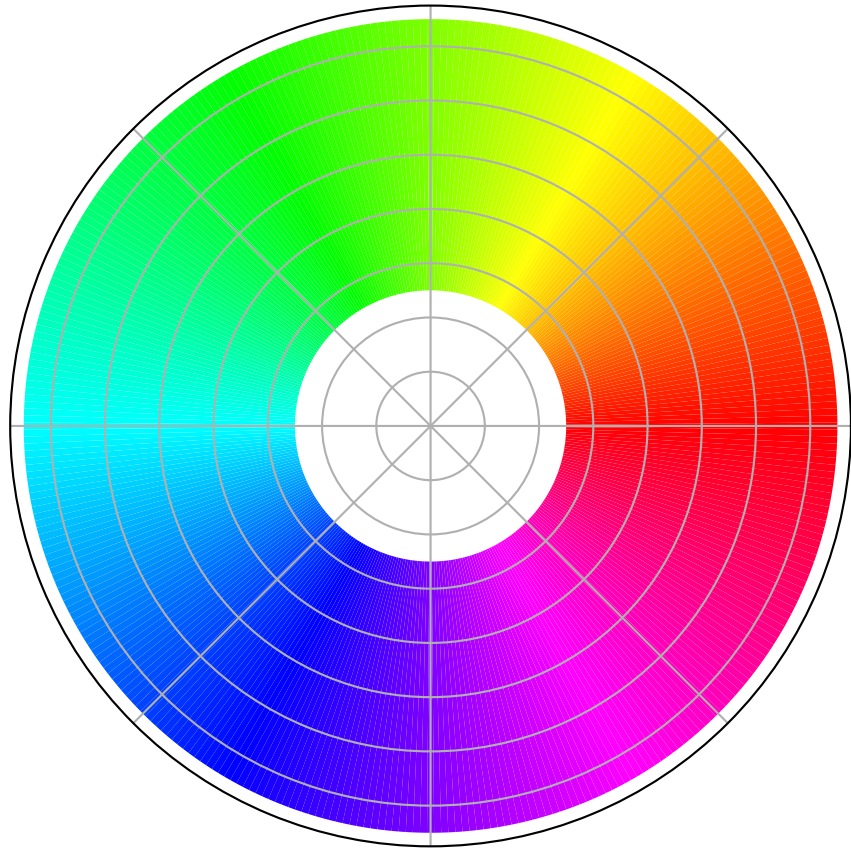


Figure 5.30: A hue wheel: wedges around a circle spanning the spectrum.

5.6.2 31 — Climate stripes

Class: named first

Canonical introduction: (Hawkins 2018)

5.6 A5. Colour — luminance, hue, the wheel, and colour-blindness

Domain: temperature as pure colour, axes removed

Stripping every axis and label leaves one coloured band per year on a blue-to-red diverging scale — warming shown as colour alone, legible with no training. Hawkins designed the warming-stripes graphic for exactly that reach (Hawkins 2018).

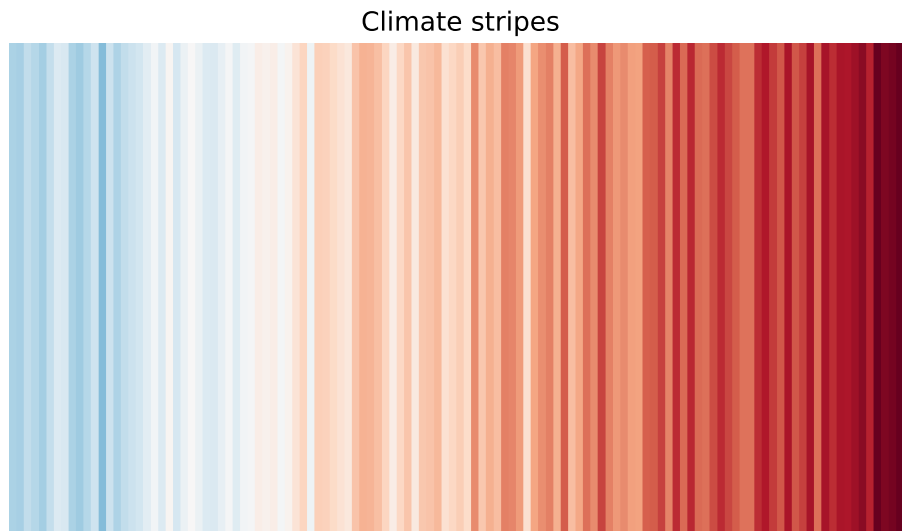


Figure 5.31: One coloured stripe per year on a diverging scale; no axes.

5.6.3 32 — CMB temperature map

Class: named first

Canonical introduction: (Smoot et al. 1992)

Domain: the oldest light as a field of fluctuations

Tiny temperature variations across the microwave sky — parts in a hundred thousand — become a field of hot and cold blotches on a diverging colour

scale. COBE's DMR, reported by Smoot and colleagues, first resolved the anisotropy (Smoot et al. 1992).

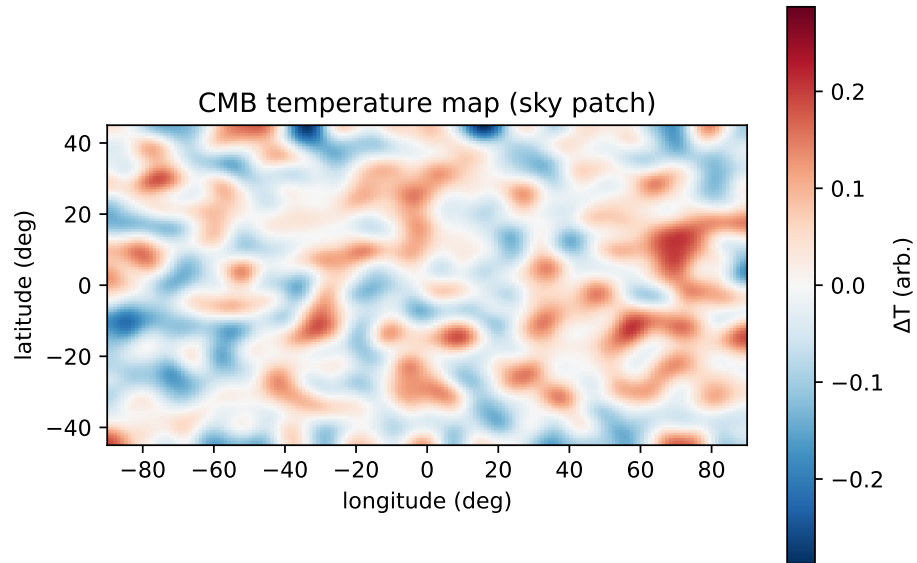


Figure 5.32: A smoothed Gaussian random field standing for CMB temperature anisotropy.

5.6.4 33 — Ozone hole

Class: named first

Canonical introduction: (Farman et al. 1985)

Domain: the Antarctic false-colour map

Total-column ozone mapped over the pole shows a deep central minimum — the hole — rendered in false colour so the depletion reads instantly. Farman, Gardiner and Shanklin's ground measurements first reported the springtime Antarctic loss (Farman et al. 1985).

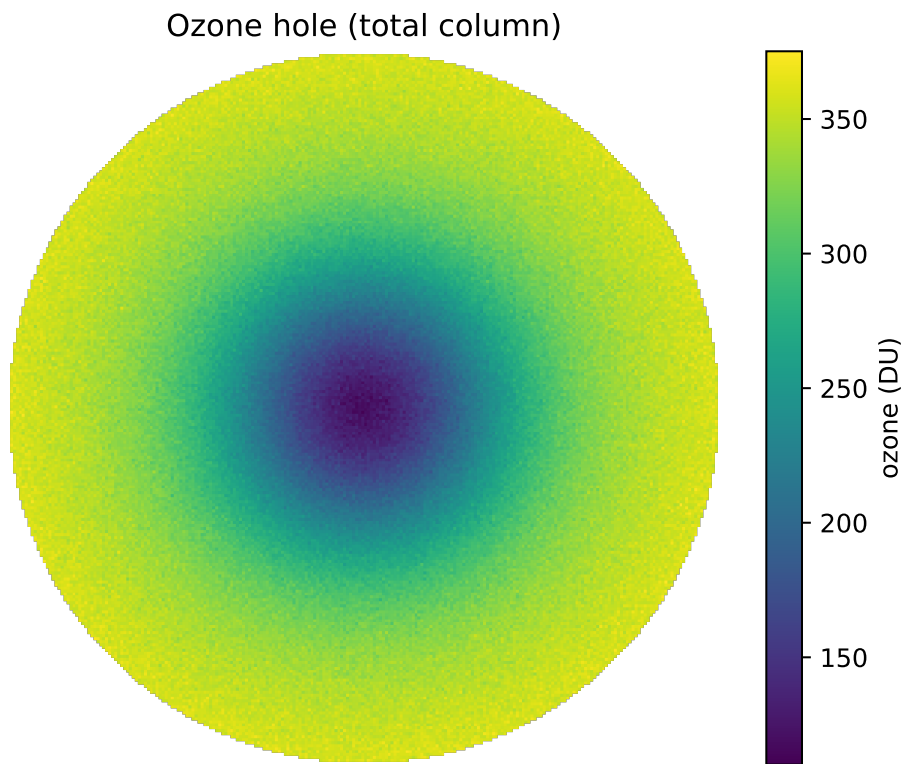


Figure 5.33: A synthetic polar ozone field with a deep central minimum in false colour.

5.6.5 34 — Choropleth + cartogram

Class: named first

Canonical introduction: (Dupin 1826)

Domain: a value carried by area

A choropleth shades each region by its value; a cartogram resizes the same regions by that value, trading true shape for true magnitude — two

answers to the problem that big areas shout and small ones vanish. Dupin's 1826 shaded map of French literacy is the first modern choropleth (Dupin 1826).

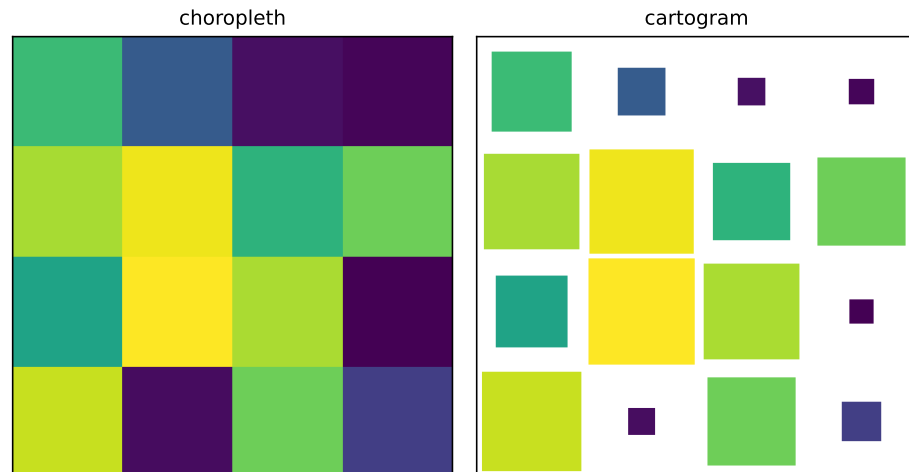


Figure 5.34: The same 4×4 regions as a choropleth (left) and a crude cartogram (right).

5.7 A6. Depth and 3D form — shading, occlusion, the volumetric

Munzner, Visualization Analysis and Design (2014), counts depth cues — shading, occlusion, perspective — as the means by which a flat display fakes the third dimension.

5.7.1 35 — 3D molecular structure

Class: shared genre idiom

Canonical introduction: (Corey and Pauling 1953)

Domain: ball-and-stick to space-filling

Atoms drawn as spheres and bonds as sticks make a molecule's geometry legible in three dimensions — here methane's carbon at the centre of a hydrogen tetrahedron. Corey and Pauling's models set the conventions for scaled atomic radii and bond sticks (Corey and Pauling 1953).

Methane (ball-and-stick)

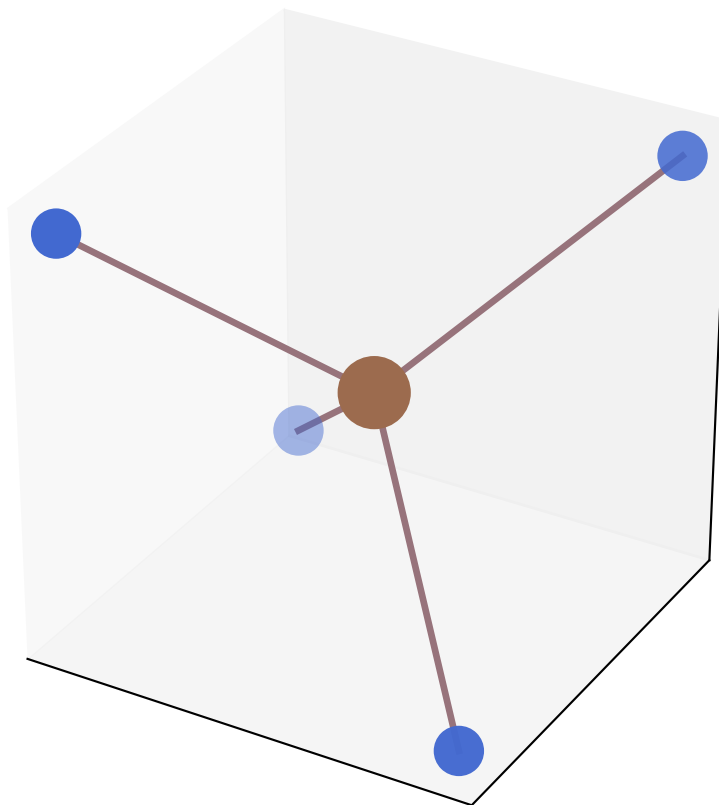


Figure 5.35: Methane as ball-and-stick: a central carbon with four tetrahedral hydrogens.

5.7.2 36 — DNA double helix

Class: named first

Canonical introduction: (Watson and Crick 1953)

Domain: 3D structure inferred from diffraction

Two antiparallel strands wind around a common axis, rungs of paired bases holding them at a fixed spacing — a shape that carried its own copying mechanism in plain sight. Watson and Crick proposed the double helix from Franklin's and Wilkins's diffraction data (Watson and Crick 1953).

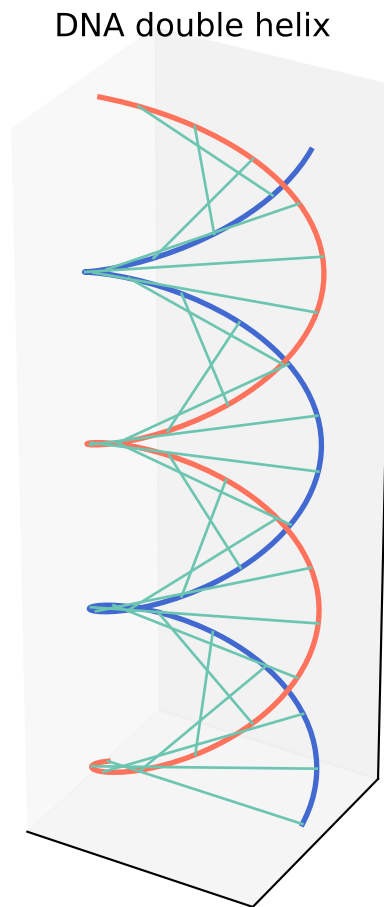


Figure 5.36: Two intertwined helices with base-pair rungs (parametric).

5.7.3 37 — Protein ribbon

Class: named first

Canonical introduction: (Richardson 1981)

5.7 A6. *Depth and 3D form — shading, occlusion, the volumetric*

Domain: the fold made legible

A single backbone traced as a ribbon coils tightly through a helix, then straightens into a strand — the schematic that lets an eye read secondary structure at a glance. Jane Richardson’s *Anatomy and Taxonomy of Protein Structure* introduced the ribbon idiom (Richardson 1981).

Protein ribbon (helix → strand)

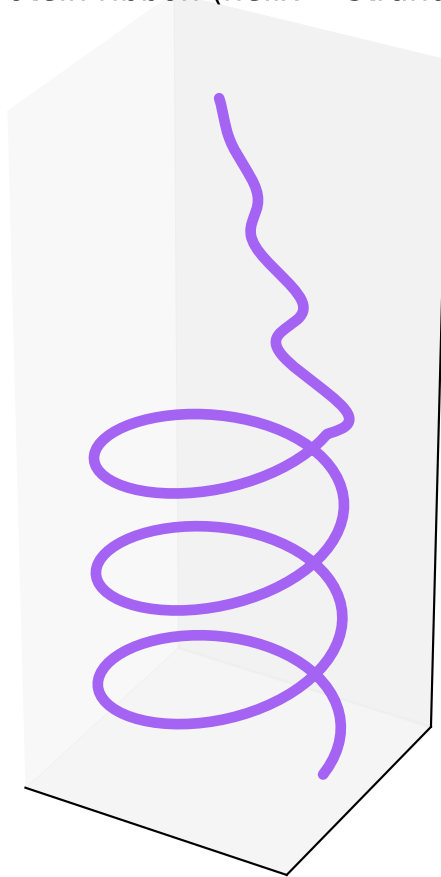


Figure 5.37: A continuous backbone coiling into a helix then extending into a strand.

5.7.4 38 — Atomic orbitals / probability clouds

Class: canonical reference

Canonical introduction: (White 1931)

Domain: continuous probability in three dimensions

An orbital is a probability density, not an orbit; the hydrogen $2p_z$ state shows as two lobes of $|\psi|^2$ above and below the nucleus, densest where the electron is most likely found. White's *Pictorial Representations of the Electron Cloud* gave these densities their canonical images (White 1931).

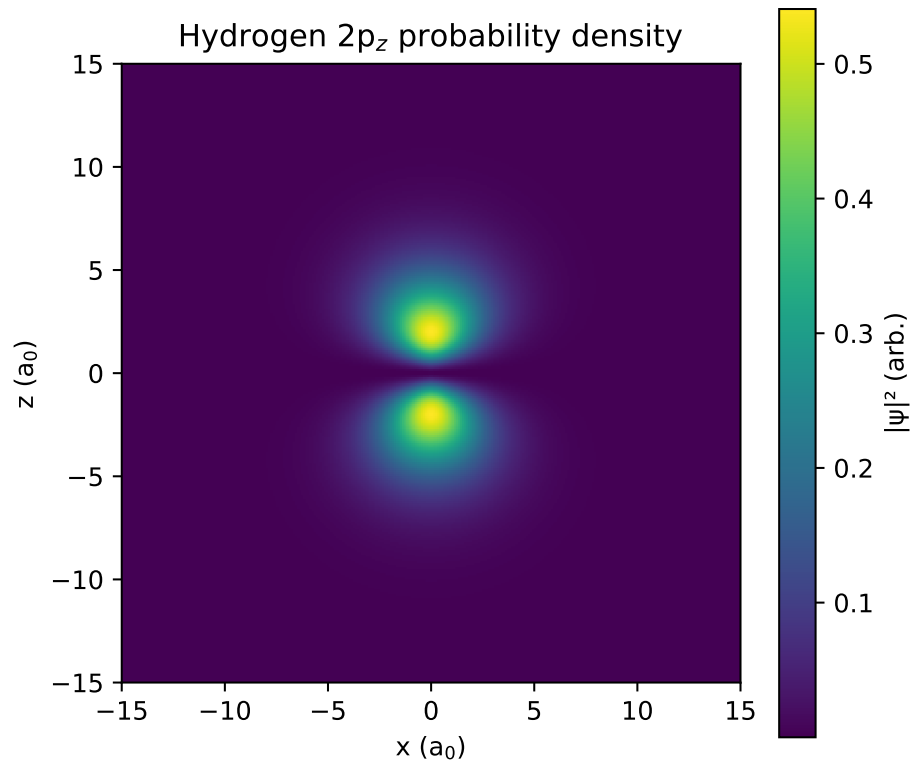


Figure 5.38: Cross-section of the hydrogen 2p_z probability density $|\psi|^2$, two lobes.

5.7.5 39 — Cosmic web

Class: shared genre idiom

Canonical introduction: (Lapparent et al. 1986)

Domain: large-scale structure as filaments and voids

Plotting galaxies by redshift and sky angle in a thin wedge reveals that

5.7 A6. Depth and 3D form — shading, occlusion, the volumetric

matter is not scattered but strung along filaments around great empty voids. De Lapparent, Geller and Huchra's slice of the CfA survey first showed the frothy structure at this scale (Lapparent et al. 1986).

Cosmic web — slice of the universe

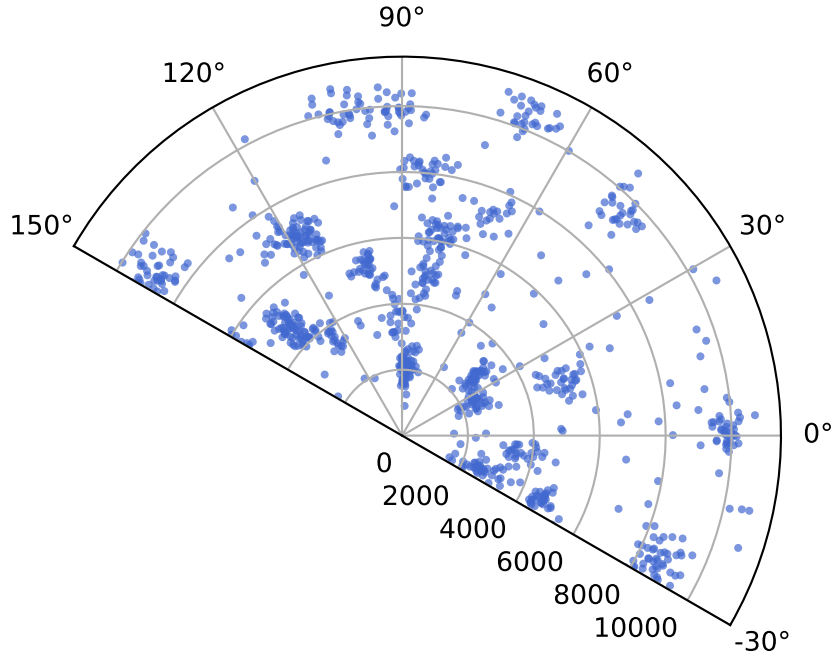


Figure 5.39: A redshift-vs-angle wedge of galaxies clustered into filaments and voids.

6 Part B — Gestalt grouping

Perception does not stop at measuring single marks; it groups them. Marks that connect read as one structure, marks that flow read as a path, marks that sit close or look alike read as a cluster, and a boundary read as enclosure turns empty space into an encoding — these Gestalt principles are the load-bearing walls of the network, the Sankey, the heatmap and the map.

6.1 B1. Connectedness — networks, trees, and designed relational structures

Munzner on network and tree layouts (2014).

6.1.1 40 — Food web

Class: uncertain attribution

Canonical introduction: (Elton 1927)

Named first: (Summerhayes and Elton 1923) **Domain:** when connection is the data

A food web draws species as nodes and feeding as directed edges, so the trophic structure of a community becomes a graph read at a glance. Elton fixed the ecological concepts of the food chain and the niche (Elton 1927), though the Bear Island survey of Summerhayes and Elton (Summerhayes and Elton 1923) had printed the connected web four years earlier.

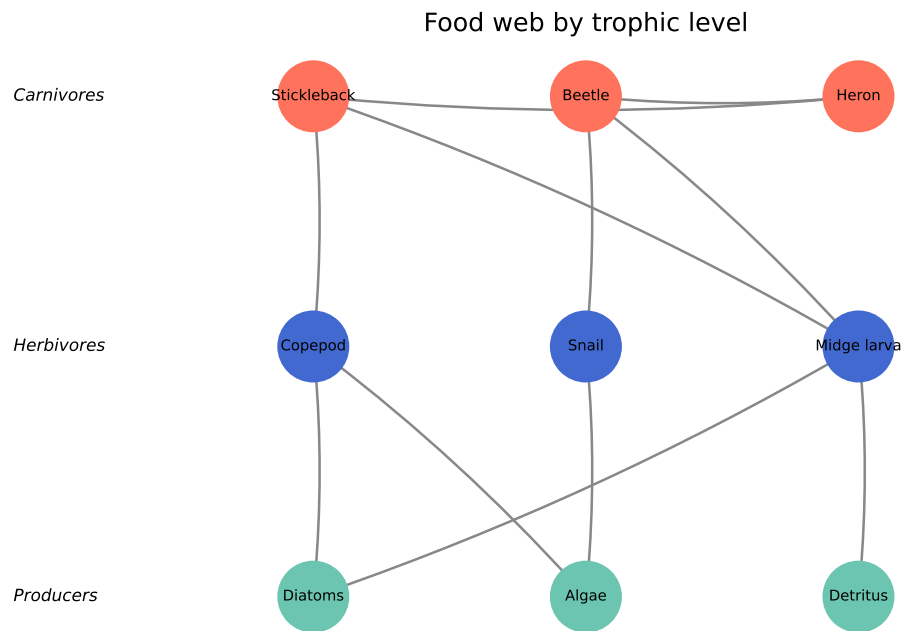


Figure 6.1: A small directed food web laid out by trophic level; arrows point from prey to predator.

6.1.2 41 — Connectome

Class: named first

Canonical introduction: (Sporns et al. 2005)

Domain: the brain as a network

The connectome renders the brain as a graph — regions as nodes, tracts or correlations as edges — whose modular community structure is the finding, not the layout. Sporns, Tononi and Kötter named the human connectome and set the network agenda for neuroscience (Sporns et al. 2005).

6.1 B1. Connectedness — networks, trees, and designed relational structures

Modular community structure

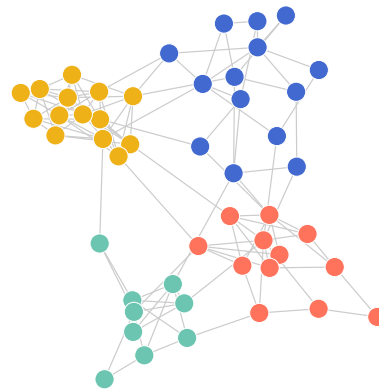


Figure 6.2: A modular graph — dense within communities, sparse between — under a spring layout, coloured by module.

6.1.3 42 — Artificial neural network

Class: canonical reference

Canonical introduction: (McCulloch and Pitts 1943)

Domain: the engineered twin

An artificial neural network diagram stacks layers of units with full connections between adjacent layers, the engineered mirror of the biological graph. McCulloch and Pitts gave the first formal model of a neuron as a logical threshold unit (McCulloch and Pitts 1943).

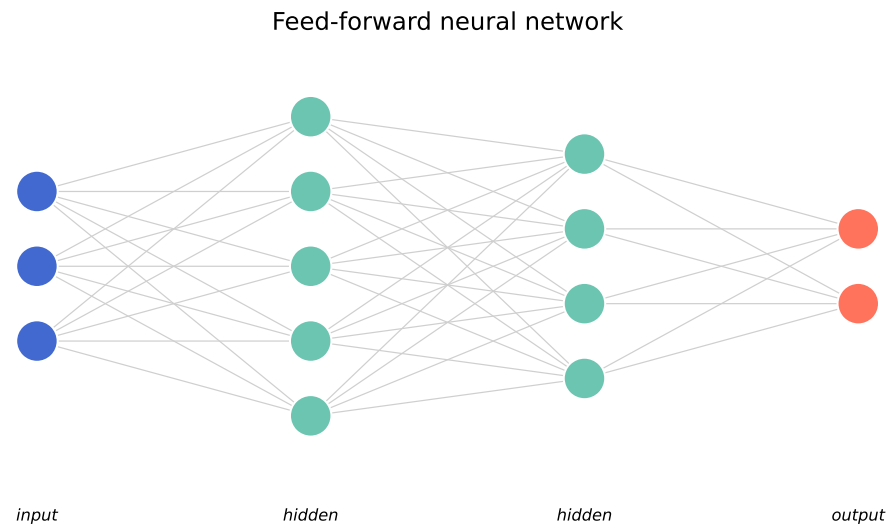


Figure 6.3: A feed-forward network: input, two hidden, and output layers, fully connected between neighbours.

6.1.4 43 — Airline / transport network

Class: shared genre idiom

Canonical introduction: (Guimerà et al. 2005)

Domain: a spatial graph

A transport network embeds nodes at their geographic coordinates and draws routes as edges, so hubs surface as high-degree points. Guimerà and colleagues analysed the worldwide airport network and exposed its scale-free, community-structured form (Guimerà et al. 2005).

6.1 B1. Connectedness — networks, trees, and designed relational structures

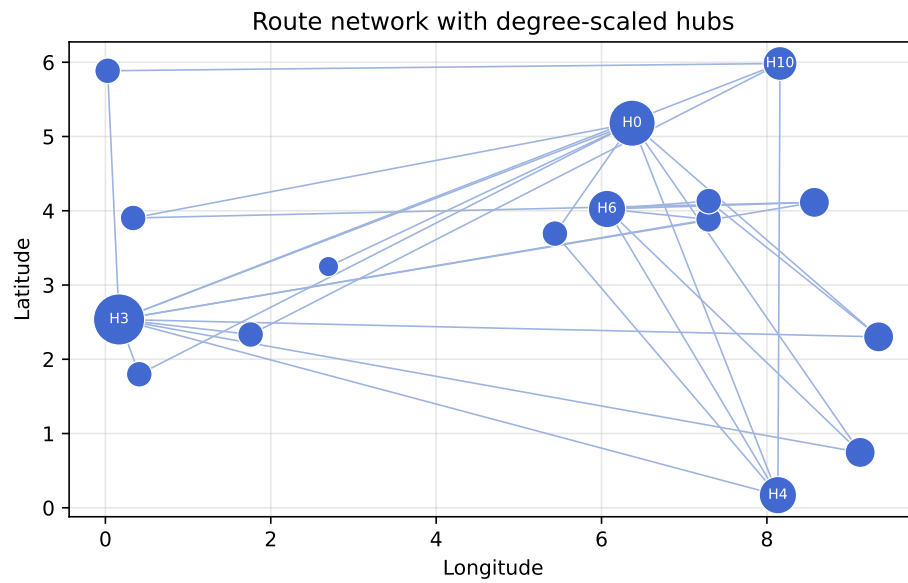


Figure 6.4: A synthetic route network on (lon, lat) positions; node size grows with degree, revealing hubs.

6.1.5 44 — Feynman diagram

Class: named first

Canonical introduction: (Feynman 1949)

Domain: a graph that is also physics

A Feynman diagram is a graph whose edges are particle worldlines and whose vertices are interactions, at once a bookkeeping device and a term in a perturbation series. Feynman introduced the space-time diagrams in his 1949 formulation of quantum electrodynamics (Feynman 1949).

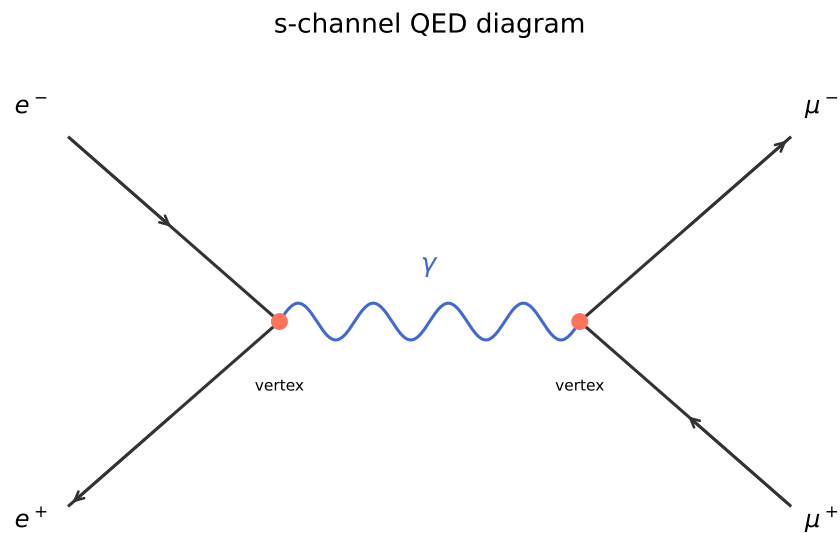


Figure 6.5: An s-channel diagram: an incoming fermion pair annihilate at a vertex, a photon propagates, an outgoing pair emerge.

6.1.6 45 — Logic gate / transistor

Class: shared genre idiom

Canonical introduction: (Shannon 1938)

Domain: the unit of computation

A logic-circuit schematic wires gate glyphs into a combinational function, each gate a node and each wire an edge carrying a Boolean value. Shannon's thesis established that switching circuits realise Boolean algebra (Shannon 1938).

6.1 B1. Connectedness — networks, trees, and designed relational structures

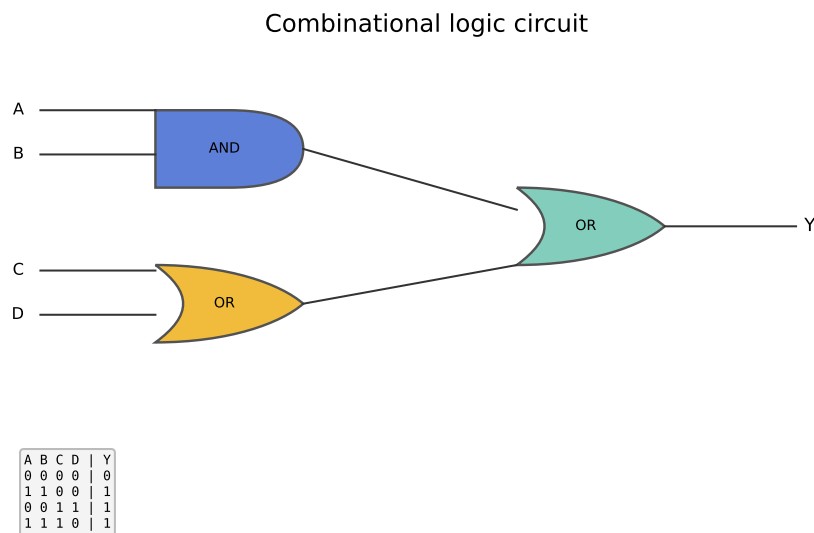


Figure 6.6: An AND gate and an OR gate feeding a final OR gate; a truth table for the output sits inset.

6.1.7 46 — Character co-occurrence network

Class: canonical reference

Canonical introduction: (Knuth 1993)

Domain: narrative as a graph

A co-occurrence network turns a narrative into a graph — characters as nodes, shared scenes as weighted edges — and lets community structure stand in for plot. The *Les Misérables* graph enters the canon through Knuth's Stanford GraphBase (Knuth 1993).

6 Part B — Gestalt grouping

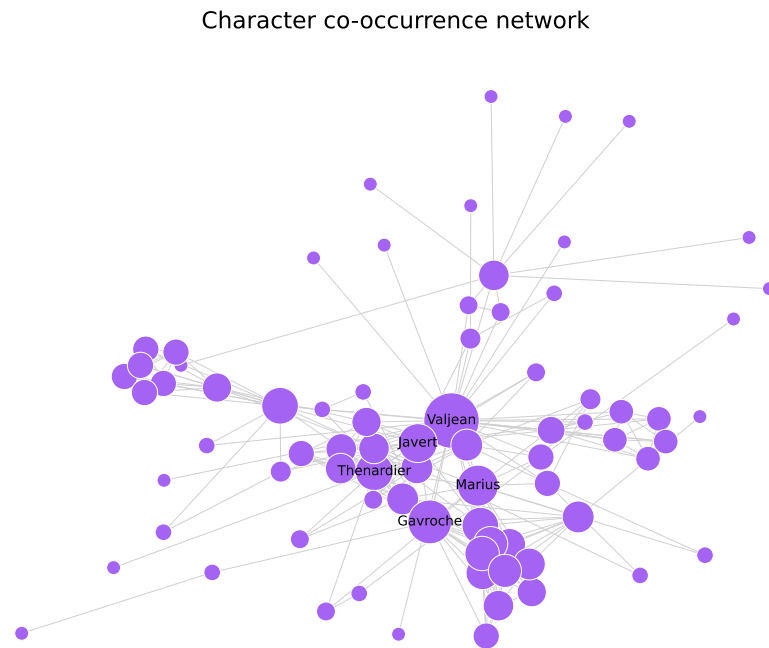


Figure 6.7: The Les Misérables co-appearance graph under a spring layout; node size scales with degree.

6.1.8 47 — Tree of life / phylogeny

Class: named first

Canonical introduction: (Darwin 1859)

Domain: the dendrogram archetype

A phylogenetic tree draws descent as a rooted branching diagram, sister clades meeting at common ancestors. The sole illustration in Darwin's *Origin* was such a tree of divergence (Darwin 1859).

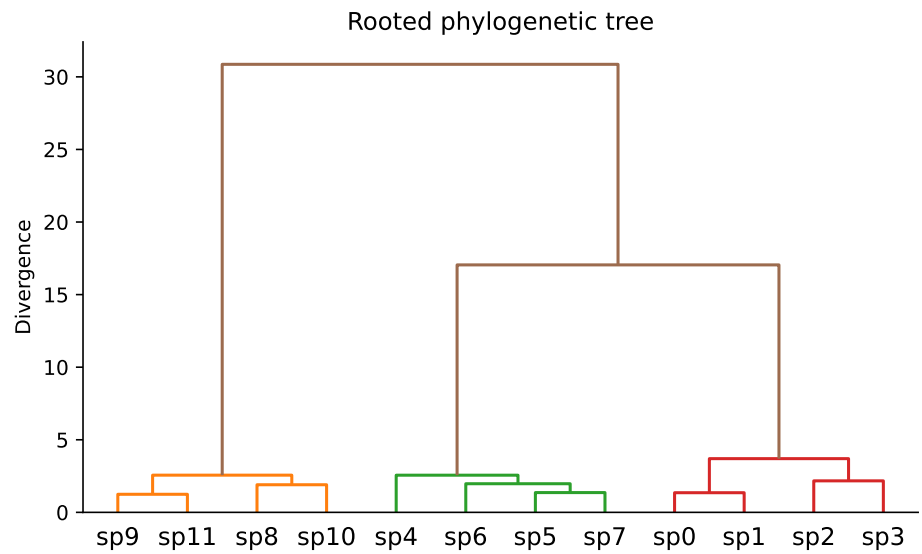


Figure 6.8: A rooted branching tree over synthetic taxa, drawn as a dendrogram from a single common ancestor.

6.1.9 48 — Molecular-clock divergence tree

Class: named first

Canonical introduction: (Zuckerkandl and Pauling 1965)

Domain: branch length as time

A phylogram lets horizontal branch length encode divergence time, turning a bare topology into a datable chronology. Zuckerkandl and Pauling proposed the molecular clock, reading sequence differences as elapsed time (Zuckerkandl and Pauling 1965).

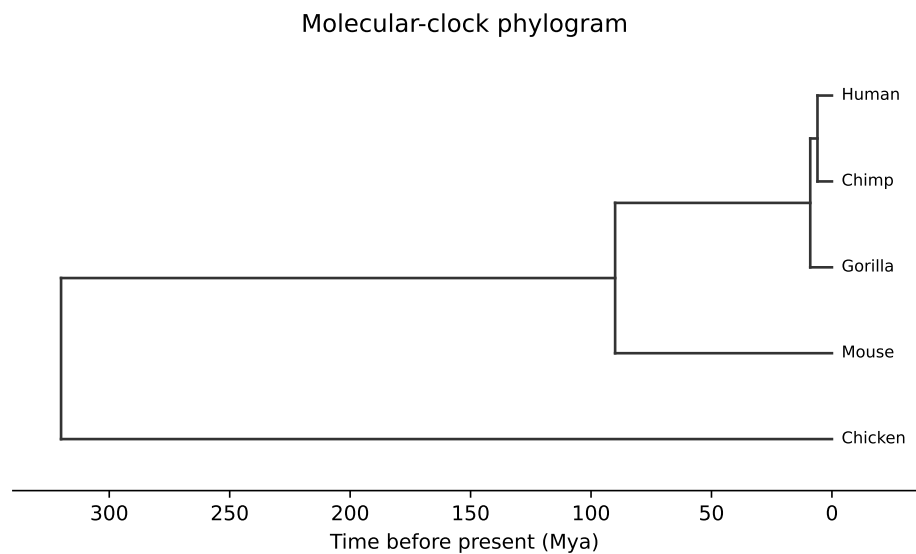


Figure 6.9: A phylogram where horizontal branch length encodes divergence time, with a Mya axis.

6.1.10 49 — Language family tree

Class: named first

Canonical introduction: (Schleicher 1863)

Domain: descent of languages

A Stammbaum draws languages as a descent tree from a reconstructed proto-language, the dendrogram idiom borrowed for historical linguistics. Schleicher imported Darwin’s branching diagram to model language families (Schleicher 1863).

6.1 B1. Connectedness — networks, trees, and designed relational structures

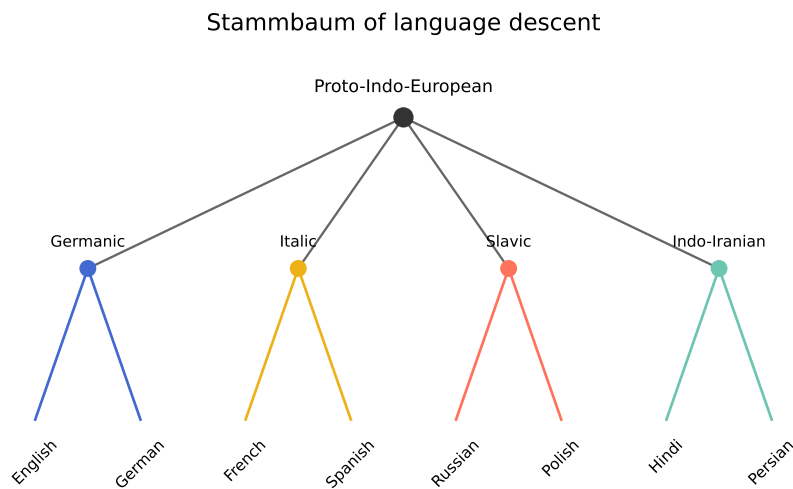


Figure 6.10: A Stammbaum: a proto-language root branching to families and then to individual languages.

6.1.11 50 — The Tonnetz

Class: named first

Canonical introduction: (Euler 1739)

Domain: a designed lattice for pitch

The Tonnetz lays pitch classes on a triangular lattice whose three axes are the perfect fifth, the major third and the minor third, so consonant triads become adjacent triangles. Euler drew the first such tone-network (Euler 1739).

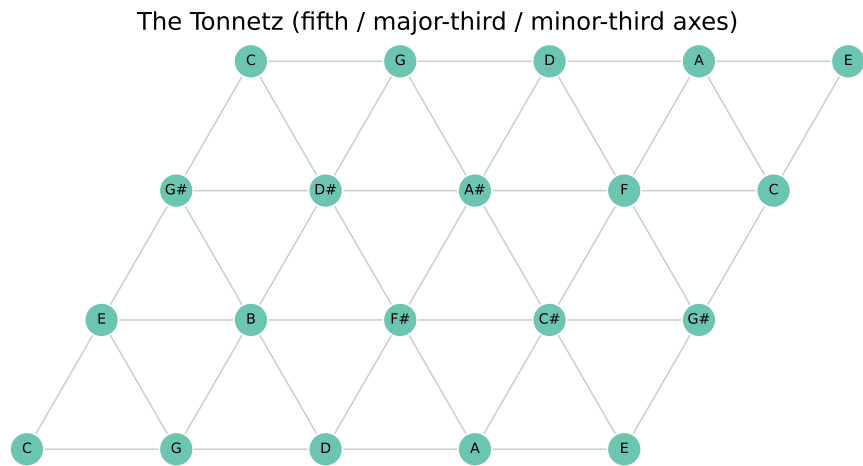


Figure 6.11: A triangular pitch lattice: nodes are pitch classes, edges run along the fifth, major-third and minor-third axes.

6.1.12 51 — cGh cube

Class: shared genre idiom

Canonical introduction: (Okun 1991)

Domain: physical theories placed by their constants

The cGh cube places physical theories at the vertices of a cube whose axes are the constants c , G and h , so each theory is fixed by which constants it takes seriously. Okun popularised the cube as a map of the fundamental theories (Okun 1991).

The cGh cube of theories

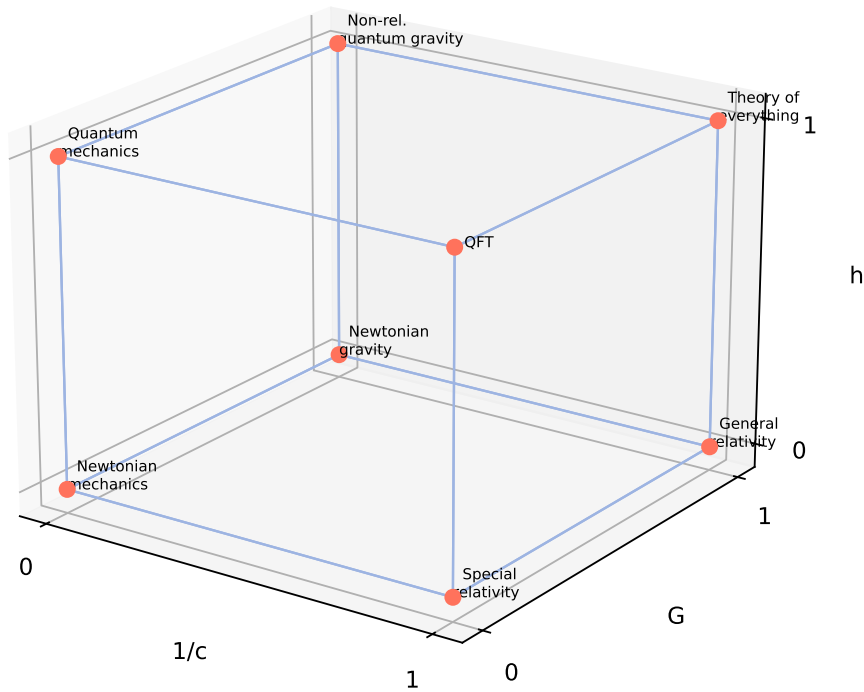


Figure 6.12: The cube of physical theories: axes $1/c$, G , h ; each vertex a theory taking those constants seriously.

6.1.13 52 — Standard Model double simplex

Class: canonical reference

Canonical introduction: (Workman and Data Group) 2022)

Domain: particles placed to show interactions

The Standard Model table arranges the fundamental particles as tiles — three generations of quarks and leptons plus the gauge bosons and the Higgs — coloured by kind so the interaction pattern reads off the grid. The Particle Data Group review is the canonical tabulation (Workman and Data Group) 2022).

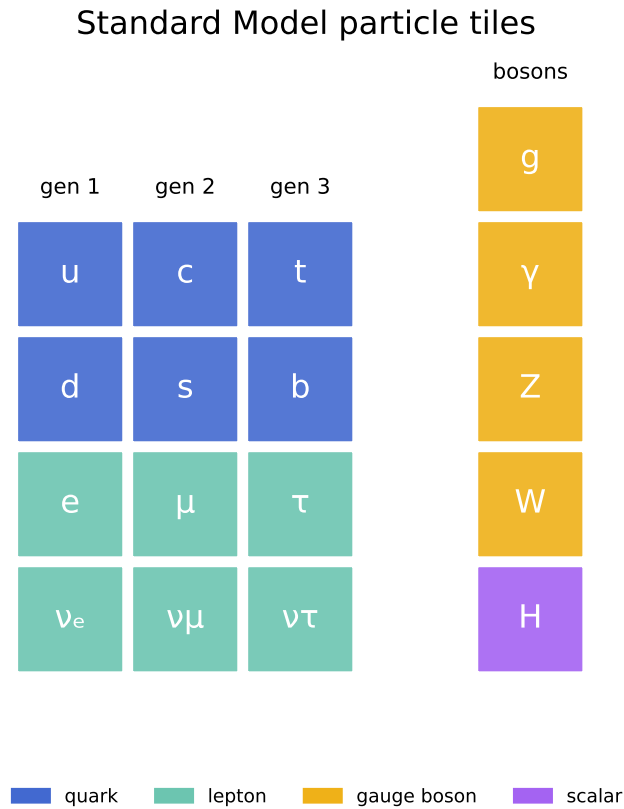


Figure 6.13: The Standard Model as a tile chart: quarks and leptons over three generations, with bosons and the Higgs aside.

6.1.14 53 — Cultural history as a network

Class: canonical reference

Canonical introduction: (Schich et al. 2014)

Domain: 150,000 lives over two millennia

A cultural-history network draws lives as directed edges from birthplace to death-place between geographic centres, edge weight tracing the migration of talent. Schich and colleagues mapped 150,000 notable lives over two millennia this way (Schich et al. 2014).

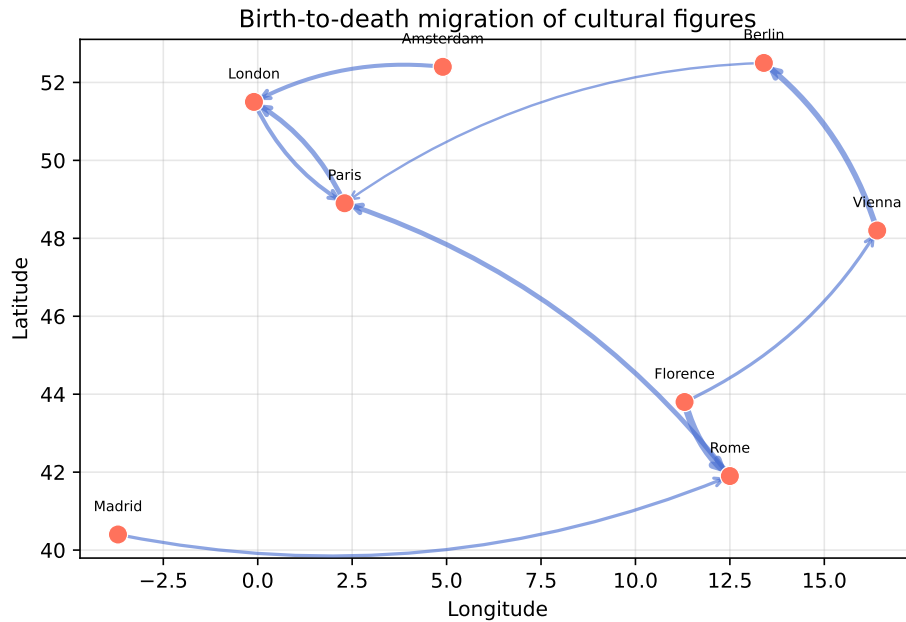


Figure 6.14: A birth-to-death migration graph over synthetic cultural centres; arrow width scales with flow.

6.2 B2. Continuity and flow — directed paths and bands

*After Tufte's reading of Minard, *The Visual Display of Quantitative Information* (1983).*

6.2.1 54 — Water & carbon cycle

Class: shared genre idiom

Canonical introduction: (Falkowski et al. 2000)

Domain: closed-loop flows

Reservoirs sit as labelled boxes and fluxes run between them as directed arrows whose width scales with the mass moved per year, so a conserved quantity is tracked around a closed loop. Falkowski and colleagues drew the carbon budget in exactly this box-and-flux idiom to weigh the anthropogenic perturbation against the natural cycle (Falkowski et al. 2000).

6.2 B2. Continuity and flow — directed paths and bands

Carbon cycle: reservoirs and fluxes (GtC yr^{-1})

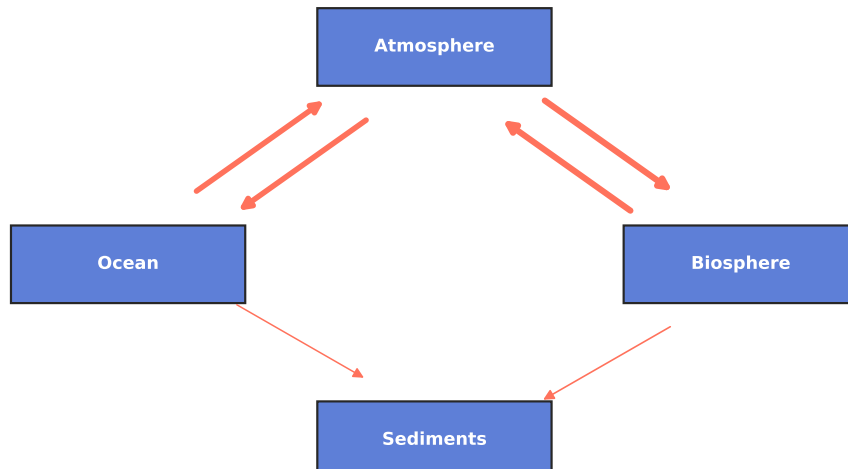


Figure 6.15: Carbon reservoirs as boxes, fluxes as arrows with width scaled to magnitude.

6.2.2 55 — Trophic / energy-flow pyramid

Class: named first + canonical reference (distinct documents)

Canonical introduction: (Lindeman 1942)

Named first: (Elton 1927)

Domain: energy through an ecosystem

Stacked tiers taper from a wide base of producers to a narrow apex, each level roughly a tenth of the one below because most energy is lost as heat at every transfer. Elton drew the pyramid of numbers (Elton 1927); Lindeman recast it as a quantitative energy budget with the $\sim 10\%$ trophic efficiency (Lindeman 1942).

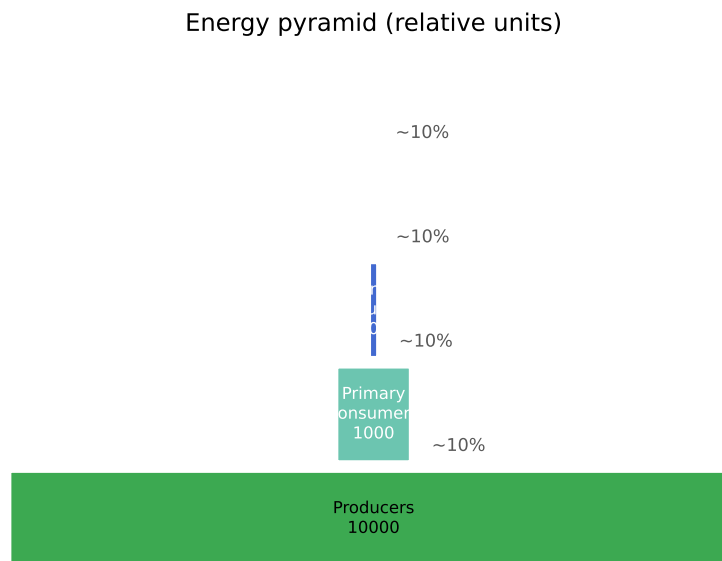


Figure 6.16: Trophic pyramid; each tier holds roughly 10% of the energy of the level below.

6.2.3 56 — Sankey of energy/material flow

Class: named first

Canonical introduction: (Sankey 1898)

Domain: magnitude along paths

A Sankey diagram splits one input stream into branches whose widths stay proportional to the quantity carried, so conservation and loss are read directly from band thickness. Sankey drew the first such diagram to account for the energy losses of a steam engine (Sankey 1898).

Ignoring fixed y limits to fulfill fixed data aspect with adjustable data line
Ignoring fixed x limits to fulfill fixed data aspect with adjustable data line

6.2 B2. Continuity and flow — directed paths and bands

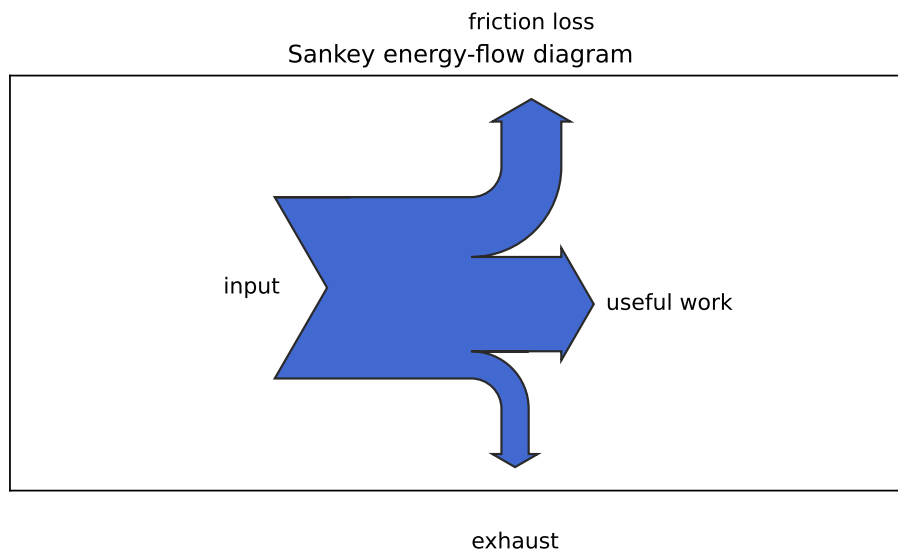


Figure 6.17: Sankey flow: one input splitting into useful work and two loss streams.

6.2.4 57 — Haber–Bosch / refinery process flow

Class: named first

Canonical introduction: (Bosch 1932)

Domain: the industrial transformation

A process-flow diagram chains unit operations as boxes and connects them with directed material streams, closing the loop where unreacted feed is recycled. Bosch described the high-pressure ammonia synthesis this diagram traces — feed compression, catalytic reaction, condensation, and recycle (Bosch 1932).

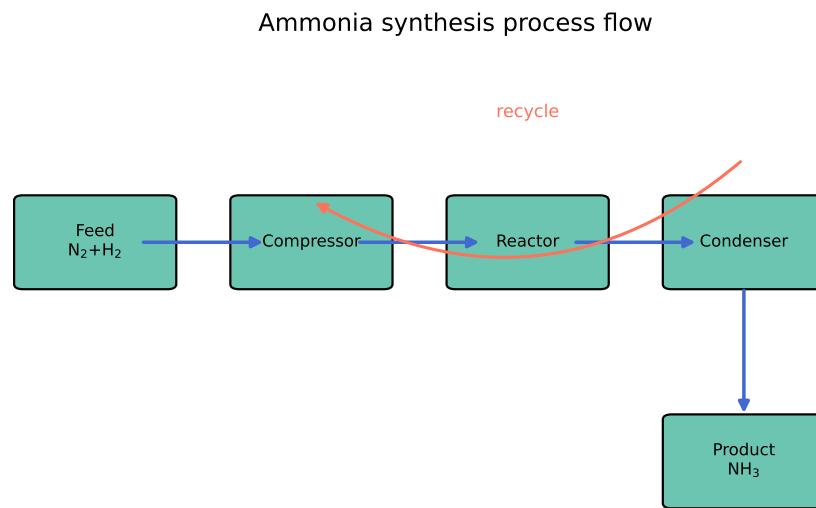


Figure 6.18: Haber–Bosch process flow: feed $\rightarrow$ compressor $\rightarrow$ reactor $\rightarrow$ condenser $\rightarrow$ product, with recycle.

6.2.5 58 — Central dogma

Class: named first + canonical reference (distinct documents)

Canonical introduction: (Crick 1970)

Named first: (Crick 1958)

Domain: directional flow of information

Information passes DNA $\rightarrow$ RNA $\rightarrow$ protein along labelled arrows, with replication looping back onto DNA, and the arrow that is absent — protein back to nucleic acid — is the claim. Crick named the central dogma (Crick 1958) and restated it precisely a decade later (Crick 1970).

6.2 B2. Continuity and flow — directed paths and bands

The central dogma of molecular biology

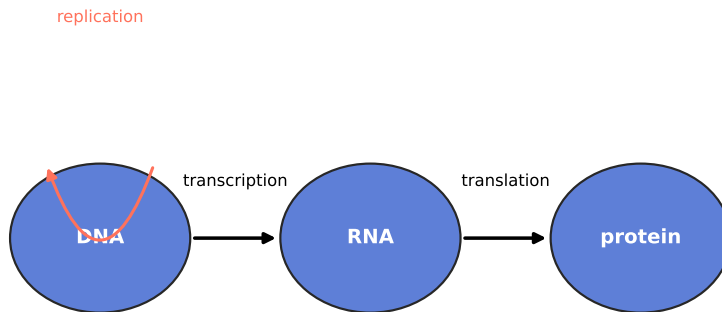


Figure 6.19: Central dogma: DNA transcribed to RNA, translated to protein; DNA self-replicates.

6.2.6 59 — Shannon channel + entropy

Class: named first

Canonical introduction: (Shannon 1948)

Domain: information through noise

The communication system is drawn as a chain — source, encoder, noisy channel, decoder, destination — and the amount of information is measured by an entropy that peaks when outcomes are equiprobable. Shannon set out both the block diagram and the entropy function in one paper (Shannon 1948).

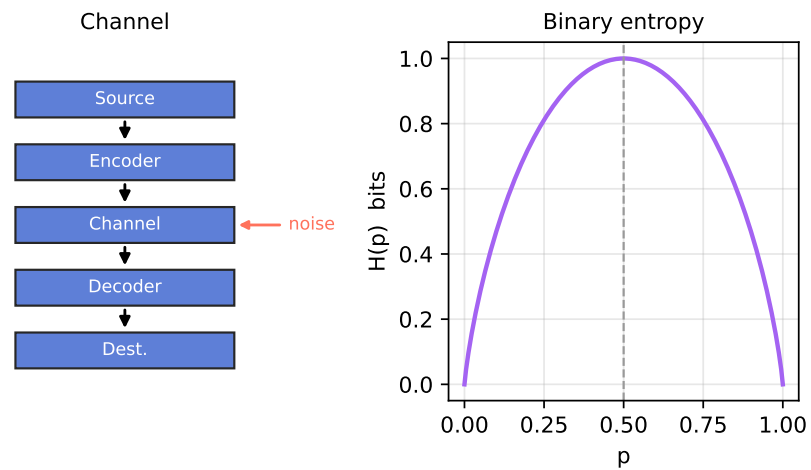


Figure 6.20: Shannon's channel block diagram (left) and binary entropy $H(p)$, maximal at $p=0.5$ (right).

6.2.7 60 — Greenhouse / energy-balance (Trenberth) diagram

Class: shared genre idiom

Canonical introduction: (Trenberth et al. 2009)

Named first: (Kiehl and Trenberth 1997)

Domain: the energy budget

Broad arrows carry incoming solar, reflected shortwave, outgoing longwave, and downward back-radiation, each tagged with a global-mean flux in W m^{-2} , so the planetary balance closes to a small residual. Kiehl and Trenberth fixed the canonical numbers (Kiehl and Trenberth 1997); Trenberth, Fasullo, and Kiehl revised them for the modern budget (Trenberth et al. 2009).

6.2 B2. Continuity and flow — directed paths and bands

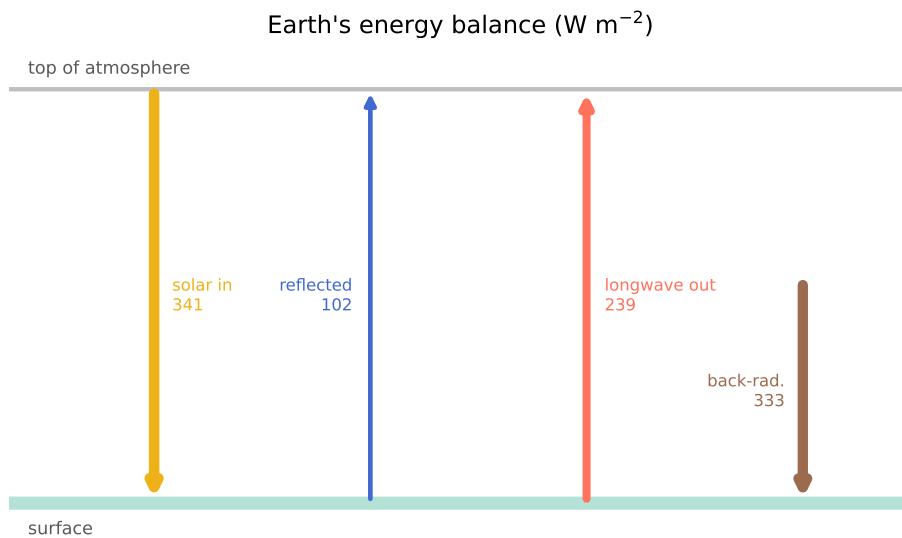


Figure 6.21: Earth's energy budget: solar in, shortwave reflected, longwave out, back-radiation (W m^{-2}).

6.2.8 61 — Out-of-Africa migration map

Class: shared genre idiom

Canonical introduction: (Cann et al. 1987)

Domain: movement across space

Arrows radiate from East Africa across an abstracted world, each annotated with an approximate arrival date, turning a genealogy into a route across space and time. Cann, Stoneking, and Wilson's mitochondrial coalescence placed the common ancestor in Africa and set this migration narrative going (Cann et al. 1987).

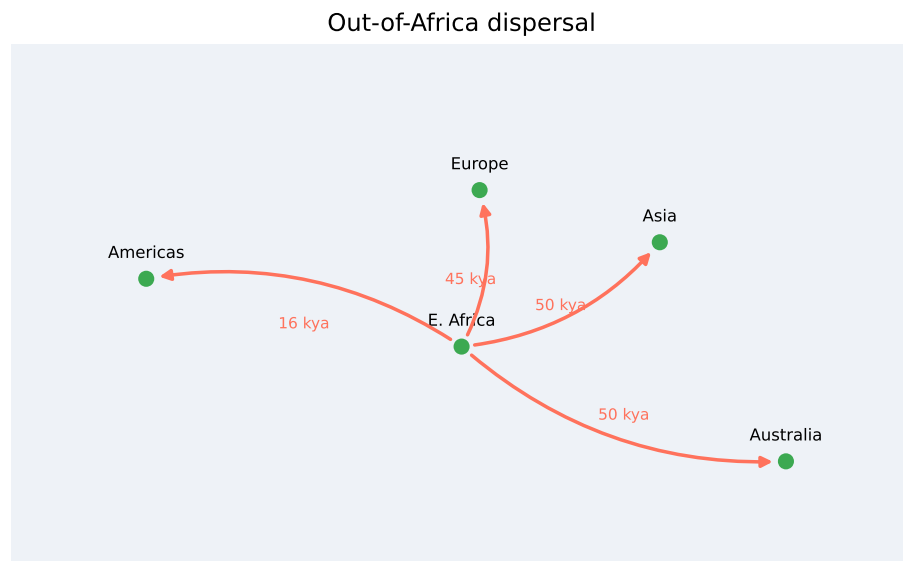


Figure 6.22: Abstract world map with dispersal arrows from East Africa, dates in thousand years ago.

6.2.9 62 — Minard's march on Moscow

Class: named first

Canonical introduction: (Minard 1869)

Domain: six variables, one frame

A band whose width tracks the surviving army thins along the advance and thins again on the retreat, while a linked panel below plots the falling temperature against longitude — six variables held in one frame. Minard's flow map of Napoleon's 1812 campaign is the canonical instance (Minard 1869).

6.3 B3. Proximity and similarity — clustering, and how machines see

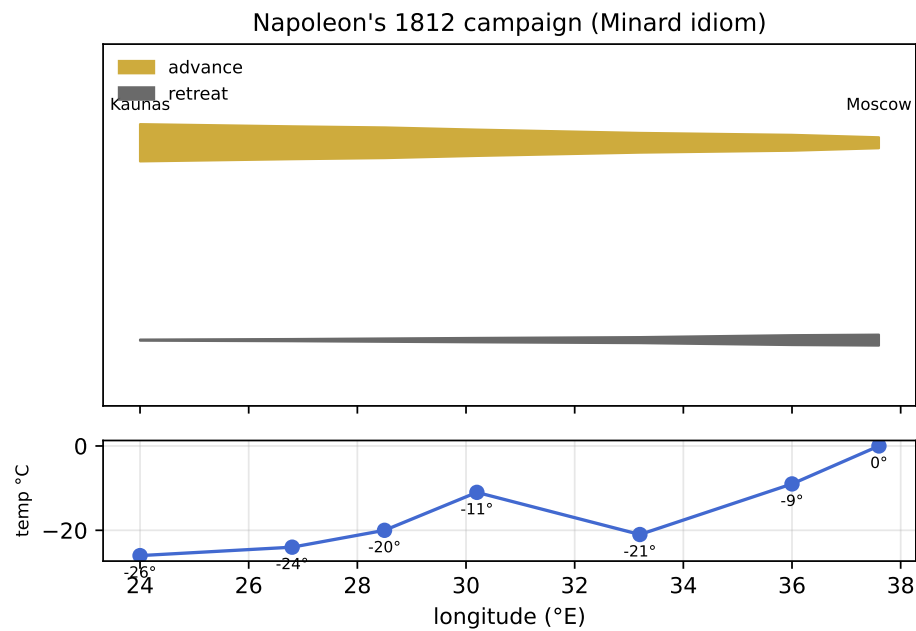


Figure 6.23: Minard idiom: army-size band (advance vs retreat) above, retreat temperature below.

6.3 B3. Proximity and similarity — clustering, and how machines see

After Cleveland & McGill on ordered comparison, and Ware on grouping by similarity.

6.3.1 63 — Clustered expression heatmap

Class: named first

Canonical introduction: (Eisen et al. 1998)

Domain: reorder, and clusters appear

Permuting the rows and columns of an expression matrix by hierarchical clustering pulls co-regulated genes into contiguous blocks that a random ordering hides. Eisen and colleagues introduced the clustered heatmap with its red–green diverging map for genome-wide expression (Eisen et al. 1998).

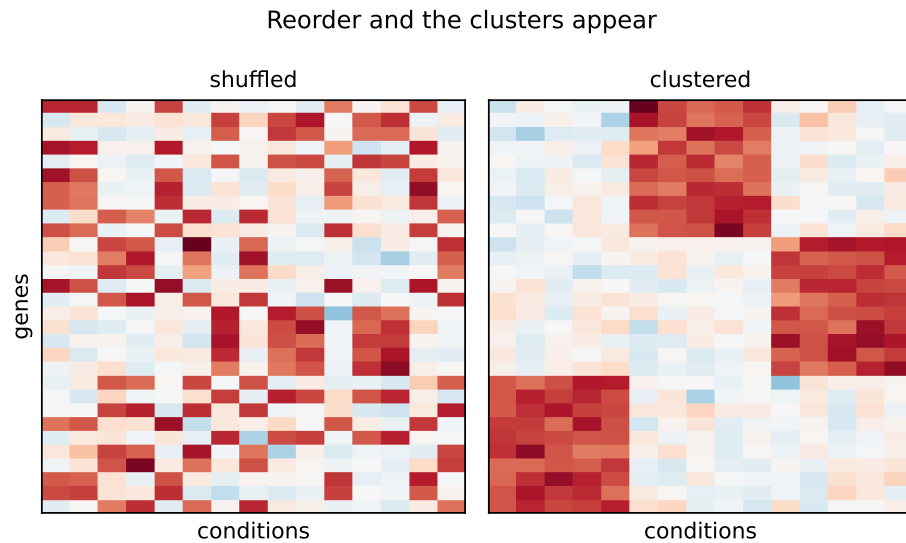


Figure 6.24: The same genes×conditions matrix: random order (left) vs clustered order (right).

6.3.2 64 — Correlation heatmap / scatter-matrix

Class: uncertain attribution

Canonical introduction: (Chambers et al. 1983)

Domain: relationships at scale

A scatterplot matrix lays every pairwise plot in a grid so structure and correlation across many variables are scanned at once, off-diagonal cells mirroring about the diagonal. Chambers and colleagues codified the scatterplot matrix as a standard tool of graphical analysis (Chambers et al. 1983).

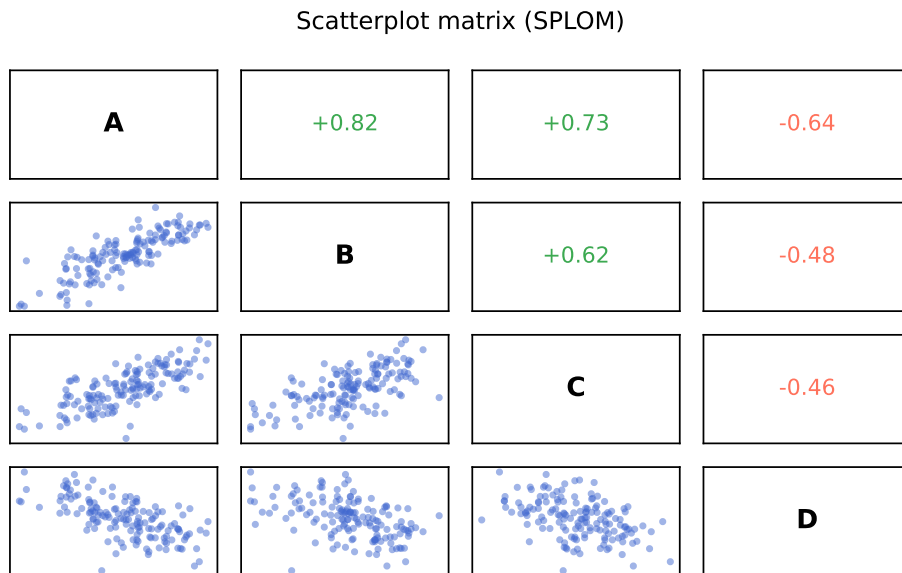


Figure 6.25: Scatterplot matrix of four variables with correlation coefficients on the diagonal.

6.3.3 65 — Cosine similarity

Class: canonical reference

Canonical introduction: (Salton et al. 1975)

Domain: the angle between vectors

Documents become vectors and their similarity is the cosine of the angle between them, so length drops out and only direction — shared term profile — counts. Salton's vector space model made cosine similarity the standard measure for information retrieval (Salton et al. 1975).

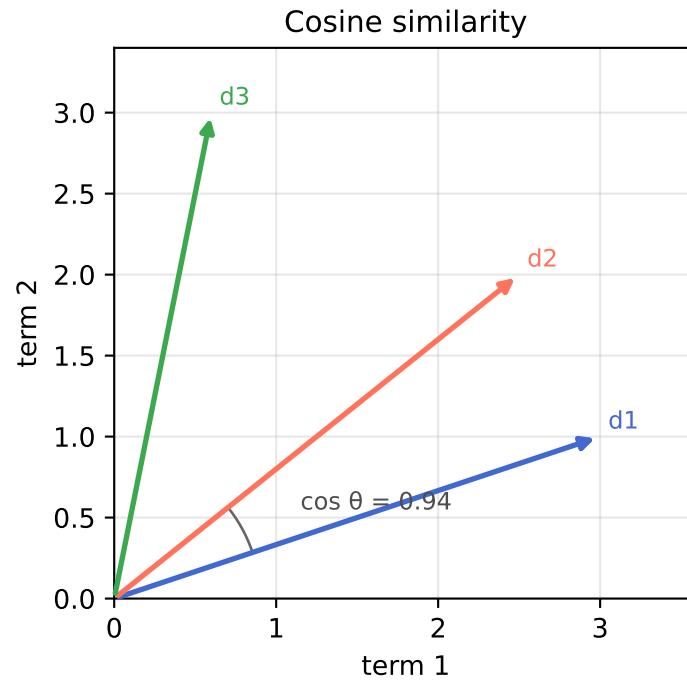


Figure 6.26: Document vectors from the origin; the angle between two gives cosine similarity.

6.3.4 66 — t-SNE / UMAP projection

Class: named first

Canonical introduction: (Maaten and Hinton 2008)

Domain: high dimensions flattened to 2D

A nonlinear embedding places high-dimensional points on the plane so that neighbours stay near and distinct groups fall into separated islands. van der Maaten and Hinton introduced t-SNE, whose crisply separated clusters became the signature look of embedding plots (Maaten and Hinton 2008).

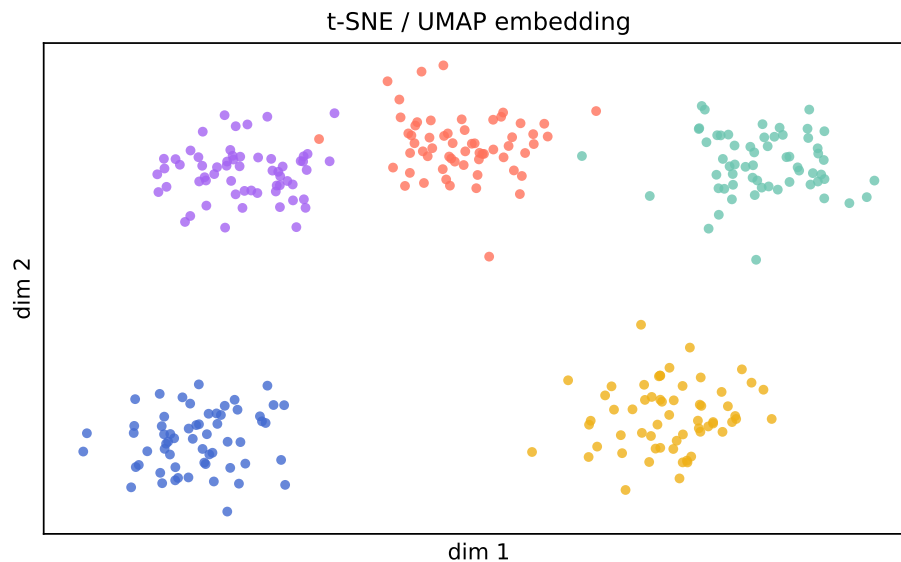


Figure 6.27: A 2D embedding: well-separated coloured clusters (simulated t-SNE output).

6.3.5 67 — Word embeddings

Class: named first

Canonical introduction: (Mikolov et al. 2013)

Domain: semantics as geometry

Words map to vectors whose differences encode relations, so an analogy becomes a parallelogram: $\text{king} - \text{man} + \text{woman}$ lands near queen. Mikolov and colleagues' word2vec produced these linear analogy structures (Mikolov et al. 2013).

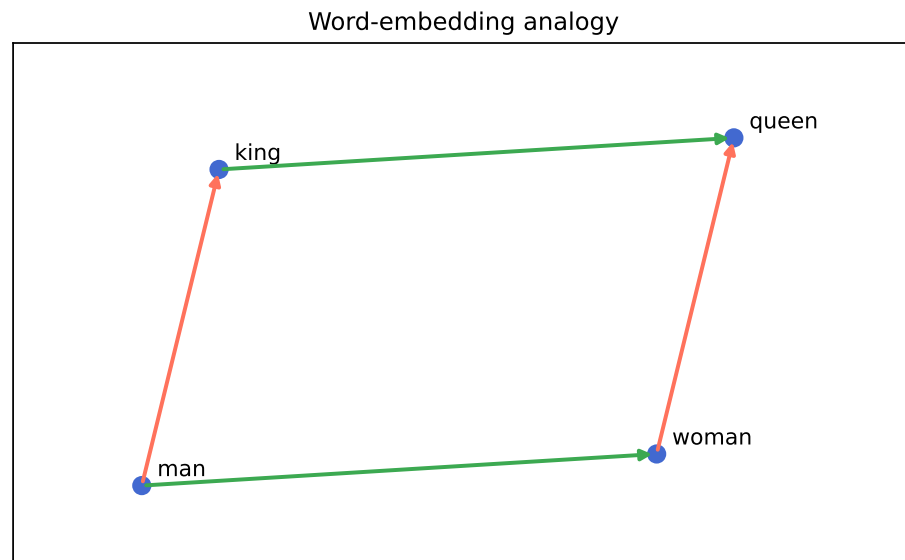


Figure 6.28: Word vectors forming the analogy parallelogram $\text{king} - \text{man} + \text{woman} \rightarrow \text{queen}$.

6.3.6 68 — Art embedding map

Class: uncertain attribution

Canonical introduction: (Karayev et al. 2014)

Domain: a collection placed by visual similarity

Every artwork becomes a point in a plane and visual style pulls similar works together, so a collection self-organises into style neighbourhoods. Karayev and colleagues learned image-style features whose embeddings cluster paintings and photographs by their look (Karayev et al. 2014).

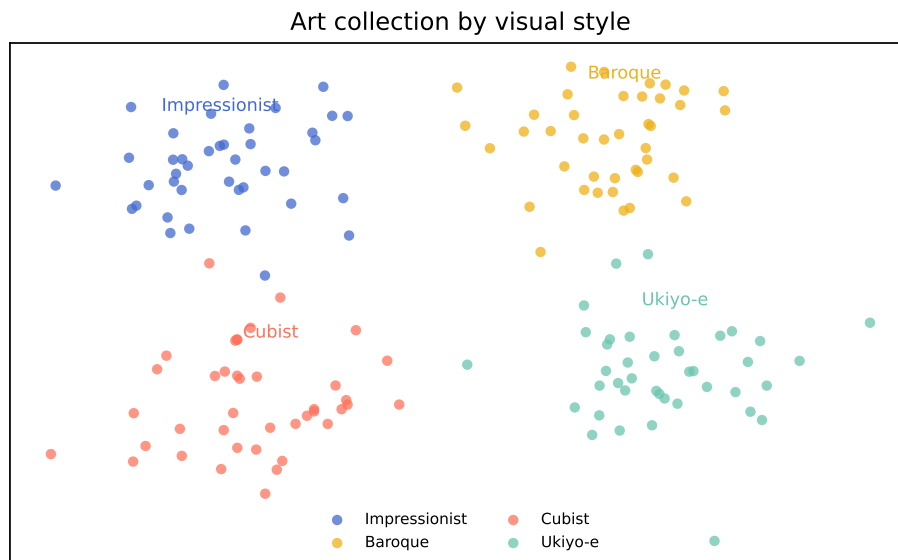


Figure 6.29: Artworks embedded in 2D, clustering by visual style (colour = style label).

6.3.7 69 — Stylometry / Burrows's Delta

Class: named first

Canonical introduction: (Burrows 2002)

Domain: authorship as distance in style-space

Each text becomes a vector of z-scored frequencies of common words, and the mean absolute difference between two such vectors — Delta — clusters texts by author. Burrows defined the Delta measure for authorship attribution (Burrows 2002).

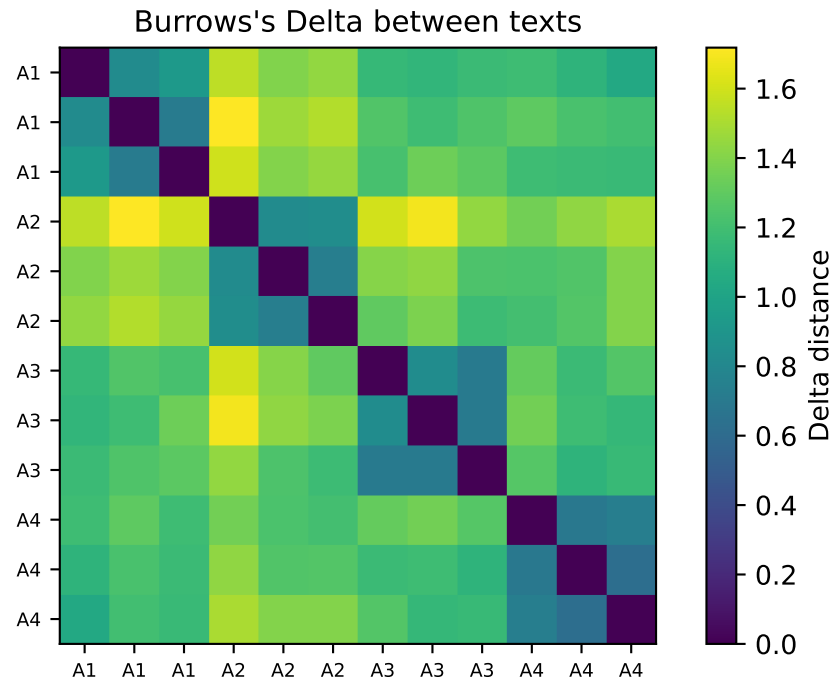


Figure 6.30: Pairwise Burrows's Delta between texts; the author blocks sit on the diagonal.

6.3.8 70 — Topic model (LDA)

Class: named first

Canonical introduction: (Blei et al. 2003)

Domain: latent themes as a reduced basis

Latent Dirichlet allocation represents each document as a mixture over a few latent topics, so a stacked bar reads off how much of every theme a document holds. Blei, Ng, and Jordan introduced LDA and its document-as-topic-mixture representation (Blei et al. 2003).

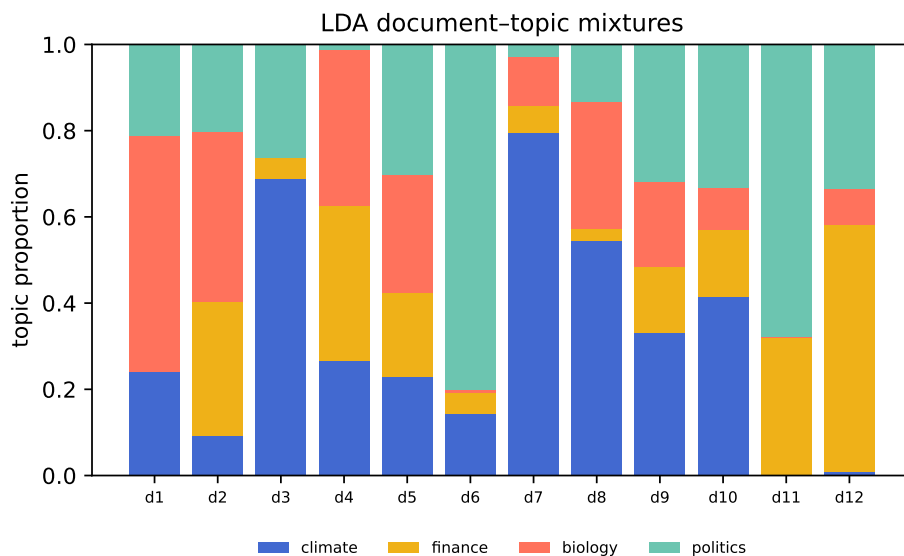


Figure 6.31: Documents as topic mixtures; each stacked bar sums to one over the latent topics.

6.3.9 71 — Emotional arcs of stories

Class: named first

Canonical introduction: (Reagan et al. 2016)

Domain: six narrative shapes from SVD

Tracking sentiment across narrative time yields a small set of recurring shapes — steady rise, fall-then-rise, rise-then-fall — that a singular value decomposition of many stories reduces to a handful of archetypes. Reagan and colleagues extracted six such emotional arcs from thousands of texts (Reagan et al. 2016).

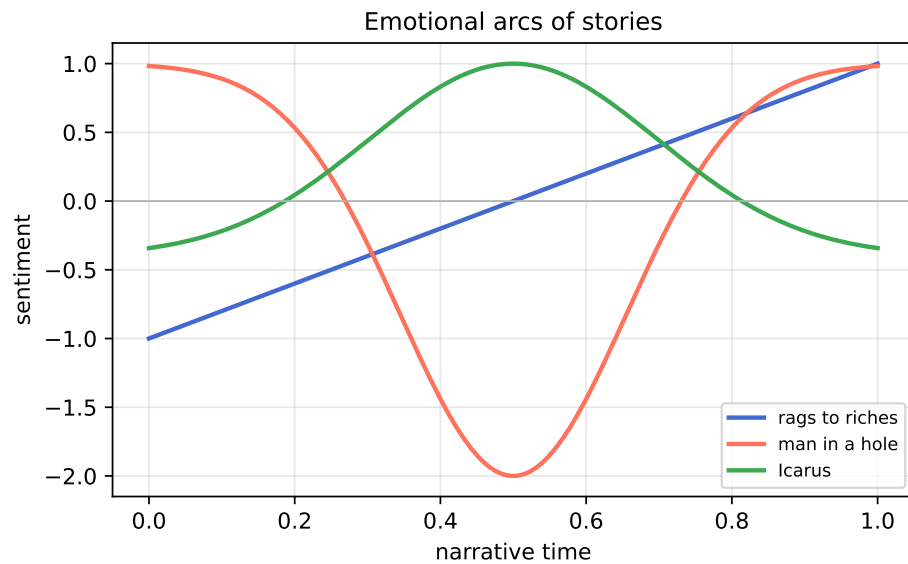


Figure 6.32: Three archetypal emotional arcs: rags to riches, man in a hole, Icarus.

6.4 B4. Enclosure, maps, and spatial reasoning

After Bertin on maps and Cairo, The Truthful Art (2016).

6.4.1 72 — Periodic table

Class: named first

Canonical introduction: (Mendelejeff 1869)

Domain: periodicity by atomic number

A tiled grid indexed by group and period turns the chemistry of an element into a position, and neighbours in a column share properties. Mendeleev arranged the known elements so that the periodic recurrence fell into columns and left gaps for elements not yet found (Mendelejeff 1869).

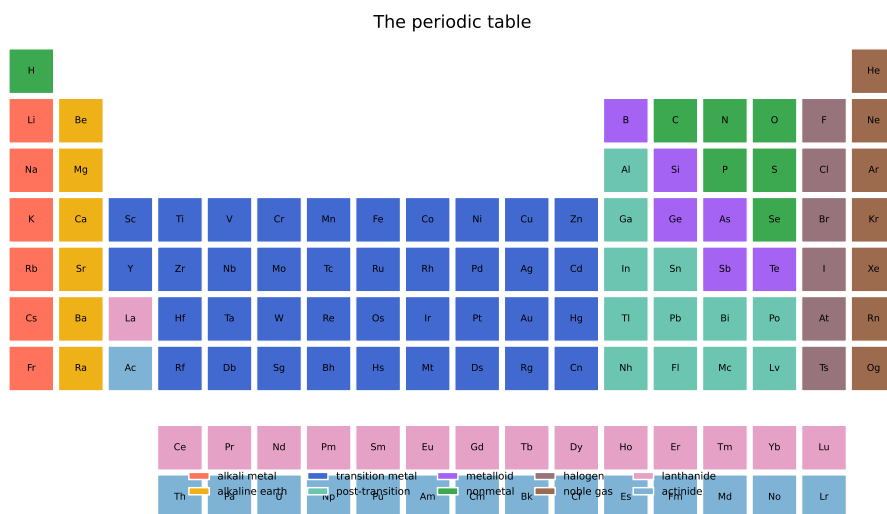


Figure 6.33: The periodic grid: tiles at (group, period), coloured by element category.

6.4.2 73 — Origin-of-the-elements table

Class: shared genre idiom

Canonical introduction: (Burbidge et al. 1957)

Domain: the grid recoloured by process

The same tiled grid, recoloured by where each nucleus is forged, turns the table into a map of cosmic history — hydrogen from the Big Bang, iron from exploding stars, the heaviest nuclei from merging neutron stars. Burbidge, Burbidge, Fowler, and Hoyle set out the stellar processes that build the elements beyond hydrogen and helium (Burbidge et al. 1957).

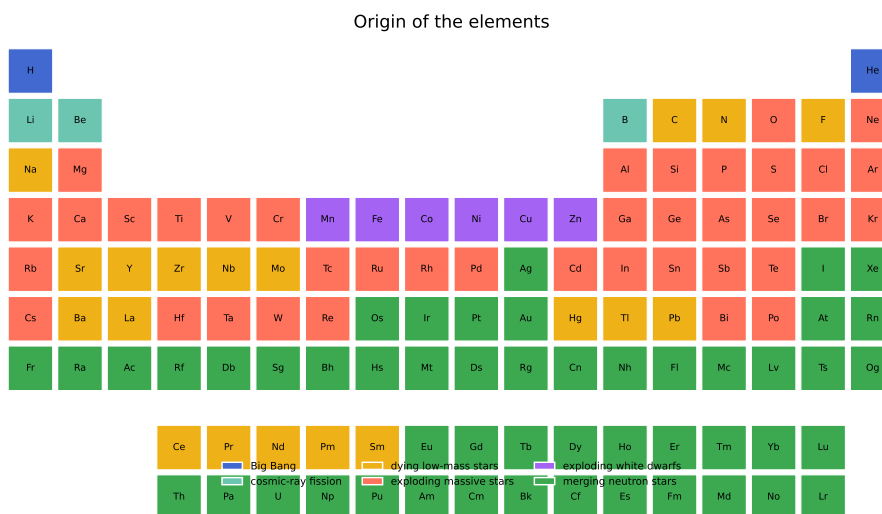


Figure 6.34: The periodic grid recoloured by nucleosynthetic origin.

6.4.3 74 — Genetic code wheel

Class: shared genre idiom

Canonical introduction: (Nirenberg and Matthaei 1961)

6.4 B4. *Enclosure, maps, and spatial reasoning*

Domain: codon to amino acid

Three concentric rings — first, second, third base — read from centre outward resolve each of the 64 codons to one amino acid, with synonymous codons falling into the same wedge. Nirenberg and Matthaei cracked the first codon by showing that a poly-uracil template directs synthesis of phenylalanine (Nirenberg and Matthaei 1961).

Genetic code wheel

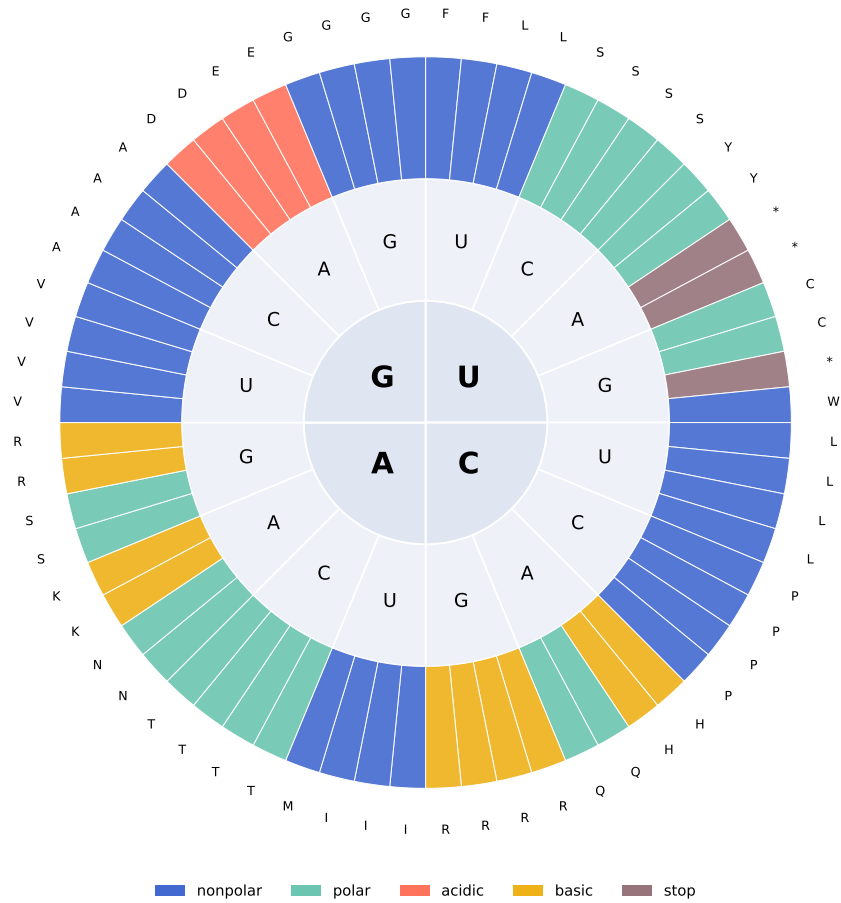


Figure 6.35: Codon wheel: rings for the 1st, 2nd, 3rd base map 64 codons to amino acids.

6.4.4 75 — Leontief input–output matrix

Class: named first

Canonical introduction: (Leontief 1936)

Domain: an economy as a matrix

Laying every sector along both axes and filling cell (i, j) with the flow of output from sector i into sector j renders a whole economy as one matrix, its diagonal thick with intra-industry use. Leontief built the first such input–output table for the United States and solved it as a linear system (Leontief 1936).

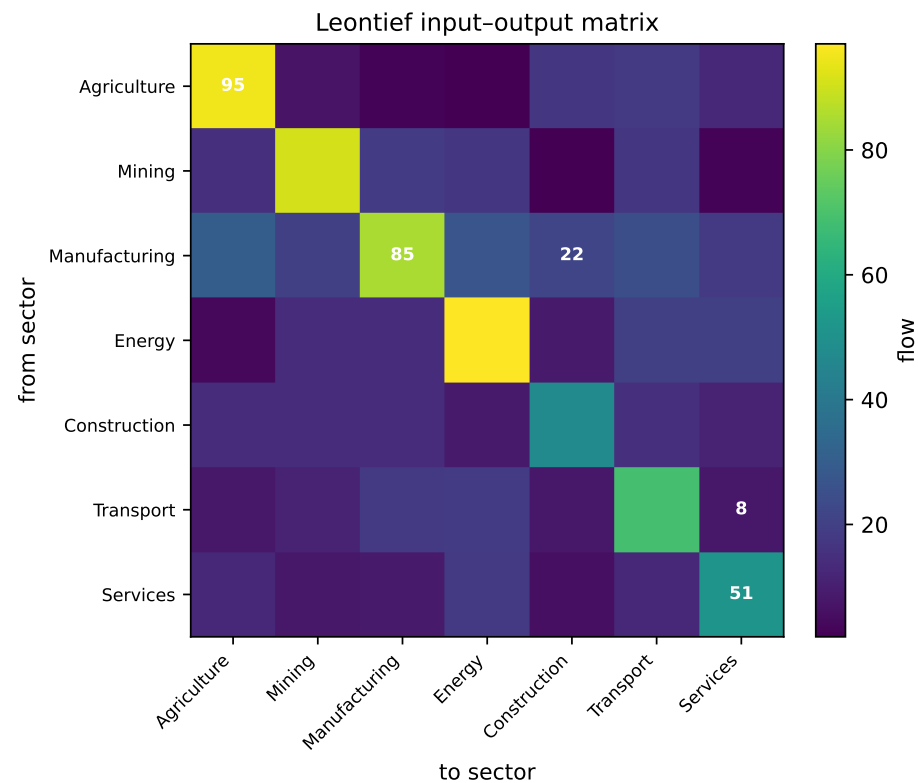


Figure 6.36: Inter-industry flow matrix: value flowing from row sector to column sector.

6.4.5 76 — Planetary interior cross-sections

Class: shared genre idiom

Canonical introduction: (Dziewonski and Anderson 1981)

Domain: layered structure

Concentric shells drawn to scale expose a planet’s radial structure — a

6.4 B4. Enclosure, maps, and spatial reasoning

solid inner core, a liquid outer core, the mantle, and a thin crust — with the density jumping at each boundary. Dziewonski and Anderson fixed the reference radii and densities of Earth's layers in the Preliminary Reference Earth Model (Dziewonski and Anderson 1981).

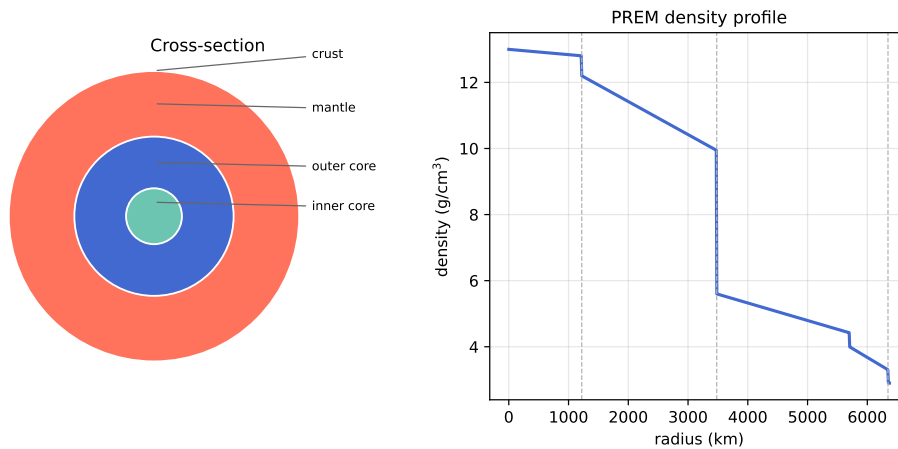


Figure 6.37: Earth's layers to scale (left) and a PREM-style density profile (right).

6.4.6 77 — Biogeography map / Wallace line

Class: named first

Canonical introduction: (Wallace 1876)

Domain: boundaries in space

A single line drawn between adjacent islands can separate two faunas that never mixed, making a deep-water barrier legible as a border on the map. Wallace traced the sharp zoological divide running between Bali and Lombok that splits Asian from Australian animals (Wallace 1876).

6 Part B — Gestalt grouping

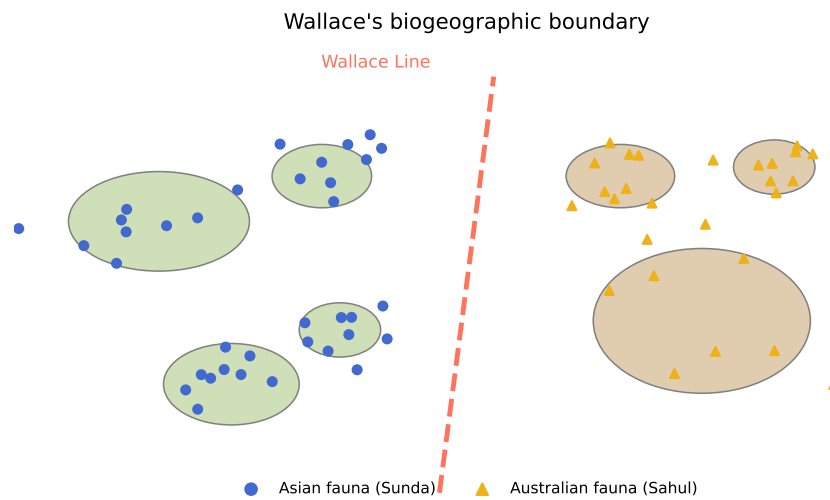


Figure 6.38: Schematic map: the Wallace Line dividing Asian (Sunda) from Australian (Sahul) fauna.

6.4.7 78 — John Snow's cholera map

Class: named first

Canonical introduction: (Snow 1855)

Domain: spatial pattern as causal argument

Plotting each death as a dot turns a street into an argument: the cases pile up around one water pump and thin with distance from it, so the spatial pattern itself names the cause. Snow mapped the 1854 Soho outbreak and traced it to the Broad Street pump (Snow 1855) — the second beat of the Day-One pairing, where a map, not a table, carries the inference.

6.4 B4. Enclosure, maps, and spatial reasoning

Snow's cholera map

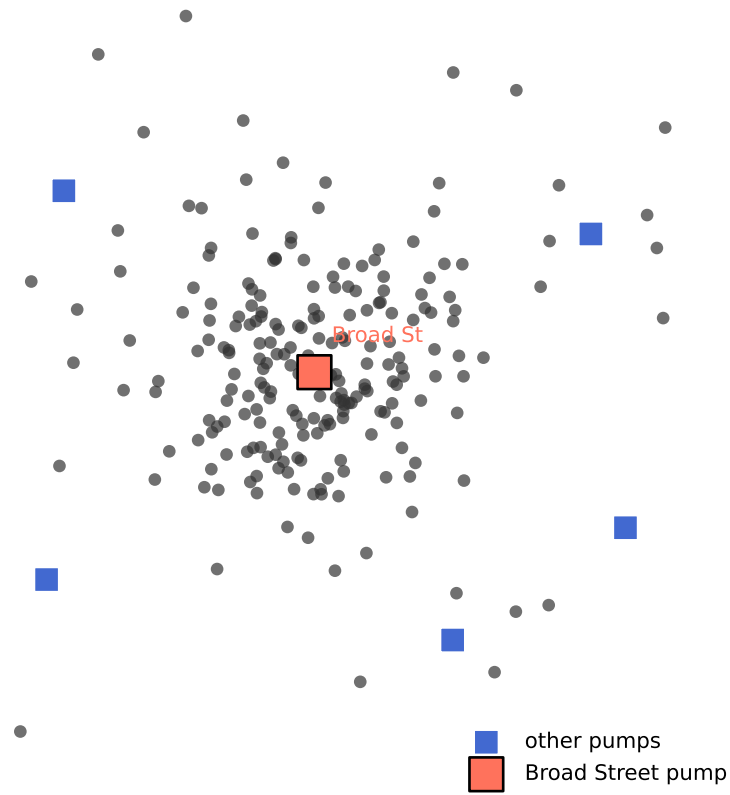


Figure 6.39: Dot map of cholera deaths clustered on the Broad Street pump.

6.4.8 79 — Gerrymander / districting

Class: named first

Canonical introduction: (Griffith 1907)

Domain: how boundaries manufacture outcomes

Holding the votes fixed and redrawing the district lines changes who wins, so the partition — not the electorate — sets the seat count. Griffith gave the first systematic geometric analysis of how district shape converts votes into representation (Griffith 1907).

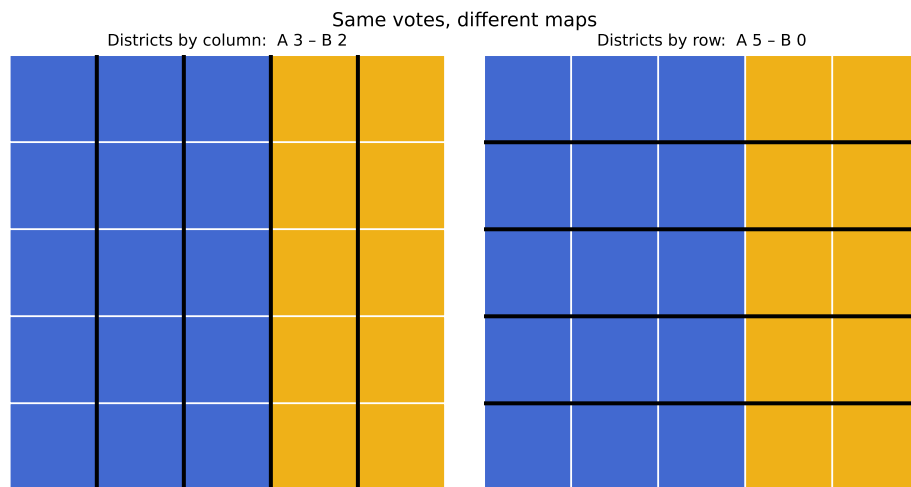


Figure 6.40: Same 60/40 grid, two partitions: 3–2 by columns versus 5–0 by rows.

6.4.9 80 — Minkowski / light cone

Class: named first

Canonical introduction: (Minkowski 1909)

Domain: the cone partitions spacetime

6.4 B4. Enclosure, maps, and spatial reasoning

Drawing space against time and marking the 45° paths of light carves space-time into a causal future, a causal past, and an unreachable “elsewhere”. Minkowski recast special relativity as geometry on this four-dimensional continuum, with the cone dividing what an event can and cannot influence (Minkowski 1909).

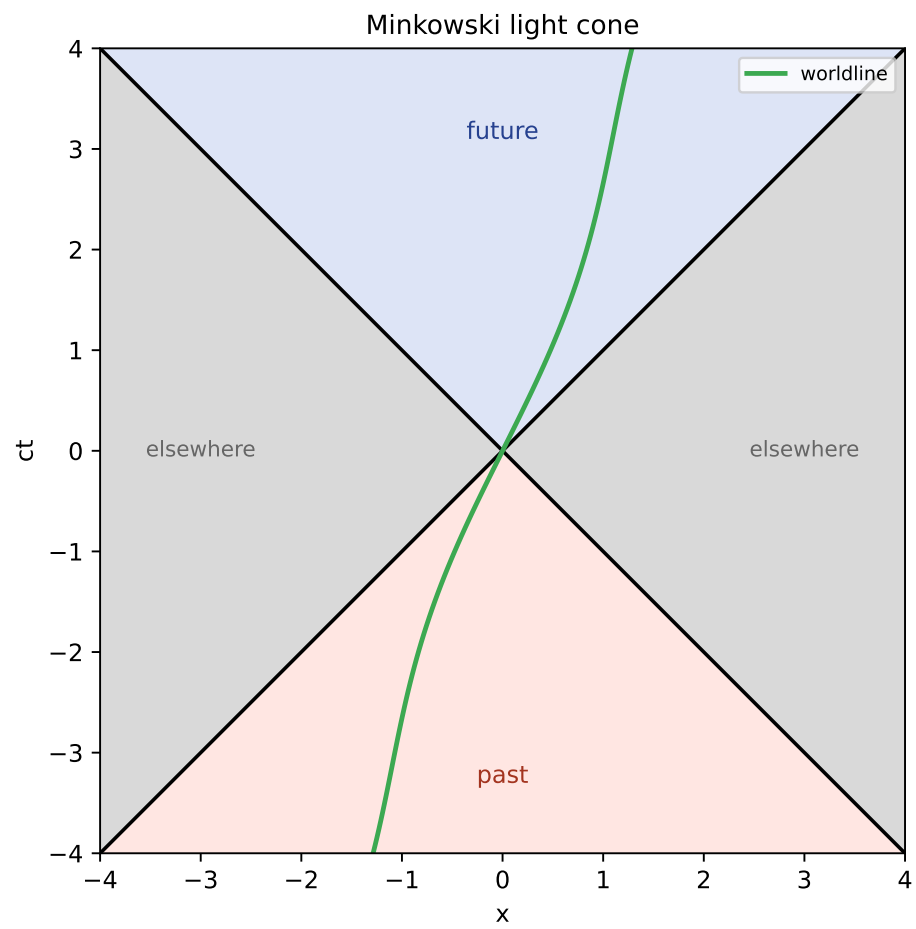


Figure 6.41: A spacetime diagram: light cones partition future, past, and spacelike elsewhere.

6.4.10 81 — Map projection problem

Class: shared genre idiom

Canonical introduction: (Tissot 1881)

Domain: every map distorts

No flat map preserves both shape and area; placing a small circle at each graticule node makes the distortion visible as the circles swell or stretch toward the poles. Tissot formalised this indicatrix as the general measure of a projection's local error (Tissot 1881) — the same tension the AIATSIS prelude raised, that no single frame renders the whole faithfully.

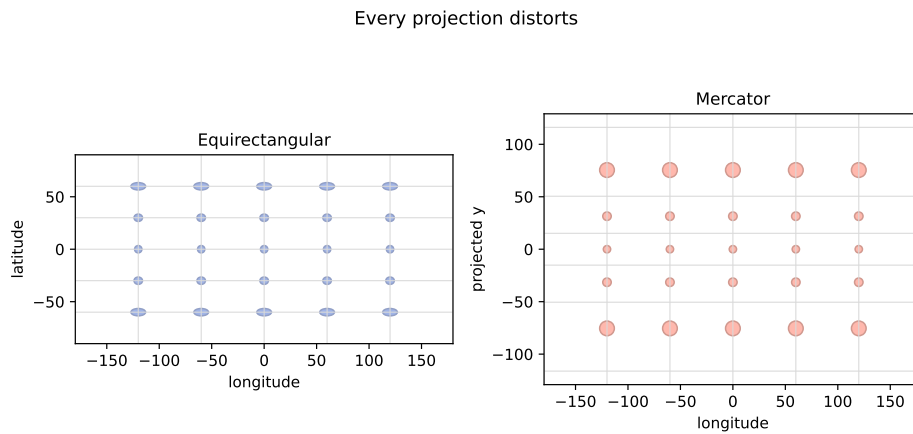


Figure 6.42: Tissot indicatrices under equirectangular (stretched) and Mercator (inflated) projections.

6.5 B5. Closure, common fate, and the diagnostic pattern

After Ware on Gestalt closure and common fate (2004).

6.5.1 82 — Blackbody curve + spectral lines

Class: named first

Canonical introduction: (Planck 1901)

Domain: lines pop against the continuum

The smooth Planck continuum sets the backdrop against which sharp absorption lines read as notches — closure fills the missing light, so the eye completes the curve and the dark lines stand out. Planck derived the radiance law by quantising the oscillators of the cavity wall (Planck 1901), and its peak shifts to shorter wavelengths as temperature climbs.

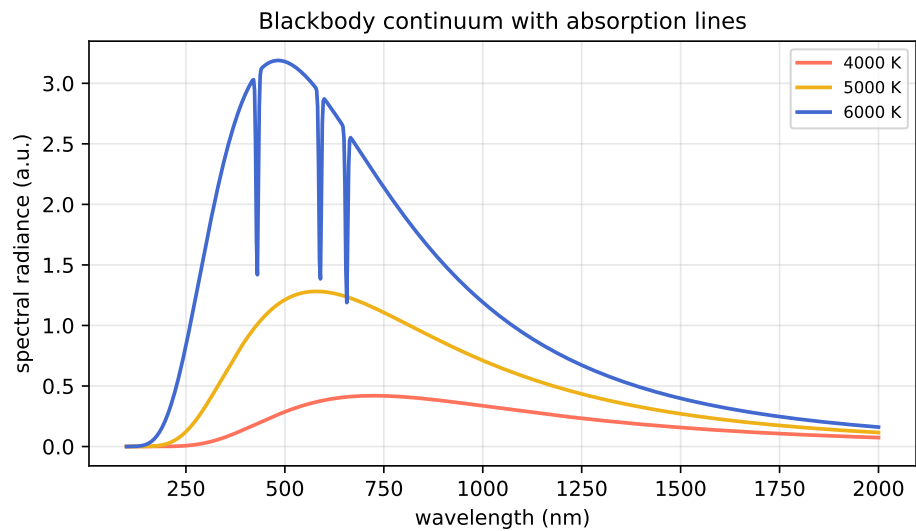


Figure 6.43: Planck curves at three temperatures; absorption lines notched into the hottest.

6.5.2 83 — CMB angular power spectrum

Class: named first

Canonical introduction: (Bernardis et al. 2000)

Domain: the acoustic peaks

Power plotted against multipole rises to a first acoustic peak near 200 and then rings down in damped harmonics — a single wave train the eye groups by common fate. The BOOMERANG flight resolved that first peak and pinned the geometry of the universe to flat (Bernardis et al. 2000).

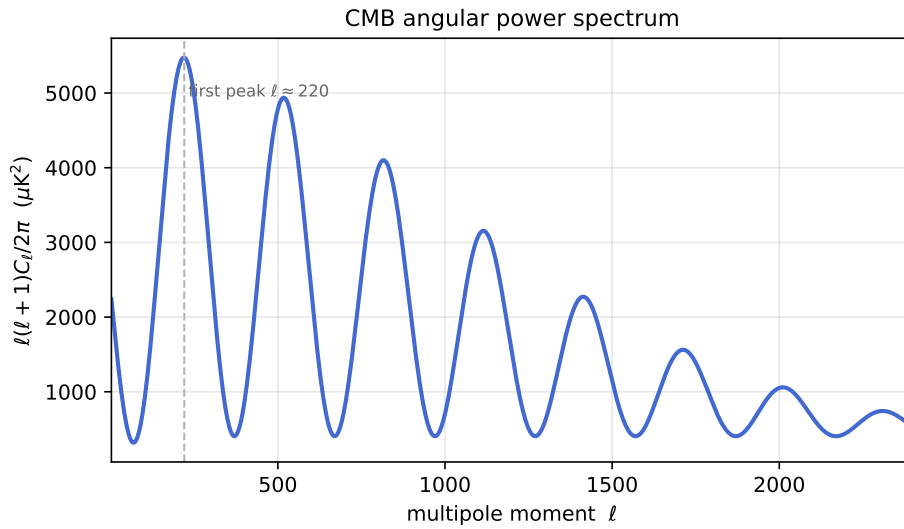


Figure 6.44: Synthesised CMB angular power spectrum with acoustic peaks and Silk damping.

6.5.3 84 — Higgs discovery bump

Class: named first

Canonical introduction: (Aad and Collaboration) 2012)

Domain: a particle as a bump

A new particle announces itself as a small Gaussian bump riding on a smooth falling background in the invariant-mass spectrum; closure separates the excess from the continuum the eye extrapolates beneath it. The ATLAS collaboration reported such an excess near 125 GeV, the signature of the Higgs boson (Aad and Collaboration) 2012).

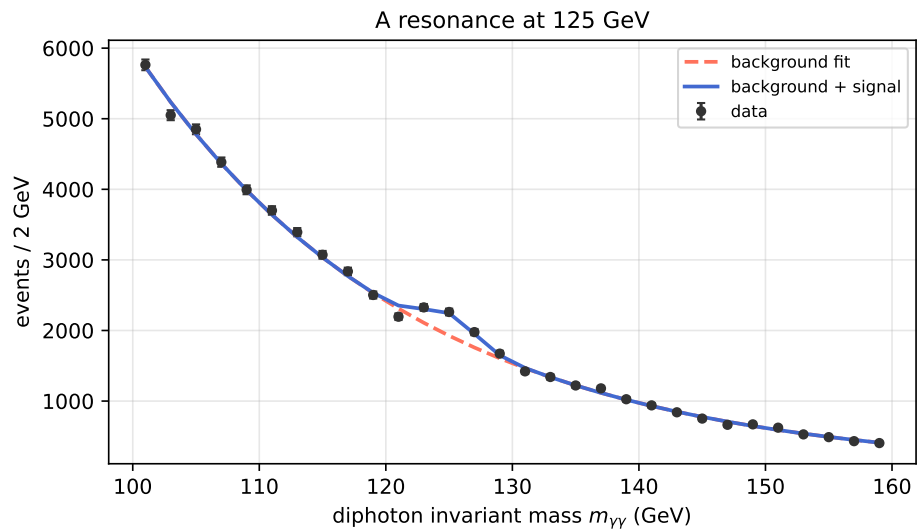


Figure 6.45: Diphoton invariant-mass spectrum: a small bump near 125 GeV over a falling background.

6.5.4 85 — Double-slit interference

Class: canonical reference

Canonical introduction: (Young 1804)

Domain: the wave pattern itself

Two slits produce a cosine fringe pattern modulated by a single-slit envelope, and the regular banding is read as one continuous wave field. Young's Bakerian lecture argued from exactly this pattern that light interferes, and so behaves as a wave (Young 1804).

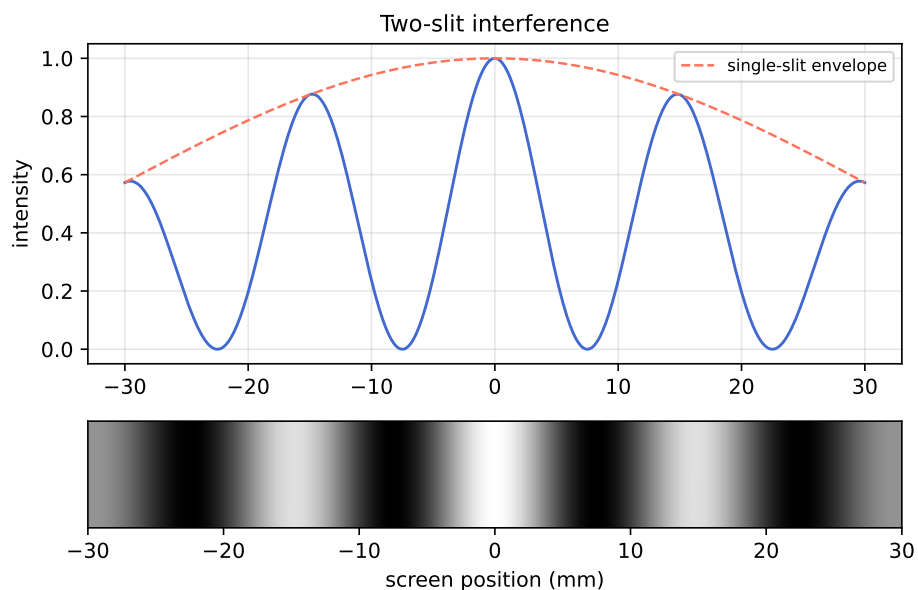


Figure 6.46: Two-slit intensity under a single-slit envelope, with the fringe strip below.

6.5.5 86 — Fourier decomposition + spectrogram

Class: named first

Canonical introduction: (Fourier 1822)

Domain: signal as frequency

A tangled time signal resolves into a handful of sharp lines once redrawn against frequency, each peak a sinusoidal component the ear and eye recover as one tone. Fourier proved that any such signal decomposes into a sum of sines and cosines (Fourier 1822).

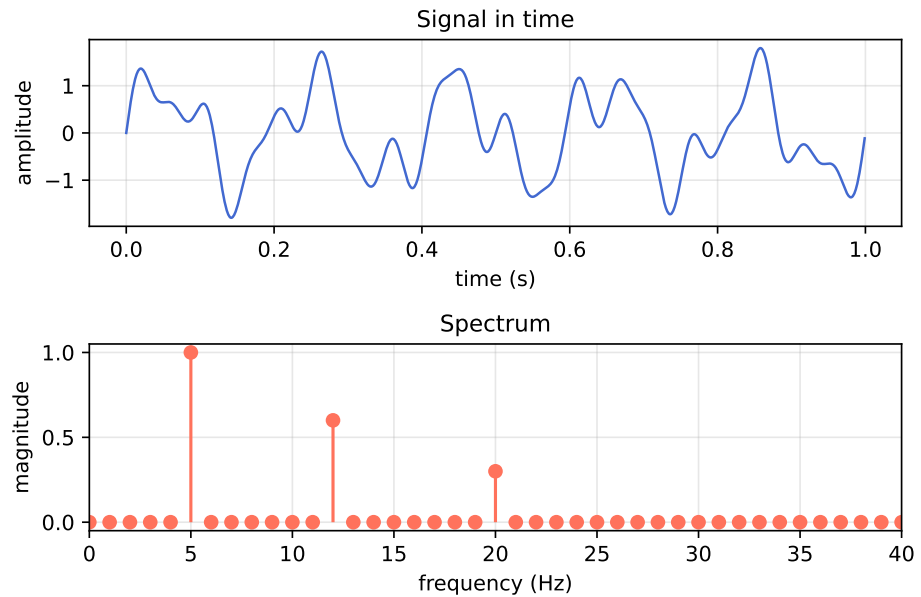


Figure 6.47: A signal built from three sines (top) and its frequency spectrum (bottom).

6.5.6 87 — Plate tectonics + seafloor-age stripes

Class: named first

Canonical introduction: (Vine and Matthews 1963)

Domain: the symmetric stripes as one pattern

Magnetic anomalies recorded in new crust form a barcode of normal and reversed stripes that mirrors exactly across the ridge axis, and the symmetry reads as a single pattern generated by spreading. Vine and Matthews argued that these matched stripes record seafloor spreading writing the field's reversals into the rock (Vine and Matthews 1963).

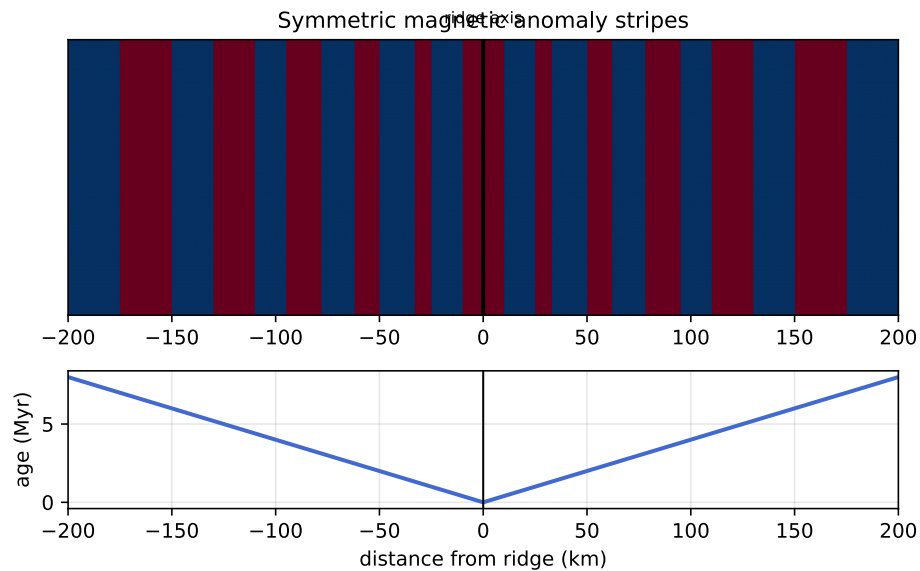


Figure 6.48: Magnetic stripes mirrored about a mid-ocean ridge, with seafloor age increasing outward.

7 Part C — Beyond the static frame

The static frame is left behind here — systems that move over time, figures whose whole point is animation, and the honest depiction of what is not known. Uncertainty is drawn rather than suppressed; where a form is genuinely animated, a print or HTML book shows it as one representative still or a short small-multiple sequence.

7.1 C1. Dynamical systems, chaos, and emergence

*After Strogatz, *Nonlinear Dynamics and Chaos* (1994).*

7.1.1 88 — Lotka–Volterra phase portrait

Class: named first

Canonical introduction: (Lotka 1925)

Domain: predator–prey limit cycle

Predator and prey abundances chase each other around a closed orbit rather than settling to a point, and plotting one against the other turns a pair of coupled rate equations into a family of nested loops. Lotka set out the equations in 1925 (Lotka 1925); each initial condition traces its own conserved cycle, and the inset time series shows the same motion as the out-of-phase peaks familiar from population records.

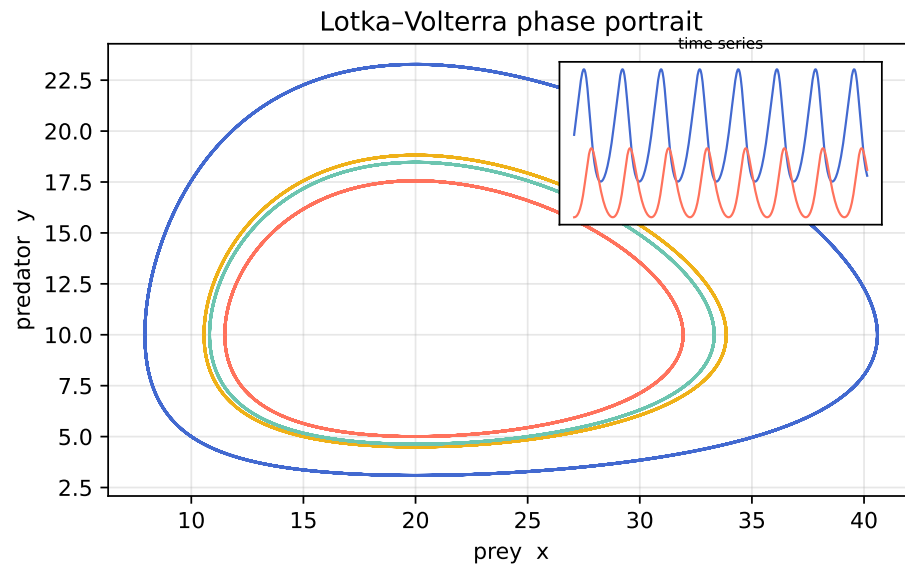


Figure 7.1: Predator–prey phase portrait: closed orbits, with an inset time series.

7.1.2 89 — The strange attractor

Class: named first

Canonical introduction: (Lorenz 1963)

Domain: deterministic chaos with structure

A deterministic system with no noise can still be unpredictable, and its trajectory need not wander freely — it settles onto a set with fixed geometry that it never leaves and never repeats. Lorenz found this in a three-variable convection model in 1963 (Lorenz 1963); the x – z projection is the double-lobed butterfly whose two wings the state orbits in an order no forecast can pin down.

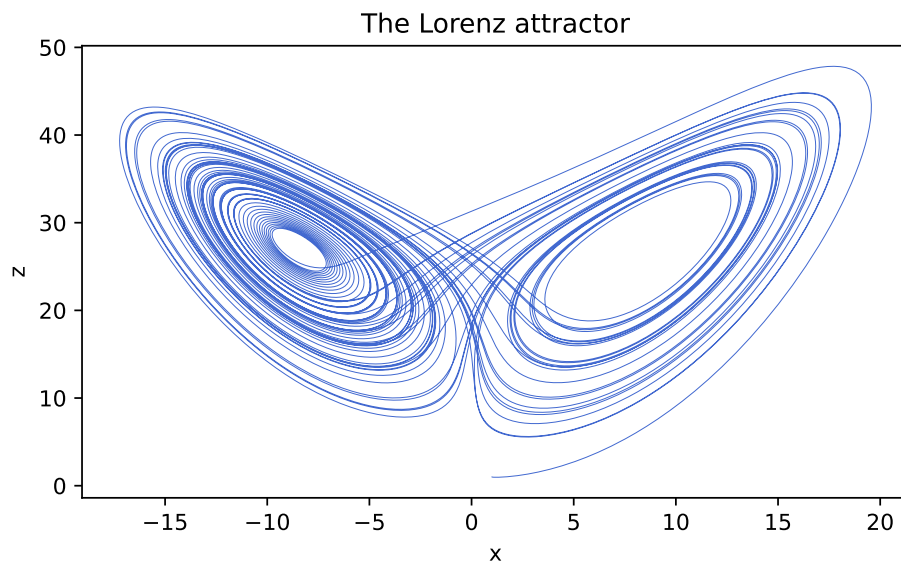


Figure 7.2: The Lorenz attractor, x-z projection ($\sigma=10$, $\rho=28$, $\beta=8/3$).

7.1.3 90 — The bifurcation diagram

Class: named first

Canonical introduction: (May 1976)

Domain: the road to chaos

Raising a single control parameter takes a system from one stable state, to an oscillation between two, then four, then a cascade of doublings that ends in chaos. May's 1976 review made the logistic map the emblem of this route (May 1976); iterating $x = r \cdot x(1-x)$ and scattering the settled values against r draws the period-doubling tree with its self-similar windows of order inside the chaos.

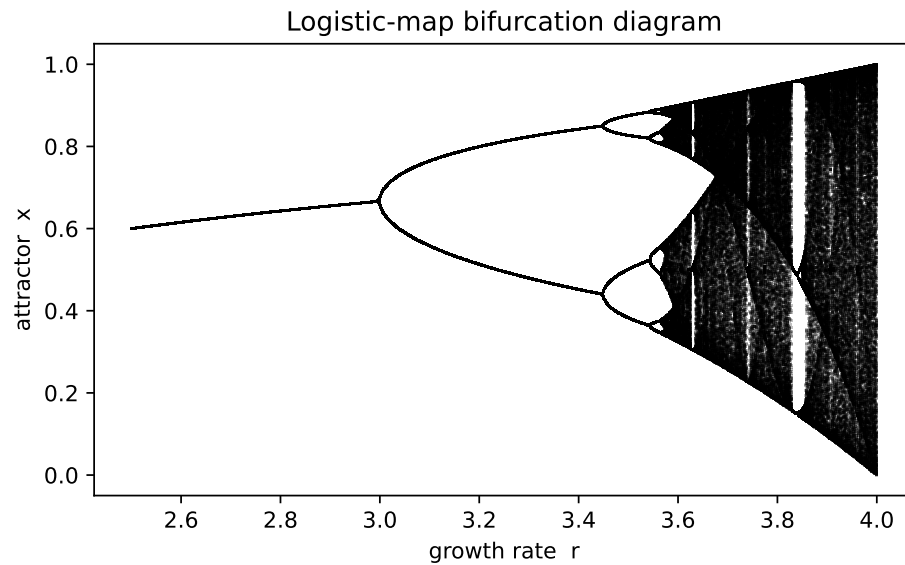


Figure 7.3: Logistic-map bifurcation diagram: attractor values versus growth rate r .

7.1.4 91 — Double pendulum

Class: shared genre idiom

Canonical introduction: (Shinbrot et al. 1992)

Domain: sensitive dependence you can watch

Two double pendulums released from all-but-identical angles keep step for a moment and then swing wildly apart, making sensitive dependence on initial conditions visible without any equations on the page. Shinbrot and colleagues used exactly this apparatus to demonstrate chaos in a table-top mechanical system (Shinbrot et al. 1992); the tip traces below start 0.01 radians apart and diverge within seconds. The physical form is animated — a single still stands in for the motion.

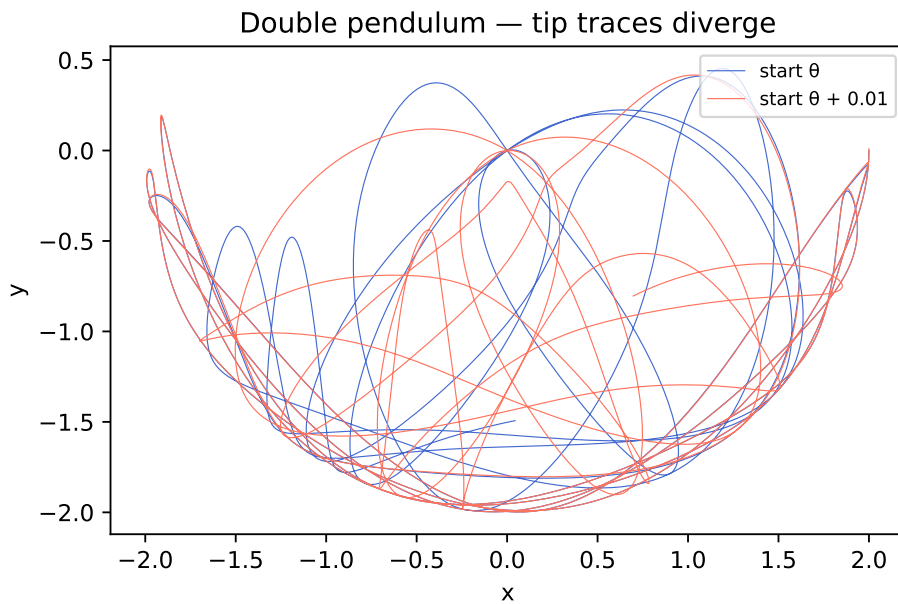


Figure 7.4: Two double-pendulum tip traces from near-identical starts, diverging.

7.1.5 92 — The Mandelbrot set / fractal

Class: named first

Canonical introduction: (Mandelbrot 1982)

Domain: self-similarity at every scale

The boundary of the Mandelbrot set has detail at every magnification, an infinitely intricate coastline generated by iterating one quadratic map. Mandelbrot's 1982 book gave fractal geometry its name and this figure its icon (Mandelbrot 1982); colouring each point c by how fast $z = z^2 + c$ escapes to infinity draws the familiar bulbed body and its filigree edge.

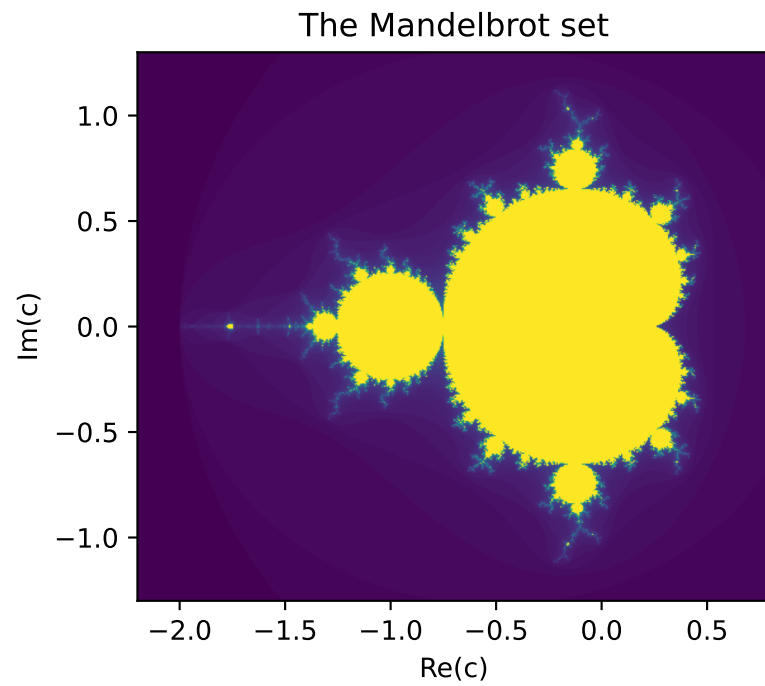


Figure 7.5: The Mandelbrot set, coloured by escape-time iteration count.

7.1.6 93 — Reaction–diffusion (Turing) patterns

Class: named first

Canonical introduction: (Turing 1952)

Domain: pattern self-organising

Two chemicals that react and diffuse at different rates can break a uniform field into spots and stripes on their own, no template required. Turing proposed this mechanism for biological pattern in 1952 (Turing 1952); a short Gray–Scott simulation started from a small seed self-organises into the maze of blobs shown here.

Reaction-diffusion (Turing) pattern

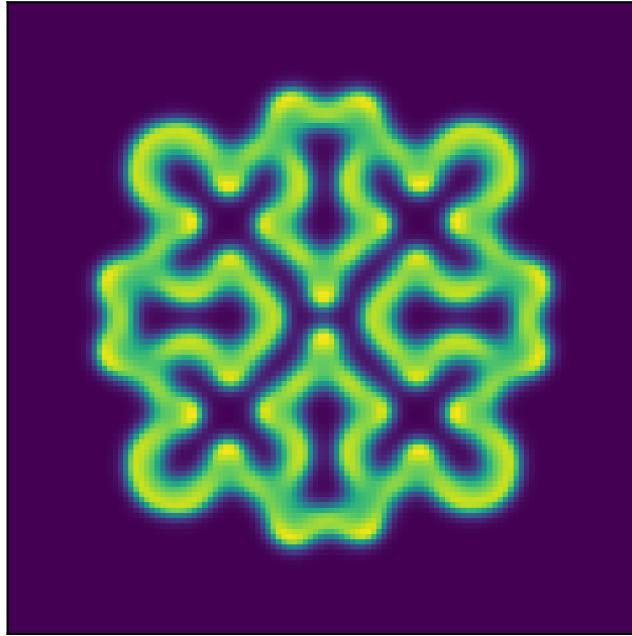


Figure 7.6: A Gray–Scott reaction–diffusion field self-organised into spots and stripes.

7.1.7 94 — Flocking / boids

Class: named first

Canonical introduction: (Reynolds 1987)

Domain: emergence from local rules

Coherent flocking needs no leader and no global plan — three local rules, keep apart, match your neighbours' heading, steer toward their centre, are enough for a crowd to move as one. Reynolds introduced this model of

“boids” in 1987 (Reynolds 1987); after a few hundred steps the velocity arrows below have aligned into a common drift.

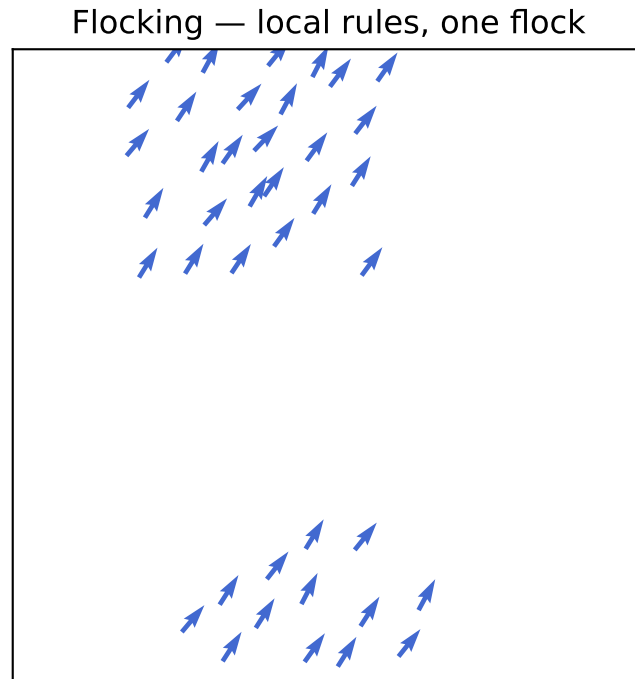


Figure 7.7: Boids after simulation: positions with velocity arrows aligned into a flock.

7.1.8 95 — SIR epidemic curve

Class: named first

Canonical introduction: (Kermack and McKendrick 1927)

Domain: contagion dynamics

7.2 C2. Interaction, animation, and narrative

Splitting a population into susceptible, infective, and recovered compartments and letting infection flow between them produces the single-humped epidemic curve, with its peak and its burnt-out tail. Kermack and McKendrick set out the compartmental equations in 1927 (Kermack and McKendrick 1927); the infective fraction rises, crests when the susceptible pool runs thin, and decays.

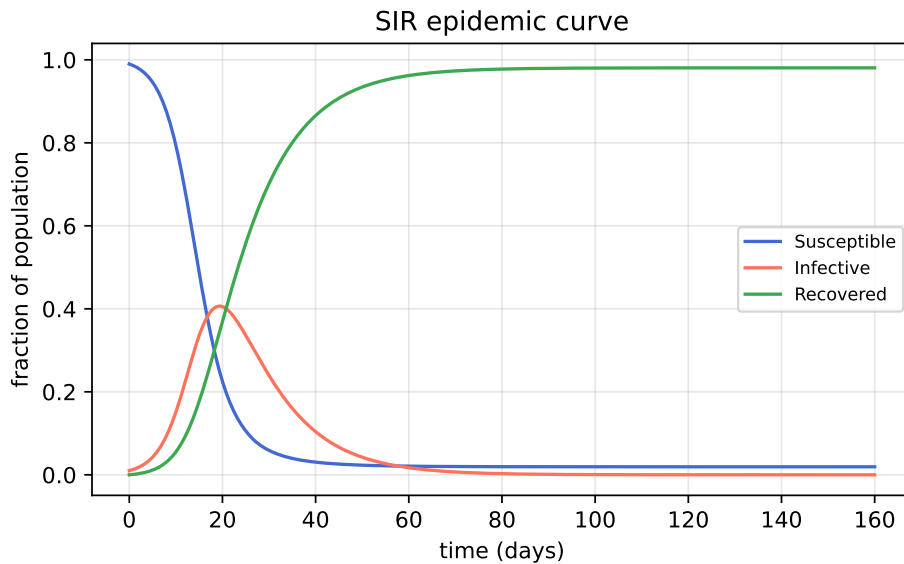


Figure 7.8: SIR compartments over time, with the classic infective peak.

7.2 C2. Interaction, animation, and narrative

After Cairo, The Functional Art (2012).

7.2.1 96 — Gapminder bubbles

Class: named first

Canonical introduction: (Rosling et al. 2007)

Domain: wealth vs health vs time, animated

Four variables ride on one scatter — income on a log x-axis, life expectancy on y, population as bubble area, region as colour — and running it across years turns development into visible motion. Rosling’s Gapminder presentations made this the definitive animated statistical graphic (Rosling et al. 2007); the frame below is one representative year, since the animation over time collapses to a single still in print.

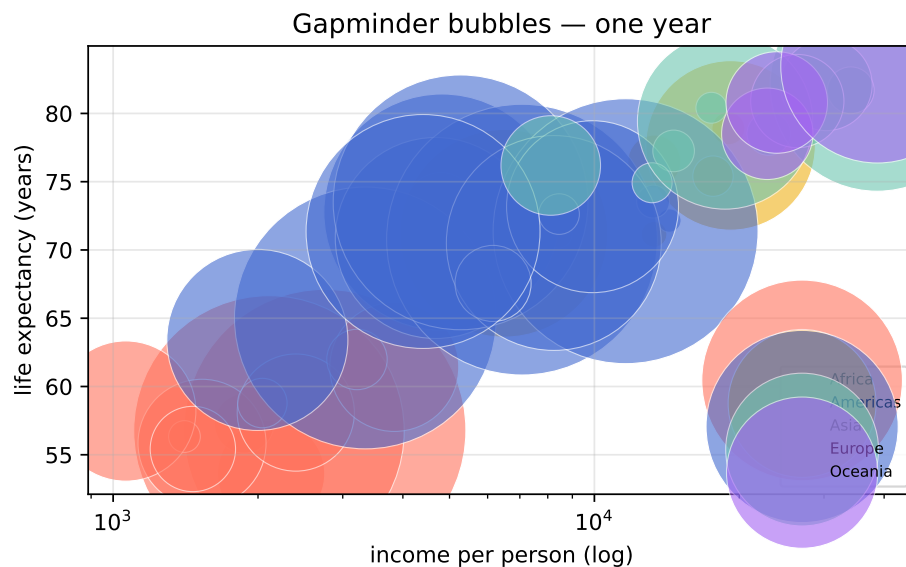


Figure 7.9: Gapminder bubbles for one representative year (synthetic country data).

7.2.2 97 — Sequence alignment / genome browser

Class: named first

Canonical introduction: (Kent et al. 2002)

Domain: zoom from chromosome to base

A genome browser stacks heterogeneous data on a shared coordinate axis — gene models, read coverage, conservation — so that any locus can be read across all tracks at once and zoomed from whole chromosome down to single base. Kent’s UCSC browser fixed this track layout for genomics in 2002 (Kent et al. 2002); genes appear as exon boxes joined by intron lines above histogram and conservation tracks.

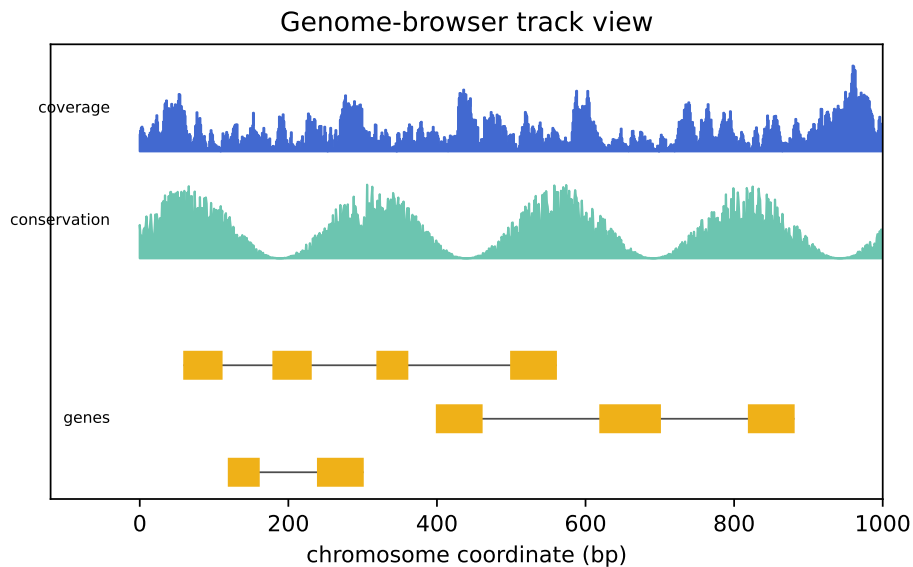


Figure 7.10: A genome-browser track view: genes, coverage, and conservation on one axis.

7.2.3 98 — **Geologic time spiral / stratigraphic column**

Class: canonical reference

Canonical introduction: (Smith 1815)

Domain: deep time compressed

A stratigraphic column stacks the eons and eras vertically with thickness scaled to duration, compressing four and a half billion years into a single legible strip that reads oldest at the bottom. William Smith's 1815 geological map established stratigraphic order as the organising principle of deep time (Smith 1815); the column below colours each interval and sizes it by the span it covers.

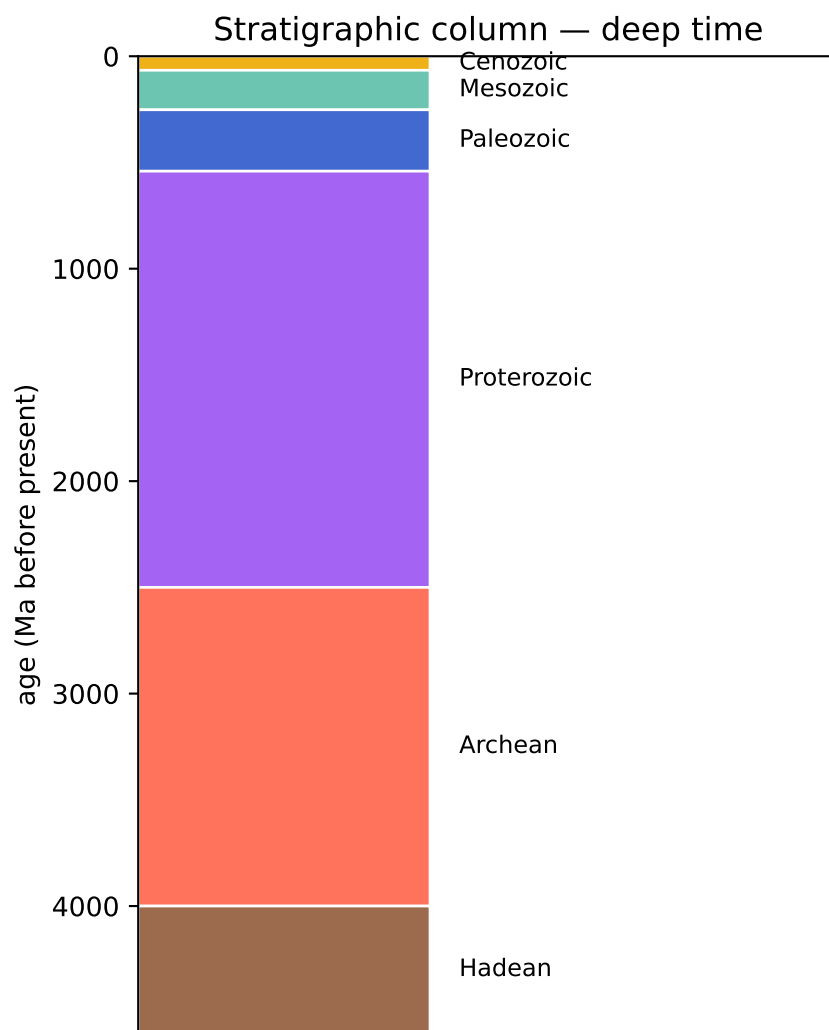


Figure 7.11: A stratigraphic column: eons and eras stacked, thickness scaled to duration.

7.3 C3. Encoding uncertainty — and reading wisely

After Spiegelhalter, The Art of Statistics (2019), and Tufte on the honest depiction of uncertainty.

7.3.1 99 — Confidence / credible intervals, error bars, fan charts

Class: shared genre idiom

Canonical introduction: (Neyman 1937)

Domain: uncertainty made visible

An estimate without its interval is half a statement; error bars and shaded bands attach the range of plausible values to every point and every fitted curve. Neyman's 1937 paper defined the confidence interval that these bars most often carry (Neyman 1937); a wide bar and a narrow one side by side say directly how much a number can be trusted.

7.3 C3. Encoding uncertainty — and reading wisely

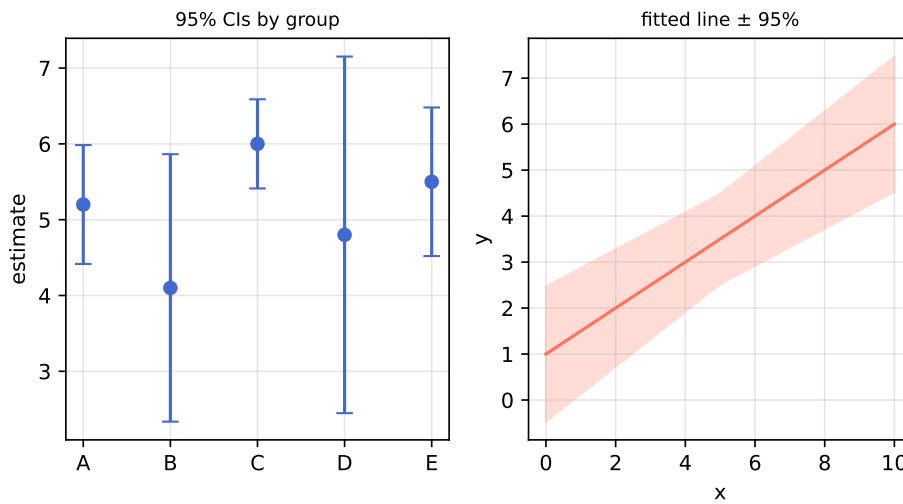


Figure 7.12: Point estimates with 95% error bars, and a fitted line with a shaded band.

7.3.2 100 — Poll-of-polls fan chart

Class: canonical reference

Canonical introduction: (Britton et al. 1998)

Domain: uncertainty as the message

A fan chart puts uncertainty at the centre of the graphic: a central projection surrounded by nested predictive bands that fan wider the further ahead the forecast reaches. Britton, Fisher and Whitley described the Bank of England's inflation fan in 1998 (Britton et al. 1998); the darkening-to-lightening bands say plainly that the future is a distribution, not a line.

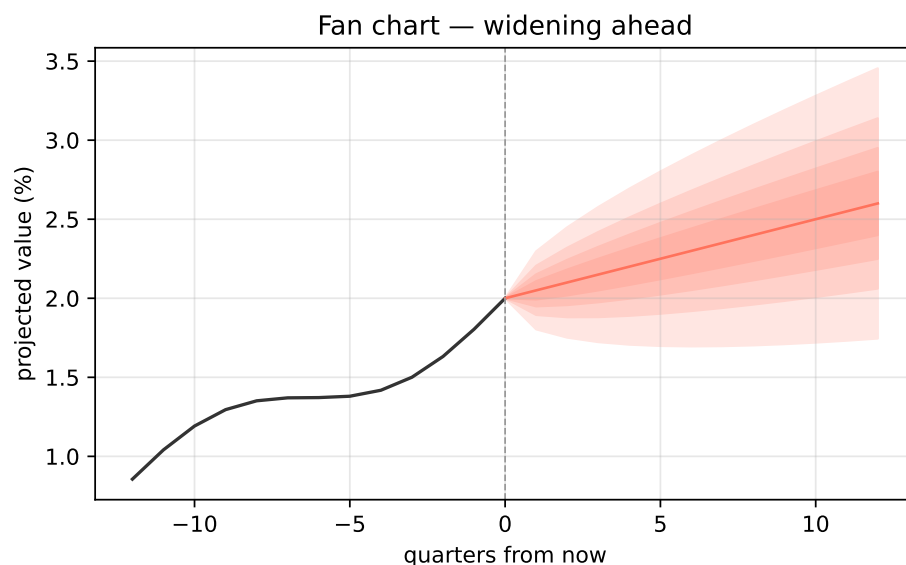


Figure 7.13: A fan chart: central projection with predictive bands widening ahead.

7.3.3 101 — Bayesian updating

Class: canonical reference

Canonical introduction: (Bayes and Price 1763)

Domain: prior to posterior

Bayesian updating is one picture: a prior belief, multiplied by the likelihood of the data, renormalised into a posterior that is sharper than either. The rule traces to Bayes's essay published in 1763 (Bayes and Price 1763); a Beta prior times a binomial likelihood gives a Beta posterior, and plotting all three on one axis shows belief concentrating as evidence arrives.

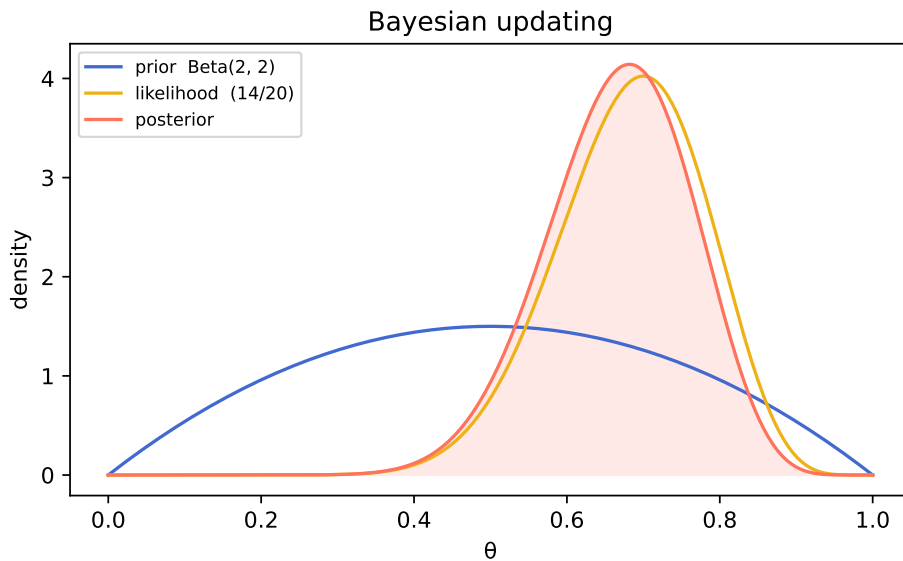


Figure 7.14: Prior, likelihood, and posterior densities — the posterior sharpens.

7.3.4 102 — ROC curve + confusion matrix

Class: named first

Canonical introduction: (Green and Swets 1966)

Domain: decision under a threshold

Any classifier that scores cases trades false alarms against missed hits as its threshold slides, and the ROC curve plots that whole trade-off while the confusion matrix pins down one chosen operating point. Green and Swets brought signal-detection theory and the ROC curve into psychophysics in 1966 (Green and Swets 1966); the area under the curve summarises separability, and the 2×2 grid counts the four outcomes at the threshold picked.

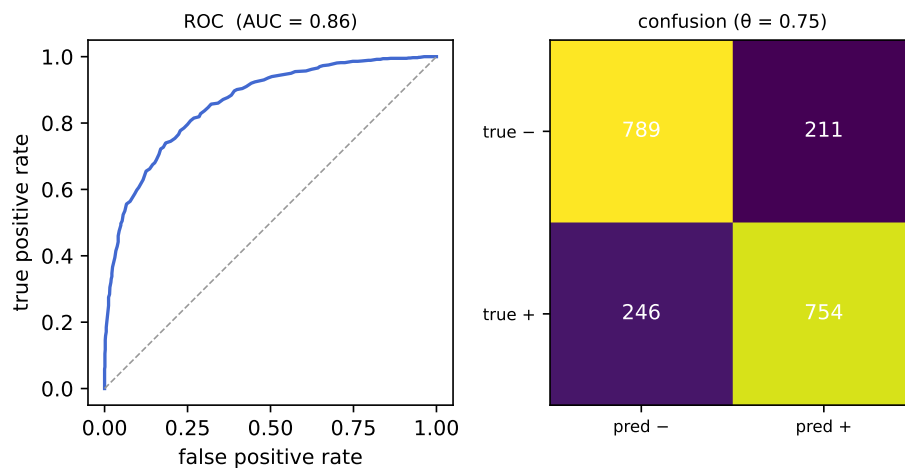


Figure 7.15: ROC curve with AUC (left) and a confusion matrix at a chosen threshold (right).

7.3.5 103 — Null distribution / permutation test

Class: canonical reference

Canonical introduction: (Fisher 1935)

Domain: what “significant” means

A permutation test makes the null hypothesis concrete: shuffle the group labels many times, recompute the statistic each time, and the spread of those values is the distribution to beat. Fisher set out this logic of randomisation in 1935 (Fisher 1935); marking the observed difference against the shuffled histogram turns the p-value into a visible tail fraction.

7.3 C3. Encoding uncertainty — and reading wisely

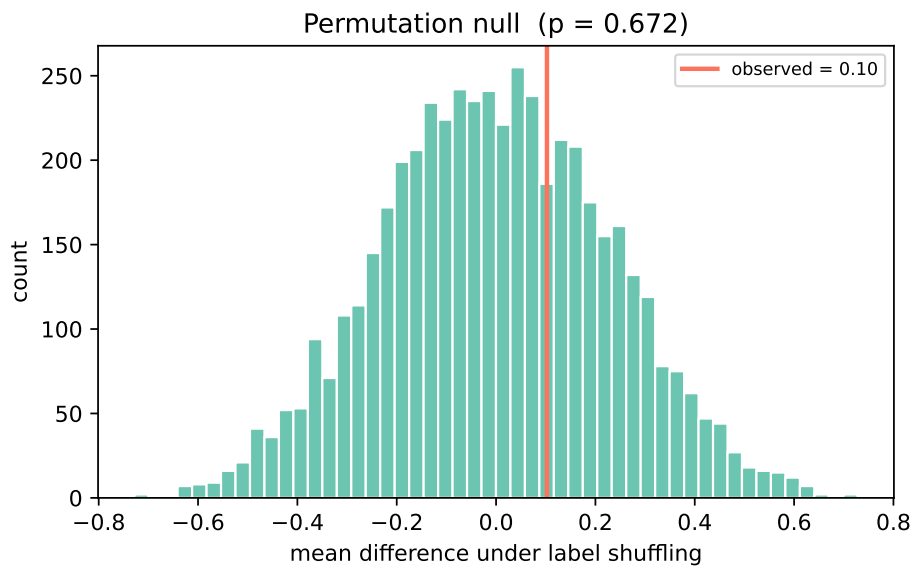


Figure 7.16: A permutation null distribution with the observed statistic marked in the tail.

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